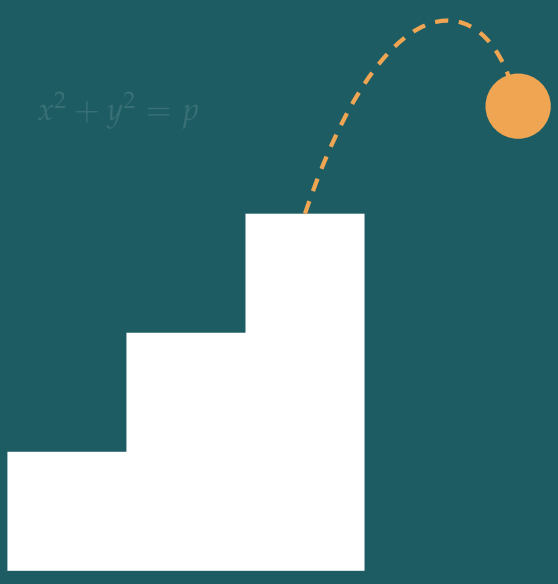
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2011 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2011 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2011 Worked Solutions

Worked solutions to all 39 questions of the 2011 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

First published online November 2026.

Everything in this PDF that is the author's own work, including the text, solutions, figures, exercises and answers, is dedicated to the public domain under the Creative Commons CC0 1.0 Universal licence: you may copy, adapt and share it for any purpose, without asking permission. See `creativecommons.org/publicdomain/zero/1.0`.

The STEP questions discussed here are the copyright of Cambridge University Press & Assessment. They are not reproduced: each one links to the question at an official source, the STEP Support Programme's question database or OCR's past papers.

Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP I

Q1 Implicit Differentiation Pure . . . 3
Q2 Integration by Substitution Pure . . . 7
Q3 Trigonometric Identities Pure . . . 10
Q4 Tangents to a Parabola Pure . . . 13
Q5 Integrals with Modulus Pure . . . 17
Q6 Binomial Series Summation Pure . . . 20
Q7 Separable Differential Equations Pure . . . 23
Q8 Diophantine Equations Pure . . . 27
Q9 Projectile Motion Mechanics . . . 30
Q10 Elastic Collisions Mechanics . . . 33
Q11 Equilibrium of a Rigid Body Mechanics . . . 37
Q12 Queues and Ballot Problem Probability and Statistics . . . 40
Q13 Continuous Random Variables Probability and Statistics . . . 44

STEP II

Q1 Equations with Square Roots Pure . . . 47
Q2 Diophantine Equations Pure . . . 51
Q3 Inequalities via Monotonicity Pure . . . 54
Q4 Trigonometric Equations Pure . . . 58
Q5 Vectors and Reflections Pure . . . 62
Q6 Integration by Parts Pure . . . 65
Q7 Sequences and Series Pure . . . 68
Q8 Involute of a Circle Pure . . . 71
Q9 Collisions and Restitution Mechanics . . . 75
Q10 Projectiles and Impulse Mechanics . . . 78
Q11 Equilibrium in Three Dimensions Mechanics . . . 81
Q12 Geometric Series of Games Probability and Statistics . . . 85
Q13 Measures of Skewness Probability and Statistics . . . 89

STEP III

Q1 Reduction of Order Pure . . . 93

Q2 Rational Roots of Polynomials Pure . . . 97

Q3 Solving Cubic Equations Pure . . . 100

Q4 Young's Inequality Pure . . . 104

Q5 Areas by Integration Pure . . . 108

Q6 Integral Substitutions Pure . . . 112

Q7 Proof by Induction Pure . . . 115

Q8 Complex Loci Pure . . . 118

Q9 Motion on a Cylinder Mechanics . . . 122

Q10 Elastic Strings and SHM Mechanics . . . 125

Q11 Rigid Body Rotation Mechanics . . . 129

Q12 Compound Distributions via PGFs Probability and Statistics . . . 133

Q13 Mode of Discrete Distributions Probability and Statistics . . . 136

WORKED SOLUTION

STEP I 2011, Question 1

Pure • Implicit Differentiation

> **STEP I 2011, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/11-s1-q1

The key idea

Implicit differentiation gives the gradient of each curve without solving for y. After that, every part is a matter of writing down the conditions at the meeting point (p, q): it lies on the line, it lies on the curve, and the gradients are related (equal for touching, product -1 for a normal). The gradient condition gives a simple link between p and q, and substituting that link into the other two equations pins down p in two different ways, which must agree.

Solution

Part (i)

Differentiating $\dfrac{a}{x} + \dfrac{b}{y} = 1$ with respect to x, treating y as a function of x,

$$-\frac{a}{x^2} - \frac{b}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Since $b \neq 0$ we can rearrange to get

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{ay^2}{bx^2},$$

as required. (Both x and y are non-zero at every point of the curve, because the equation involves a/x and b/y.)

Equal gradients. The line $ax + by = 1$ has gradient $-\frac{a}{b}$. At (p, q) the curve has gradient $-\dfrac{aq^2}{bp^2}$, so

$$-\frac{aq^2}{bp^2} = -\frac{a}{b} \quad \Longrightarrow \quad \frac{q^2}{p^2} = 1 \quad \Longrightarrow \quad q^2 = p^2,$$

where we cancelled a/b, which is allowed because $a \neq 0$. Hence $p = \pm q$.

The two cases. The point (p, q) also satisfies

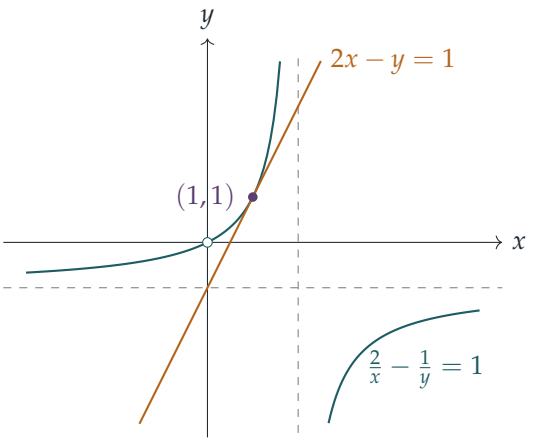
$$ap + bq = 1 \qquad \text{and} \qquad \frac{a}{p} + \frac{b}{q} = 1.$$

- If $q = p$: the line gives $p(a+b) = 1$, so $p = \dfrac{1}{a+b}$ (in particular $a + b \neq 0$). The curve gives $\dfrac{a+b}{p} = 1$, so $p = a + b$. Comparing, $a + b = \dfrac{1}{a+b}$, that is, $(a+b)^2 = 1$.
- If $q = -p$: the line gives $p(a-b) = 1$, so $p = \dfrac{1}{a-b}$. The curve gives $\dfrac{a-b}{p} = 1$, so $p = a - b$. Comparing, $(a-b)^2 = 1$.

So either $(a-b)^2 = 1$ or $(a+b)^2 = 1$, as required.

> **Ask yourself** Do such points really exist, or have we only found a necessary condition?

They do exist. Take $a = 2$, $b = -1$, so $a + b = 1$. The point $(1, 1)$ lies on the line $2x - y = 1$ and on the curve $\frac{2}{x} - \frac{1}{y} = 1$, and both have gradient 2 there. Here the curve is $y = \dfrac{x}{2-x}$, a rectangular hyperbola with the origin removed. Equal gradients at a common point mean the line is a tangent:



The open circle marks the origin, which lies on the hyperbola $y = \frac{x}{2-x}$ but not on the curve itself.

Part (ii)

Differentiating $\dfrac{a}{x} - \dfrac{b}{y} = 1$ implicitly,

$$-\frac{a}{x^2} + \frac{b}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{ay^2}{bx^2}.$$

Suppose the line is normal to the curve at (p, q). Then $p \neq 0$ and $q \neq 0$, and

$$ap + bq = 1 \qquad \text{and} \qquad \frac{a}{p} - \frac{b}{q} = 1.$$

The tangent gradient $\dfrac{aq^2}{bp^2}$ and the line's gradient $-\dfrac{a}{b}$ are both finite and non-zero, because a, b, p, q are all non-zero. So "normal" means their product is -1:

$$\frac{aq^2}{bp^2} \cdot \left(-\frac{a}{b}\right) = -1 \quad \Longrightarrow \quad a^2q^2 = b^2p^2 \quad \Longrightarrow \quad aq = \pm bp.$$

Ask yourself Which sign is possible?

Test each sign in the curve equation.

- If $aq = bp$, then $\dfrac{b}{q} = \dfrac{a}{p}$, so $\dfrac{a}{p} - \dfrac{b}{q} = 0 \neq 1$. This sign is impossible.
- If $aq = -bp$, then $\dfrac{b}{q} = -\dfrac{a}{p}$, so the curve equation becomes $\dfrac{2a}{p} = 1$, giving $p = 2a$, and then $q = -\dfrac{bp}{a} = -2b$.

Substituting $p = 2a$, $q = -2b$ into the line equation gives

$$2a^2 - 2b^2 = 1, \qquad \text{that is,} \qquad a^2 - b^2 = \tfrac{1}{2},$$

as required. The foot of the normal is the point $(2a, -2b)$.

Checks

- For part (i), the example $a = 2$, $b = -1$ above has $(a + b)^2 = 1$ and the line touches at $(1, 1) = (p, p)$. Similarly $a = 2$, $b = 1$ has $(a - b)^2 = 1$, and the line $2x + y = 1$ touches $\frac{2}{x} + \frac{1}{y} = 1$ at $(1, -1)$.
- For part (ii), take $a = \frac{\sqrt{3}}{2}$, $b = \frac{1}{2}$, so that $a^2 - b^2 = \frac{1}{2}$, and the point $(2a, -2b) = (\sqrt{3}, -1)$. The line gives $\frac{3}{2} - \frac{1}{2} = 1$ and the curve gives $\frac{1}{2} + \frac{1}{2} = 1$. The tangent gradient there is $\dfrac{aq^2}{bp^2} = \frac{1}{\sqrt{3}} \approx 0.577$ and the line's gradient is $-\sqrt{3} \approx -1.732$, with product -1.
- Numerically, the gradient formulae agree with finite-difference gradients at hundreds of random points, and a random search over (a, b) finds normals only when $a^2 - b^2$ is within rounding of 0.5.

Common slips

- Dropping the case $q = -p$ in part (i): it is this case that gives $(a - b)^2 = 1$.

- In part (ii), stopping at $aq = \pm bp$ without checking which sign the curve allows.

Beyond STEP

Implicit differentiation is a first glimpse of the implicit function theorem: near any point where $\partial F/\partial y \neq 0$, the equation $F(x, y) = 0$ defines y as a differentiable function of x, with $\mathrm{d}y/\mathrm{d}x = -F_x/F_y$. The tangency argument here also has an algebraic twin: a line touches a conic exactly when the quadratic for their intersections has a repeated root, and you can rederive part (i) that way. For the analysis behind all this, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2011, Question 2

Pure • Integration by Substitution

STEP I 2011, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q2

The key idea

The integral E has no elementary closed form, so we never try to evaluate it. Instead, every other integral is to be manipulated until E itself appears. For the first two, this means splitting the fraction by algebraic division so that the troublesome $\frac{1}{1+x}$ is isolated. For (i) and (ii), it means choosing a substitution that turns the integrand back into $\frac{\mathrm{e}^u}{1+u}$ on $[0,1]$: look at the exponent and ask what would make it the new variable.

Solution

The first two integrals

Writing $x = (1+x) - 1$,

$$\frac{x\,\mathrm{e}^x}{1+x} = \mathrm{e}^x - \frac{\mathrm{e}^x}{1+x},$$

so

$$\int_0^1 \frac{x\,\mathrm{e}^x}{1+x}\,\mathrm{d}x = \int_0^1 \mathrm{e}^x\,\mathrm{d}x - E = \mathrm{e} - 1 - E,$$

as required.

For the second, divide x^2 by $1+x$: since $x^2 = (x-1)(1+x) + 1$,

$$\frac{x^2\,\mathrm{e}^x}{1+x} = x\,\mathrm{e}^x - \mathrm{e}^x + \frac{\mathrm{e}^x}{1+x}.$$

Integrating by parts with $u = x$, $\mathrm{d}v = \mathrm{e}^x\,\mathrm{d}x$,

$$\int_0^1 x\,\mathrm{e}^x\,\mathrm{d}x = \left[x\,\mathrm{e}^x\right]_0^1 - \int_0^1 \mathrm{e}^x\,\mathrm{d}x = \mathrm{e} - (\mathrm{e} - 1) = 1.$$

Hence

$$\int_0^1 \frac{x^2\,\mathrm{e}^x}{1+x}\,\mathrm{d}x = 1 - (\mathrm{e} - 1) + E = 2 - \mathrm{e} + E.$$

(Alternatively, write $\frac{x^2}{1+x} = x - \frac{x}{1+x}$ and use the first result directly: $1 - (\mathrm{e} - 1 - E)$ gives the same answer.)

Part (i)

The exponent is the awkward part, so we make it the new variable:

$$u = \frac{1-x}{1+x}.$$

Solving for x: $u(1+x) = 1-x$, so $x = \dfrac{1-u}{1+u}$. Then

$$1+x = \frac{2}{1+u}, \qquad \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{-(1+u)-(1-u)}{(1+u)^2} = -\frac{2}{(1+u)^2}.$$

When $x = 0$, $u = 1$; when $x = 1$, $u = 0$. Therefore

$$\begin{aligned}\int_0^1 \frac{\mathrm{e}^{\frac{1-x}{1+x}}}{1+x}\,\mathrm{d}x &= \int_1^0 \mathrm{e}^u \cdot \frac{1+u}{2} \cdot \left(-\frac{2}{(1+u)^2}\right)\mathrm{d}u \\ &= \int_0^1 \frac{\mathrm{e}^u}{1+u}\,\mathrm{d}u = E.\end{aligned}$$

The minus sign from $\mathrm{d}x/\mathrm{d}u$ was used to swap the limits back into increasing order.

Ask yourself Why should we expect the answer to be exactly E, with no e at all?

The map $x \mapsto \frac{1-x}{1+x}$ sends $[0,1]$ to itself (it swaps the endpoints), and the factor $\frac{1}{1+x}$ combines with $\mathrm{d}x$ to give exactly $\frac{\mathrm{d}u}{1+u}$. Indeed $\frac{\mathrm{d}x}{1+x} = -\frac{\mathrm{d}u}{1+u}$, which is a neat way to remember the whole computation.

Part (ii)

Now the target interval $[0,1]$ must come from $x \in [1, \sqrt{2}]$, where x^2 runs from 1 to 2. So we try

$$u = x^2 - 1, \qquad \mathrm{d}u = 2x\,\mathrm{d}x,$$

with $u = 0$ at $x = 1$ and $u = 1$ at $x = \sqrt{2}$. Then

$$\frac{\mathrm{d}x}{x} = \frac{x\,\mathrm{d}x}{x^2} = \frac{\mathrm{d}u}{2(1+u)}, \qquad \mathrm{e}^{x^2} = \mathrm{e}^{1+u} = \mathrm{e}\cdot\mathrm{e}^u,$$

so

$$\int_1^{\sqrt{2}} \frac{\mathrm{e}^{x^2}}{x}\,\mathrm{d}x = \frac{\mathrm{e}}{2}\int_0^1 \frac{\mathrm{e}^u}{1+u}\,\mathrm{d}u = \frac{\mathrm{e}E}{2}.$$

Ask yourself Why not substitute $u = x^2$?

That would work too, giving $\frac{1}{2}\int_1^2 \frac{\mathrm{e}^u}{u}\,\mathrm{d}u$, but the interval is $[1,2]$ rather than $[0,1]$. Shifting by one, $u \mapsto u-1$, lands on the same answer; $u = x^2 - 1$ simply does both steps at once.

Checks

Simpson's rule gives $E \approx 1.125386$. Numerically, the first integral is $0.592896 = \mathrm{e} - 1 - E$, the second is $0.407104 = 2 - \mathrm{e} + E$, the integral in (i) is $1.125386 = E$, and the integral in (ii) is $1.529558 = \frac{1}{2}\mathrm{e}E$. A quick sanity check without a computer: the first two integrands are both positive, and indeed $\mathrm{e} - 1 - E \approx 0.59 > 0$ and $2 - \mathrm{e} + E \approx 0.41 > 0$; also the second is smaller than the first, as it should be since $x^2 \leq x$ on $[0, 1]$.

Common slips

- Forgetting to reverse the limits (or dropping the minus sign) in (i), which gives $-E$, an impossible answer for a positive integrand.
- In (ii), losing the factor e from $\mathrm{e}^{x^2} = \mathrm{e}^{1+u}$, or the factor $\frac{1}{2}$ from $\mathrm{d}u = 2x\,\mathrm{d}x$.

Beyond STEP

Integrals like E are tied to the *exponential integral* $\mathrm{Ei}(x) = \int_{-\infty}^{x} \frac{\mathrm{e}^t}{t}\,\mathrm{d}t$: substituting $t = 1 + x$ shows $E = \mathrm{e}^{-1}\big(\mathrm{Ei}(2) - \mathrm{Ei}(1)\big)$. Liouville's theory shows that $\int \frac{\mathrm{e}^x}{x}\,\mathrm{d}x$ has no elementary antiderivative, which is exactly why the question treats E as a named constant and asks us to reduce everything to it. Siklos, *Advanced Problems in Mathematics*, has many more questions in this spirit.

WORKED SOLUTION

STEP I 2011, Question 3

Pure • Trigonometric Identities

STEP I 2011, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q3

The key idea

The product of two sines at $\frac{1}{3}\pi \mp \theta$ collapses neatly by a difference-of-squares trick, which proves $(*)$ in a line. After that, the cotangent sums come from differentiating a product: the derivative of a product of three factors, divided by the product itself, is a sum of three "logarithmic derivatives", and for sines those are cotangents. Doing the same to the cosine version gives tangents, and $\tan x + \cot x$ is a cosecant.

Solution

The identity $(*)$

Write $a = \frac{1}{3}\pi$. By the compound-angle formulae,

$$\begin{aligned}\sin(a-\theta)\sin(a+\theta) &= (\sin a\cos\theta - \cos a\sin\theta)(\sin a\cos\theta + \cos a\sin\theta)\\ &= \sin^2 a\cos^2\theta - \cos^2 a\sin^2\theta\\ &= \sin^2 a\,(1-\sin^2\theta) - (1-\sin^2 a)\sin^2\theta = \sin^2 a - \sin^2\theta.\end{aligned}$$

With $\sin^2 a = \frac{3}{4}$,

$$4\sin\theta\sin(\tfrac{1}{3}\pi-\theta)\sin(\tfrac{1}{3}\pi+\theta) = 4\sin\theta\left(\tfrac{3}{4}-\sin^2\theta\right) = 3\sin\theta - 4\sin^3\theta = \sin 3\theta,$$

by the triple-angle formula. This proves $(*)$, for every θ.

Part (i)

Since $(*)$ holds for all θ, we may differentiate both sides. By the product rule,

$$\begin{aligned}4\Big[\cos\theta\sin(a-\theta)\sin(a+\theta) - \sin\theta\cos(a-\theta)\sin(a+\theta)\\ + \sin\theta\sin(a-\theta)\cos(a+\theta)\Big] = 3\cos 3\theta,\end{aligned}$$

where the minus sign in the middle term comes from the chain rule applied to $a-\theta$.

Ask yourself How do we turn this into cotangents?

Divide both sides by $\sin 3\theta$, which by $(*)$ equals $4\sin\theta\sin(a-\theta)\sin(a+\theta)$. Each term on the left loses all its sines except the one that was differentiated. This is allowed whenever $\sin 3\theta \neq 0$, and then all three sines are non-zero too. We get

$$\cot\theta - \cot(\tfrac{1}{3}\pi-\theta) + \cot(\tfrac{1}{3}\pi+\theta) = 3\cot 3\theta \qquad (\sin 3\theta \neq 0). \tag{A}$$

Now put $\theta = \frac{1}{9}\pi$, where $\sin 3\theta = \sin\frac{1}{3}\pi \neq 0$. Since $\frac{1}{3}\pi - \frac{1}{9}\pi = \frac{2}{9}\pi$ and $\frac{1}{3}\pi + \frac{1}{9}\pi = \frac{4}{9}\pi$,

$$\cot\tfrac{1}{9}\pi - \cot\tfrac{2}{9}\pi + \cot\tfrac{4}{9}\pi = 3\cot\tfrac{1}{3}\pi = \frac{3}{\sqrt{3}} = \sqrt{3},$$

as required.

Part (ii)

Put $\theta = \frac{1}{6}\pi - \phi$ in $(*)$. The four sines become

$$\begin{aligned}
\sin\theta = \sin(\tfrac{1}{6}\pi-\phi) &= \cos(\tfrac{1}{3}\pi+\phi),\\
\sin(\tfrac{1}{3}\pi-\theta) = \sin(\tfrac{1}{6}\pi+\phi) &= \cos(\tfrac{1}{3}\pi-\phi),\\
\sin(\tfrac{1}{3}\pi+\theta) = \sin(\tfrac{1}{2}\pi-\phi) &= \cos\phi,\\
\sin 3\theta = \sin(\tfrac{1}{2}\pi-3\phi) &= \cos 3\phi,
\end{aligned}$$

using $\sin(\frac{1}{2}\pi - x) = \cos x$ each time. So, renaming ϕ as θ,

$$4\cos\theta\cos(\tfrac{1}{3}\pi-\theta)\cos(\tfrac{1}{3}\pi+\theta) = \cos 3\theta \qquad \text{for all } \theta. \tag{**}$$

Dividing $(*)$ by $(**)$, the factors of 4 cancel and each sine over cosine is a tangent; taking reciprocals,

$$\cot\theta\cot(\tfrac{1}{3}\pi-\theta)\cot(\tfrac{1}{3}\pi+\theta) = \cot 3\theta,$$

valid wherever both sides are defined, that is, where $\sin 3\theta \neq 0$ and $\cos 3\theta \neq 0$ (each of these makes the three corresponding factors non-zero).

For the cosecant sum, we differentiate $(**)$ just as in part (i):

$$\begin{aligned}
4\Big[&-\sin\theta\cos(a-\theta)\cos(a+\theta) + \cos\theta\sin(a-\theta)\cos(a+\theta)\\
&\qquad - \cos\theta\cos(a-\theta)\sin(a+\theta)\Big] = -3\sin 3\theta.
\end{aligned}$$

Dividing by $\cos 3\theta = 4\cos\theta\cos(a-\theta)\cos(a+\theta)$ (when it is non-zero) and multiplying by -1,

$$\tan\theta - \tan(\tfrac{1}{3}\pi-\theta) + \tan(\tfrac{1}{3}\pi+\theta) = 3\tan 3\theta. \tag{B}$$

The signs in (A) and (B) match, so we can add them.

Ask yourself Why would adding tangents and cotangents help?

Because $\tan x + \cot x$ is a single cosecant:

$$\tan x + \cot x = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{2}{\sin 2x} = 2\operatorname{cosec} 2x.$$

So adding (A) and (B), whenever $\sin 6\theta \neq 0$, gives

$$2\operatorname{cosec} 2\theta - 2\operatorname{cosec}(\tfrac{2}{3}\pi - 2\theta) + 2\operatorname{cosec}(\tfrac{2}{3}\pi + 2\theta) = 6\operatorname{cosec} 6\theta.$$

Now take $\theta = \frac{1}{18}\pi$, so that $2\theta = \frac{1}{9}\pi$ and $6\theta = \frac{1}{3}\pi$ (with $\sin 6\theta \neq 0$). Since $\frac{2}{3}\pi - \frac{1}{9}\pi = \frac{5}{9}\pi$ and $\frac{2}{3}\pi + \frac{1}{9}\pi = \frac{7}{9}\pi$, dividing by 2 gives

$$\operatorname{cosec} \tfrac{1}{9}\pi - \operatorname{cosec} \tfrac{5}{9}\pi + \operatorname{cosec} \tfrac{7}{9}\pi = 3\operatorname{cosec} \tfrac{1}{3}\pi = 3 \cdot \frac{2}{\sqrt{3}} = 2\sqrt{3},$$

as required.

Checks

- At $\theta = \frac{1}{6}\pi$, $(*)$ reads $4 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = 1 = \sin \frac{1}{2}\pi$, and $(**)$ at $\theta = 0$ reads $4 \cdot 1 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1 = \cos 0$.
- Numerically, $\cot 20^\circ - \cot 40^\circ + \cot 80^\circ \approx 2.7475 - 1.1918 + 0.1763 = 1.7321 \approx \sqrt{3}$, and $\operatorname{cosec} 20^\circ - \operatorname{cosec} 100^\circ + \operatorname{cosec} 140^\circ \approx 2.9238 - 1.0154 + 1.5557 = 3.4641 \approx 2\sqrt{3}$.
- Identities (A), (B) and the cotangent product can be tested at any random θ on a calculator; we checked them at a few thousand random values.

Common slips

- Forgetting the chain-rule minus sign when differentiating $\sin(\frac{1}{3}\pi - \theta)$, which spoils the alternating signs in (A).
- Dividing by $\sin 3\theta$ without noting that it must be non-zero at the value substituted.
- Trying to get the cosecant sum from the cotangent *product*: it is the *sums* (A) and (B) that combine.

Beyond STEP

The trick of dividing a derivative by the function is logarithmic differentiation: $\frac{\mathrm{d}}{\mathrm{d}\theta} \ln |f| = f'/f$ turns products into sums. Pushed further, sin itself can be written as an infinite product over its zeros, and logarithmic differentiation then gives the famous partial-fraction expansion $\pi \cot \pi x = \sum_n \frac{1}{x-n}$; identity $(*)$ is a finite shadow of this. These ideas are developed in complex analysis; Needham's *Visual Complex Analysis* is a beautiful place to start.

WORKED SOLUTION

STEP I 2011, Question 4

Pure • Tangents to a Parabola

> **STEP I 2011, Question 4**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/11-s1-q4

The key idea

Write the tangent at a general point of the parabola in a form with no fractions, and the intersection of two tangents drops out by subtraction. For the angle, use the scalar product of $\overrightarrow{FT}$ and $\overrightarrow{FP}$. The lengths are kinder than they look: FP is the focal distance, and FT^2 factorises. Once $\cos\phi$ is found, notice that the answer is symmetric in p and q. That symmetry is the whole of the bisection argument.

Solution

The point T

Differentiating $y^2 = 4ax$ implicitly gives $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 4a$, so at P, where $y = 2ap \neq 0$ (because $p \neq 0$), the gradient is $\dfrac{4a}{4ap} = \dfrac{1}{p}$. The tangent at P is therefore

$$y - 2ap = \frac{x - ap^2}{p} \quad \Longleftrightarrow \quad py = x + ap^2,$$

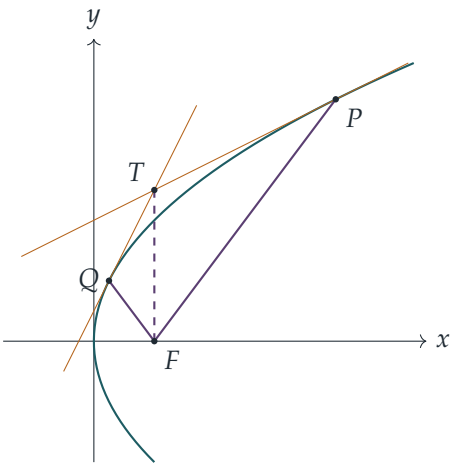
and likewise the tangent at Q is $qy = x + aq^2$. Subtracting,

$$(p - q)\,y = a(p^2 - q^2) = a(p - q)(p + q).$$

Since P and Q are distinct, $p \neq q$, so we may divide by $p - q$ to get $y = a(p + q)$. Then

$$x = py - ap^2 = ap^2 + apq - ap^2 = apq,$$

so T is the point $\big(apq,\, a(p + q)\big)$, as required.



The figure has $a = 1$, $p = 2$, $q = \frac{1}{2}$, where FT happens to be vertical.

The angle ϕ

Note first that $T \neq F$: that would need $p + q = 0$ and $pq = 1$, so $p^2 = -1$, which is impossible. So the angle TFP makes sense. With $F = (a, 0)$,

$$\overrightarrow{FP} = a\begin{pmatrix} p^2 - 1 \\ 2p \end{pmatrix}, \qquad \overrightarrow{FT} = a\begin{pmatrix} pq - 1 \\ p + q \end{pmatrix}.$$

The lengths are

$$FP^2 = a^2\left[(p^2-1)^2 + 4p^2\right] = a^2(p^2+1)^2,$$
$$FT^2 = a^2\left[(pq-1)^2 + (p+q)^2\right] = a^2\left[p^2q^2 + p^2 + q^2 + 1\right] = a^2(p^2+1)(q^2+1),$$

since the cross terms $-2pq$ and $+2pq$ cancel. So $FP = |a|(p^2+1)$ and $FT = |a|\sqrt{(p^2+1)(q^2+1)}$.

Ask yourself How do we spot that the scalar product will factorise?

We expect (p^2+1) to cancel with FP, so we look for it. Expanding,

$$\begin{aligned}\overrightarrow{FP}\cdot\overrightarrow{FT} &= a^2\left[(p^2-1)(pq-1) + 2p(p+q)\right] \\ &= a^2\left[p^3q - p^2 - pq + 1 + 2p^2 + 2pq\right] \\ &= a^2\left[p^3q + pq + p^2 + 1\right] \\ &= a^2(p^2+1)(pq+1).\end{aligned}$$

Therefore, using $|a|^2 = a^2$,

$$\cos\phi = \frac{\overrightarrow{FP}\cdot\overrightarrow{FT}}{FP\cdot FT} = \frac{a^2(p^2+1)(pq+1)}{a^2(p^2+1)\sqrt{(p^2+1)(q^2+1)}} = \frac{pq+1}{\sqrt{(p^2+1)(q^2+1)}},$$

which is the required result.

The bisection

The coordinates of T are symmetric in p and q, so swapping the roles of P and Q in the calculation above shows that the angle TFQ has

$$\cos\angle TFQ = \frac{qp+1}{\sqrt{(q^2+1)(p^2+1)}} = \cos\phi.$$

Both angles lie in $[0,\pi]$, where cosine is one-to-one, so $\angle TFP = \angle TFQ = \phi$.

So the rays FP and FQ each make the angle ϕ with the ray FT. In the plane there are only two rays from F making angle ϕ with FT, and they are mirror images in the line FT. The rays FP and FQ are not the same ray: a line through the focus meets the parabola in two points on opposite sides of F (or only once), so one ray from F cannot pass through both P and Q. Hence FP and FQ are reflections of each other in the line FT, which is exactly the statement that the line FT bisects the angle PFQ.

One detail explains why the question says the *line* FT. If $pq+1>0$ then $\phi<\frac{\pi}{2}$, and the ray FT itself lies inside the angle $PFQ = 2\phi$. If $pq+1<0$ then $\phi>\frac{\pi}{2}$, the angle PFQ is $2\pi-2\phi$, and it is the extension of TF beyond F that bisects it. If $pq=-1$ then PQ is a focal chord, $\phi=\frac{\pi}{2}$, and FT is perpendicular to it, bisecting the straight angle. In every case the line FT bisects the angle PFQ.

Checks

Take $a=1$, $p=2$, $q=\frac{1}{2}$ as in the figure: $T=(1,\frac{5}{2})$, $\overrightarrow{FP}=(3,4)$, $\overrightarrow{FQ}=(-\frac{3}{4},1)$ and $\overrightarrow{FT}=(0,\frac{5}{2})$. Then $\cos\angle TFP=\frac{4}{5}$ and $\cos\angle TFQ=\frac{1}{5/4}=\frac{4}{5}$, matching $\dfrac{2}{\sqrt{5\cdot\frac{5}{4}}}=\dfrac{4}{5}$.

A computer testing 10^5 random choices of a (of either sign), p and q confirmed the coordinates of T, the formula for $\cos\phi$ and the bisection to within 10^{-10}. It included many cases with $pq<-1$, where the backward extension of FT does the bisecting.

Common slips

- Dividing by $p-q$ without saying why it is non-zero (the points are distinct).
- Writing $FP=a(p^2+1)$, which is wrong if $a<0$. Only a^2 appears in $\cos\phi$, so the sign of a does not matter in the end.
- Saying "the cosines are equal, so FT bisects" without explaining why FP and FQ lie on opposite sides of the line FT.

Beyond STEP

This is the reflective property of the parabola in disguise: the focus sees the two tangent segments from any external point at equal angles. The same holds for the ellipse and hyperbola, and all three come from the focus–directrix definition of a conic, which

first-year geometry and projective geometry courses develop much further. For more questions in this spirit, Siklos's *Advanced Problems in Mathematics* is free and well worth working through.

WORKED SOLUTION

STEP I 2011, Question 5

Pure • Integrals with Modulus

STEP I 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q5

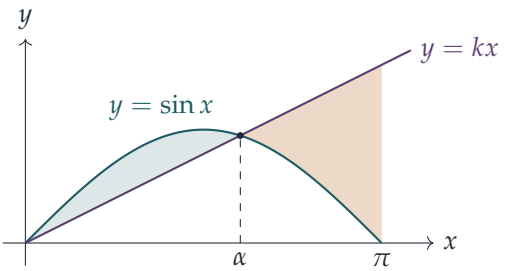
The key idea

The modulus sign is only awkward until we know where $\sin x - kx$ changes sign, and the sketch tells us: once, at $x = \alpha$. Split the integral there and the modulus disappears. The last step is to stop thinking of k as the variable: since $k = \sin\alpha/\alpha$, everything can be written in terms of α, and the derivative of I then factorises neatly.

Solution

The unique root

Let $g(x) = \sin x - kx$. The line $y = kx$ passes through the origin with gradient $k < 1$, while $y = \sin x$ leaves the origin with gradient $\cos 0 = 1$. So just to the right of 0 the sine curve is above the line. At $x = \pi$ the sine curve is back at 0 but the line is at height $k\pi > 0$, so there the line is above. Since $y = \sin x$ bends downwards on $(0, \pi)$ and the line is straight, they cross exactly once in between.



To make "exactly once" watertight: $g(0) = 0$, $g'(x) = \cos x - k$ and $g''(x) = -\sin x < 0$ on $(0, \pi)$. So g' is strictly decreasing, from $1 - k > 0$ at 0 to $-1 - k < 0$ at π, and vanishes at exactly one point β. Hence g is strictly increasing on $(0, \beta]$, so positive there (it starts at 0), and strictly decreasing on $[\beta, \pi)$, from a positive value to $g(\pi) = -k\pi < 0$.

It therefore has exactly one zero α in $(0, \pi)$, and

$$\sin x > kx \text{ for } 0 < x < \alpha, \qquad \sin x < kx \text{ for } \alpha < x < \pi.$$

Evaluating I

Using these signs (the teal and amber regions in the sketch),

$$I = \int_0^{\alpha} (\sin x - kx)\,\mathrm{d}x - \int_{\alpha}^{\pi} (\sin x - kx)\,\mathrm{d}x.$$

An antiderivative of $\sin x - kx$ is $F(x) = -\cos x - \frac{1}{2}kx^2$, so

$$\begin{aligned} I &= \big(F(\alpha) - F(0)\big) - \big(F(\pi) - F(\alpha)\big) = 2F(\alpha) - F(0) - F(\pi) \\ &= \big(-2\cos\alpha - k\alpha^2\big) + 1 - \big(1 - \tfrac{1}{2}k\pi^2\big) \\ &= \tfrac{1}{2}k\pi^2 - 2\cos\alpha - k\alpha^2. \end{aligned}$$

Since α solves $(*)$, $k = \dfrac{\sin\alpha}{\alpha}$, so $k\alpha^2 = \alpha\sin\alpha$ and

$$I = \frac{\pi^2 \sin\alpha}{2\alpha} - 2\cos\alpha - \alpha\sin\alpha,$$

as required.

The stationary value

? **Ask yourself** What values can α take?

Each $k \in (0, 1)$ gives an $\alpha \in (0, \pi)$. Conversely, each $\alpha \in (0, \pi)$ comes from $k = \sin\alpha/\alpha$, which lies in $(0, 1)$ because $0 < \sin\alpha < \alpha$ there. So we must study $I(\alpha)$ for all α in the open interval $(0, \pi)$.

Differentiating, with the quotient rule on the first term,

$$\begin{aligned} \frac{\mathrm{d}I}{\mathrm{d}\alpha} &= \frac{\pi^2(\alpha\cos\alpha - \sin\alpha)}{2\alpha^2} + 2\sin\alpha - (\sin\alpha + \alpha\cos\alpha) \\ &= \frac{\pi^2(\alpha\cos\alpha - \sin\alpha)}{2\alpha^2} - (\alpha\cos\alpha - \sin\alpha) \\ &= (\alpha\cos\alpha - \sin\alpha)\left(\frac{\pi^2}{2\alpha^2} - 1\right). \end{aligned}$$

The first factor is never zero on $(0, \pi)$. Let $h(\alpha) = \alpha\cos\alpha - \sin\alpha$. Then $h(0) = 0$ and $h'(\alpha) = \cos\alpha - \alpha\sin\alpha - \cos\alpha = -\alpha\sin\alpha < 0$ on $(0, \pi)$, so $h(\alpha) < 0$ throughout. (Equivalently, $\tan\alpha > \alpha$ on $(0, \frac{1}{2}\pi)$, and $\cos\alpha \le 0$ beyond.)

So $\mathrm{d}I/\mathrm{d}\alpha = 0$ exactly when $\pi^2 = 2\alpha^2$, that is, at $\alpha = \pi/\sqrt{2}$, which does lie in $(0, \pi)$. This is the unique stationary value. Since $h < 0$, the sign of $\mathrm{d}I/\mathrm{d}\alpha$ is opposite to the sign of $\frac{\pi^2}{2\alpha^2} - 1$:

- for $0 < \alpha < \pi/\sqrt{2}$, $\frac{\pi^2}{2\alpha^2} > 1$, so $\mathrm{d}I/\mathrm{d}\alpha < 0$;
- for $\pi/\sqrt{2} < \alpha < \pi$, $\frac{\pi^2}{2\alpha^2} < 1$, so $\mathrm{d}I/\mathrm{d}\alpha > 0$.

So I decreases and then increases: the stationary value is a minimum, and since it is the only stationary point on the interval, it is the smallest value of I.

At $\alpha = \pi/\sqrt{2}$ we have $\dfrac{\pi^2}{2\alpha} = \dfrac{\pi^2}{\sqrt{2}\,\pi} = \dfrac{\pi}{\sqrt{2}} = \alpha$, so the first and third terms of I cancel:

$$I_{\min} = \alpha \sin\alpha - 2\cos\alpha - \alpha\sin\alpha = -2\cos\frac{\pi}{\sqrt{2}}.$$

This is positive, as it must be, since $\pi/\sqrt{2} \approx 2.22$ lies between $\frac{1}{2}\pi$ and π, where cosine is negative.

Checks

- As $k \to 0$, $\alpha \to \pi$ and the formula gives $I \to 2$, which is $\int_0^\pi \sin x\,\mathrm{d}x$. As $k \to 1$, $\alpha \to 0$ and $I \to \frac{1}{2}\pi^2 - 2$, which is $\int_0^\pi (x - \sin x)\,\mathrm{d}x$.
- Numerically, $-2\cos(\pi/\sqrt{2}) \approx 1.2114$, which is less than both end values 2 and $\frac{1}{2}\pi^2 - 2 \approx 2.935$. Direct midpoint quadrature of $\int_0^\pi |\sin x - kx|\,\mathrm{d}x$ with $k = \sin\alpha/\alpha \approx 0.3582$ also gives 1.2114.

Common slips

- Differentiating with respect to α while treating k as a constant: k depends on α, which is exactly why we substitute $k = \sin\alpha/\alpha$ first.
- Showing $\mathrm{d}I/\mathrm{d}\alpha = 0$ only at $\alpha = \pi/\sqrt{2}$ but forgetting to show that $\alpha\cos\alpha - \sin\alpha$ has no zero in $(0, \pi)$.

Beyond STEP

The argument for the unique root uses the intermediate value theorem together with strict monotonicity, and the minimisation quietly uses the fact that $\alpha \mapsto k$ is a bijection between $(0, \pi)$ and $(0, 1)$, so that minimising over α is the same as minimising over k. Making such reparametrisations and existence arguments rigorous is the business of a first course in real analysis; Abbott's *Understanding Analysis* is an excellent place to start.

WORKED SOLUTION

STEP I 2011, Question 6

Pure • Binomial Series Summation

STEP I 2011, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q6

The key idea

Once we know the coefficients of $(1-x)^{-3}$, multiplying by a polynomial just mixes a few neighbouring coefficients together, so coefficients of more complicated rational functions come almost for free. Reading the process backwards, a series whose terms are (polynomial in r) $\times\, x^r$ can be summed by spotting which rational function generates it and then substituting the right value of x, here $x = \frac{1}{2}$.

Solution

The opening result

For $|x| < 1$ the binomial expansion with index -3 gives

$$(1-x)^{-3} = \sum_{r=0}^{\infty} \frac{(-3)(-4)\cdots(-3-r+1)}{r!}\,(-x)^r.$$

The numerator has r factors, each negative, so their product is $(-1)^r\,\dfrac{(r+2)!}{2!}$, and the $(-1)^r$ cancels with the one from $(-x)^r$. Hence the coefficient of x^r is

$$a_r = \frac{(r+2)!}{2!\,r!} = \tfrac{1}{2}(r+1)(r+2),$$

as required. (So $a_0 = 1$, $a_1 = 3$, $a_2 = 6$, $a_3 = 10$, …)

Part (i)

Write

$$\frac{1-x+2x^2}{(1-x)^3} = (1-x+2x^2)\sum_{k\geq 0} a_k x^k.$$

A term x^r arises from $1 \cdot a_r x^r$, from $-x \cdot a_{r-1}x^{r-1}$ and from $2x^2 \cdot a_{r-2}x^{r-2}$, so for $r \geq 2$ the coefficient of x^r is

$$\begin{aligned} a_r - a_{r-1} + 2a_{r-2} &= \tfrac{1}{2}(r+1)(r+2) - \tfrac{1}{2}r(r+1) + (r-1)r \\ &= (r+1) + r^2 - r \\ &= r^2 + 1. \end{aligned}$$

Ask yourself Does the formula still hold for $r = 0$ and $r = 1$, where a_{r-1} or a_{r-2} does not exist?

Yes. If we set $a_{-1} = a_{-2} = 0$, the formula $\frac{1}{2}(r+1)(r+2)$ also gives 0 at $r = -1$ and $r = -2$, so the calculation above is valid for every $r \geq 0$. Directly: the constant term is $a_0 = 1 = 0^2 + 1$, and the coefficient of x is $a_1 - a_0 = 2 = 1^2 + 1$. So the coefficient of x^r is $r^2 + 1$ for all $r \geq 0$, as required.

For the series, the numerators $1, 2, 5, 10, 17, 26, 37, 50, \ldots$ are $r^2 + 1$ for $r = 0, 1, 2, \ldots$ and the denominators are 2^r. So the series is

$$\sum_{r=0}^{\infty} (r^2 + 1) \left(\tfrac{1}{2}\right)^r,$$

which is the expansion above at $x = \frac{1}{2}$. This is legitimate because the binomial expansion of $(1-x)^{-3}$ converges for $|x| < 1$. Therefore

$$1 + \frac{2}{2} + \frac{5}{4} + \frac{10}{8} + \cdots = \frac{1 - \frac{1}{2} + 2 \cdot \frac{1}{4}}{\left(\frac{1}{2}\right)^3} = \frac{1}{1/8} = 8.$$

Part (ii)

The first job is to find the pattern. Rewriting the terms with denominators $1, 2, 4, 8, 16, 32, 64$:

$$1, \frac{4}{2}, \frac{9}{4}, \frac{16}{8}, \frac{25}{16}, \frac{36}{32}, \frac{49}{64}, \cdots$$

so the series is $\displaystyle\sum_{r=0}^{\infty} \frac{(r+1)^2}{2^r}$.

Ask yourself Which numerator over $(1-x)^3$ gives coefficients $(r+1)^2$?

Part (i) suggests trying a numerator $1 + x$ (a short polynomial again). The coefficient of x^r in $(1+x)(1-x)^{-3}$ is

$$a_r + a_{r-1} = \tfrac{1}{2}(r+1)(r+2) + \tfrac{1}{2}r(r+1) = \tfrac{1}{2}(r+1)(2r+2) = (r+1)^2,$$

valid for all $r \geq 0$ (with $a_{-1} = 0$ as before). Putting $x = \frac{1}{2}$,

$$1 + 2 + \frac{9}{4} + 2 + \frac{25}{16} + \cdots = \frac{1 + \frac{1}{2}}{\left(\frac{1}{2}\right)^3} = \frac{3}{2} \cdot 8 = 12.$$

Alternatively, using part (i). Shift the index with $s = r+1$, so that $2^{-r} = 2\cdot 2^{-s}$:

$$\sum_{r\geq 0}\frac{(r+1)^2}{2^r} = 2\sum_{s\geq 1}\frac{s^2}{2^s} = 2\left(\sum_{s\geq 0}\frac{s^2+1}{2^s} - \sum_{s\geq 0}\frac{1}{2^s}\right) = 2(8-2) = 12,$$

using the answer to part (i) and the geometric series $\sum 2^{-s} = 2$.

Checks

- Expanding $(1-x)^{-3}$ by hand as $(1+x+x^2+\cdots)^3$, or by multiplying out a few terms, gives $1+3x+6x^2+10x^3+\cdots$, matching $\frac{1}{2}(r+1)(r+2)$.
- The partial sums of the part (i) series approach 8: after 200 terms the sum agrees with 8 to machine precision. The same holds for part (ii) with limit 12.
- The two routes in part (ii) give the same answer, 12.

Common slips

- Getting the sign wrong in the binomial coefficient: there are r minus signs from $(-3)(-4)\cdots$ and r from $(-x)^r$, and they cancel.
- In part (ii), misreading the pattern because some fractions are given in lowest terms: $2 = \frac{4}{2} = \frac{16}{8}$ and $\frac{9}{8} = \frac{36}{32}$.
- Substituting $x = \frac{1}{2}$ without noting that the expansion converges there ($|x| < 1$).

Beyond STEP

What we have been doing is using *generating functions*: a sequence $c_0, c_1, c_2, \ldots$ is encoded as the power series $\sum c_r x^r$, and operations on sequences become algebra on functions. In fact any polynomial in r times x^r sums to a rational function with denominator a power of $(1-x)$, which is why these sums always come out rational. Justifying the substitution of a value of x rests on the theory of power series and their radius of convergence, covered in Abbott, *Understanding Analysis*; for generating functions as a problem-solving tool, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2011, Question 7

Pure • Separable Differential Equations

STEP I 2011, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q7

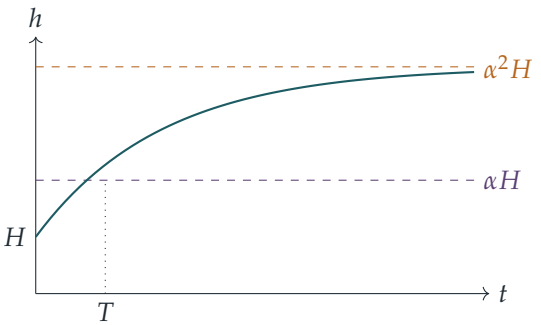
The key idea

Volume equals (constant area) $\times$ height, so the rate of change of height is just (inflow $-$ outflow) divided by the area. The phrase "the height remains constant" is the clue that pins down the inflow: at that height, inflow and outflow balance exactly. After that, each part is a separable differential equation, and in part (ii) the substitution $u = \sqrt{h}$ turns the integral into something routine.

Solution

Let the constant cross-sectional area be A and the constant inflow rate be q (volume per unit time). The volume of water is Ah, so

$$A\frac{\mathrm{d}h}{\mathrm{d}t} = q - (\text{outflow rate}).$$



The height rises from H towards the level $\alpha^2 H$ at which inflow and outflow balance, getting ever closer but never reaching it; the question is how long it takes to get to αH, which lies between them because $1 < \alpha < \alpha^2$. (The sketch shows part (i).)

Part (i)

The outflow rate is λh for some constant $\lambda > 0$, so

$$A\frac{\mathrm{d}h}{\mathrm{d}t} = q - \lambda h.$$

When $h = \alpha^2 H$ the height stays constant, so $\mathrm{d}h/\mathrm{d}t = 0$ there, giving $q = \lambda\alpha^2 H$. Substituting,

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\lambda}{A}\left(\alpha^2 H - h\right) = k\left(\alpha^2 H - h\right), \qquad k = \frac{\lambda}{A} > 0,$$

as required.

Since $H \le h < \alpha^2 H$ throughout the rise, $\alpha^2 H - h > 0$ and we may separate the variables:

$$\begin{aligned} kT &= \int_H^{\alpha H} \frac{\mathrm{d}h}{\alpha^2 H - h} = \Big[-\ln\left(\alpha^2 H - h\right) \Big]_H^{\alpha H} = \ln\frac{\alpha^2 H - H}{\alpha^2 H - \alpha H} \\ &= \ln\frac{(\alpha-1)(\alpha+1)}{\alpha(\alpha-1)} = \ln\frac{\alpha+1}{\alpha} = \ln\left(1 + \frac{1}{\alpha}\right), \end{aligned}$$

cancelling $\alpha - 1 \neq 0$ because $\alpha > 1$. This is the required result.

For large α, $x = 1/\alpha$ is small, so

$$kT \approx \frac{1}{\alpha} - \frac{1}{2\alpha^2} \approx \frac{1}{\alpha},$$

since the second term is negligible compared with the first when α is large.

Part (ii)

Now the outflow rate is $\mu\sqrt{h}$ for some constant $\mu > 0$, so $A\,\mathrm{d}h/\mathrm{d}t = q - \mu\sqrt{h}$. The height is constant at $h = \alpha^2 H$, so $q = \mu\sqrt{\alpha^2 H} = \mu\alpha\sqrt{H}$ (using $\alpha > 0$). Hence

$$\frac{\mathrm{d}h}{\mathrm{d}t} = c\left(\alpha\sqrt{H} - \sqrt{h}\right), \qquad c = \frac{\mu}{A} > 0.$$

Again the bracket is positive while $h < \alpha^2 H$, so

$$cT' = \int_H^{\alpha H} \frac{\mathrm{d}h}{\alpha\sqrt{H} - \sqrt{h}}.$$

? **Ask yourself** How do we deal with the $\sqrt{h}$ in the denominator?

Substitute $u = \sqrt{h}$, so $h = u^2$ and $\mathrm{d}h = 2u\,\mathrm{d}u$; the limits become $u = \sqrt{H}$ and $u = \sqrt{\alpha H}$. Writing $a = \alpha\sqrt{H}$ for short, the integrand is a rational function, and dividing out,

$$\frac{2u}{a-u} = -2 + \frac{2a}{a-u}.$$

Therefore

$$\begin{aligned} cT' &= \int_{\sqrt{H}}^{\sqrt{\alpha H}} \left(-2 + \frac{2a}{a-u}\right) \mathrm{d}u = \Big[-2u - 2a\ln(a-u)\Big]_{\sqrt{H}}^{\sqrt{\alpha H}} \\ &= -2\left(\sqrt{\alpha H} - \sqrt{H}\right) + 2a\sqrt{H}\,\ln\frac{a\sqrt{H} - \sqrt{H}}{a\sqrt{H} - \sqrt{a}\sqrt{H}} \\ &= 2\sqrt{H}\left(1 - \sqrt{\alpha} + \alpha\ln\frac{\alpha-1}{\alpha-\sqrt{\alpha}}\right). \end{aligned}$$

The logarithms are of positive quantities because $u \le \sqrt{\alpha H} < \alpha\sqrt{H} = a$ (as $\sqrt{\alpha} < \alpha$ for $\alpha > 1$). Finally, factorising $\alpha - 1 = (\sqrt{\alpha}-1)(\sqrt{\alpha}+1)$ and $\alpha - \sqrt{\alpha} = \sqrt{\alpha}(\sqrt{\alpha}-1)$,

$$\frac{\alpha-1}{\alpha-\sqrt{\alpha}} = \frac{\sqrt{\alpha}+1}{\sqrt{\alpha}} = 1 + \frac{1}{\sqrt{\alpha}},$$

so

$$cT' = 2\sqrt{H}\left(1 - \sqrt{\alpha} + \alpha\ln\left(1 + \frac{1}{\sqrt{\alpha}}\right)\right),$$

as required.

For large α, $1/\sqrt{\alpha}$ is small, so

$$\alpha\ln\left(1 + \frac{1}{\sqrt{\alpha}}\right) \approx \alpha\left(\frac{1}{\sqrt{\alpha}} - \frac{1}{2\alpha}\right) = \sqrt{\alpha} - \frac{1}{2}.$$

The large terms $\pm\sqrt{\alpha}$ cancel, leaving

$$cT' \approx 2\sqrt{H}\left(1 - \sqrt{\alpha} + \sqrt{\alpha} - \tfrac{1}{2}\right) = \sqrt{H}.$$

Notice that we needed the x^2 term of the logarithm here: with only $\ln(1+x) \approx x$, the bracket would be 1 and we would get $2\sqrt{H}$, which is wrong. That is why the question gives the two-term approximation.

Checks

- As $\alpha \to 1^+$, (i) gives $kT \to \ln 2$, not 0. That is right: the gap to equilibrium starts at $(\alpha^2-1)H \approx 2(\alpha-1)H$ and must shrink to $(\alpha^2-\alpha)H \approx (\alpha-1)H$, so the gap halves, and an exponential decay at rate k halves in time $(\ln 2)/k$. In (ii), $cT' \to 2\sqrt{H}\ln 2$, which matches the same halving with the effective rate $c/(2\sqrt{H})$ obtained by linearising $\sqrt{h}$ near $h = H$.
- With $\alpha = 4$ and $H = 1$: $kT = \ln 1.25 \approx 0.2231$ and $cT' = 2(-1 + 4\ln 1.5) \approx 1.2437$. Stepping both differential equations numerically (Runge–Kutta) for several random H, α, k, c reproduces both formulae to 10^{-9}.
- For $\alpha = 10^4$, $H = 2$: $\alpha\ln(1 + 1/\alpha) \approx 0.99995$ and $cT'/\sqrt{H} \approx 1.0066$, consistent with the limits (the next term gives $cT' \approx \sqrt{H}\,(1 + \frac{2}{3\sqrt{\alpha}})$).

Common slips

- Forgetting to use the equilibrium condition to eliminate the inflow rate, and so being left with too many constants.
- Using only $\ln(1+x) \approx x$ in part (ii), which loses the crucial $-\frac{1}{2}$.
- Writing $\sqrt{\alpha^2 H}$ as $\alpha^2\sqrt{H}$.

Beyond STEP

Both models are first-order autonomous differential equations $\dot{h} = f(h)$ with a stable equilibrium at $h = \alpha^2 H$; the linear case is Newton's law of cooling in disguise, while the $\sqrt{h}$ case is Torricelli's law for flow through a hole. Studying equilibria and their stability without solving explicitly is the start of the qualitative theory of ODEs. Boyce and DiPrima, *Elementary Differential Equations*, covers autonomous equations and tank-mixing models in its early chapters.

WORKED SOLUTION

STEP I 2011, Question 8

Pure • Diophantine Equations

STEP I 2011, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q8

The key idea

The cube function is strictly increasing, so comparing m^3 with n^3 and $(n+1)^3$ is the same as comparing m with n and $n+1$. If m lies *strictly* between two consecutive integers it cannot be an integer. That leaves only a short list of values of n to try by hand. Part (ii) uses the same squeeze on a new equation.

Solution

Throughout we use the fact that $x \mapsto x^3$ is strictly increasing on all of $\mathbb{R}$, so for real a, b,

$$a < b \iff a^3 < b^3.$$

Part (i)(a)

From $(*)$,

$$m^3 - n^3 = n^2 + 1 > 0,$$

since $n^2 \geq 0$. So $m^3 > n^3$, and therefore $m > n$.

Next, $m < n+1$ if and only if $m^3 < (n+1)^3$. Using $(*)$ and expanding,

$$\begin{aligned} m^3 < (n+1)^3 &\iff n^3 + n^2 + 1 < n^3 + 3n^2 + 3n + 1 \\ &\iff 0 < 2n^2 + 3n, \end{aligned}$$

which is the required equivalence. Every step is reversible, so the "if and only if" holds.

Now $2n^2 + 3n = n(2n+3)$. This is a quadratic with roots $n = -\frac{3}{2}$ and $n = 0$ and a positive leading coefficient, so it is positive exactly when $n < -\frac{3}{2}$ or $n > 0$. For any such n we have both $m > n$ and $m < n+1$. Hence

$$n < m < n+1 \quad \text{unless} \quad -\tfrac{3}{2} \leq n \leq 0.$$

Part (i)(b)

Suppose m and n are integers satisfying $(*)$. If $n < -\frac{3}{2}$ or $n > 0$, part (a) gives $n < m < n+1$. No integer lies strictly between the consecutive integers n and $n+1$, so this is impossible.

? **Ask yourself** Part (a) holds for all real n. Where does being an integer come in?

It comes in twice. It rules out every n outside the interval, as above, and it cuts the interval $-\frac{3}{2} \le n \le 0$ down to $n \in \{-1, 0\}$. We check both values:

- $n = 0$: $m^3 = 0 + 0 + 1 = 1$, so $m = 1$.
- $n = -1$: $m^3 = -1 + 1 + 1 = 1$, so $m = 1$.

Both give integer m. So the only integer solutions are $(m, n) = (1, 0)$ and $(1, -1)$, as required.

Part (ii)

We copy part (i). First compare p with q:

$$p^3 - q^3 = 2q^2 - 1,$$

which is positive whenever $q^2 > \frac{1}{2}$, so for every integer $q \neq 0$. For those q we have $p > q$.

Next compare p with $q+1$:

$$\begin{aligned} p^3 < (q+1)^3 &\iff q^3 + 2q^2 - 1 < q^3 + 3q^2 + 3q + 1 \\ &\iff 0 < q^2 + 3q + 2 = (q+1)(q+2). \end{aligned}$$

This holds exactly when $q > -1$ or $q < -2$.

Put the two together. If q is an integer with $q \neq 0$, $q \neq -1$ and $q \neq -2$, then $q < p < q+1$, so p cannot be an integer. (If $q \ge 1$ or $q \le -3$, both conditions hold.) That leaves $q \in \{-2, -1, 0\}$:

- $q = 0$: $p^3 = -1$, so $p = -1$.
- $q = -1$: $p^3 = -1 + 2 - 1 = 0$, so $p = 0$.
- $q = -2$: $p^3 = -8 + 8 - 1 = -1$, so $p = -1$.

All three give integer p, so the integer solutions are

$$(p, q) = (-1, 0), \quad (0, -1), \quad (-1, -2).$$

Checks

Substitute each answer back in. For example, $(-1)^3 = -1 = (-2)^3 + 2(-2)^2 - 1 = -8 + 8 - 1$. A computer search over all integers $|n|, |q| < 10^5$ finds exactly the two

solutions of $(*)$ and the three solutions of part (ii), and no others. You can also test the squeeze numerically. For $n = 5$, $m = \sqrt[3]{151} \approx 5.32$, which lies strictly between 5 and 6.

Common slips

- Forgetting $q = 0$ in part (ii). There, $p^3 - q^3 = -1 < 0$, so the lower bound $p > q$ fails, and the case needs checking separately.
- Writing "$m^3 > n^3$ so $m > n$" with no reason. It is true because cubing is strictly increasing (which fails for squaring with negative numbers), and that is worth saying.
- Stopping at $n < m < n + 1$ without saying why it means m is not an integer.

Beyond STEP

Equations such as $y^3 = f(x)$ asking for integer points are *Diophantine equations*. Trapping the unknown between consecutive powers is a standard first attack on them. Harder versions, such as $y^2 = x^3 + k$, lead to elliptic curves and to Baker's theory of linear forms in logarithms. These are deep areas, yet many cases still give way to elementary bounding like this. Davenport's *The Higher Arithmetic* is a good introduction to elementary number theory.

WORKED SOLUTION

STEP I 2011, Question 9

Mechanics • Projectile Motion

STEP I 2011, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

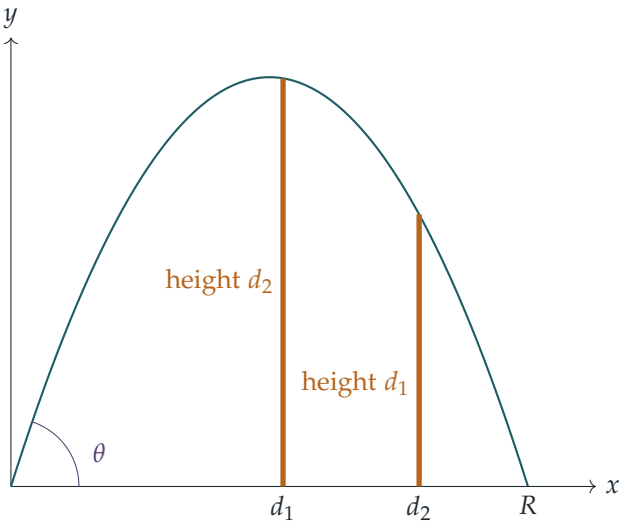
step.maths.org/questions/11-s1-q9

The key idea

Write the trajectory as a single equation $y = x\tan\theta - ax^2$, where the constant a packs together g, the speed and the angle. "Just passes over" a wall means the path goes through the top corner of the wall, so each wall gives one linear equation in the two unknowns $\tan\theta$ and a. Solving that pair gives the angle, and the range is then just the other root of $y = 0$.

Solution

We model the particle as a point moving under gravity alone (no air resistance), projected from the origin O with speed u at angle θ, with x horizontal and y vertically up. The walls have different heights, so we assume $d_1 \neq d_2$.



(The picture uses $d_1 < d_2$; nothing in the algebra depends on which is larger.)

The trajectory

At time t we have $x = ut\cos\theta$ and $y = ut\sin\theta - \frac{1}{2}gt^2$. Eliminating $t = x/(u\cos\theta)$,

$$y = x\tan\theta - ax^2, \qquad \text{where } a = \frac{g}{2u^2\cos^2\theta} > 0.$$

Write $T = \tan\theta$. The particle just clears each wall, so it passes through the tops (d_1, d_2) and (d_2, d_1):

$$d_2 = d_1 T - a d_1^2, \tag{1}$$
$$d_1 = d_2 T - a d_2^2. \tag{2}$$

Finding $\tan\theta$

To eliminate a, multiply (1) by d_2^2 and (2) by d_1^2, so that the a terms are both $a d_1^2 d_2^2$:

$$d_2^3 = d_1 d_2^2\, T - a d_1^2 d_2^2,$$
$$d_1^3 = d_1^2 d_2\, T - a d_1^2 d_2^2.$$

Subtracting,

$$d_2^3 - d_1^3 = d_1 d_2 (d_2 - d_1)\, T.$$

Ask yourself Can we divide by $d_2 - d_1$?

Yes: the walls have different heights d_2 and d_1, so $d_1 \neq d_2$. (If they were equal there would be only one wall, and one point cannot fix both θ and u.) Since $d_2^3 - d_1^3 = (d_2 - d_1)(d_2^2 + d_1 d_2 + d_1^2)$, dividing by $d_1 d_2 (d_2 - d_1)$ gives

$$\tan\theta = \frac{d_1^2 + d_1 d_2 + d_2^2}{d_1 d_2},$$

as required.

The range

The particle lands where $y = 0$ with $x \neq 0$, that is at $x = T/a$. So we need a as well. Dividing (1) by d_1 and (2) by d_2,

$$\frac{d_2}{d_1} = T - a d_1, \qquad \frac{d_1}{d_2} = T - a d_2.$$

Subtracting the first from the second removes T:

$$a(d_2 - d_1) = \frac{d_2}{d_1} - \frac{d_1}{d_2} = \frac{(d_2 - d_1)(d_2 + d_1)}{d_1 d_2}, \qquad \text{so} \qquad a = \frac{d_1 + d_2}{d_1 d_2},$$

again using $d_1 \neq d_2$. Therefore the range is

$$R = \frac{T}{a} = \frac{d_1^2 + d_1 d_2 + d_2^2}{d_1 d_2} \cdot \frac{d_1 d_2}{d_1 + d_2} = \frac{d_1^2 + d_1 d_2 + d_2^2}{d_1 + d_2}.$$

As a sanity check, $R - d_2 = \dfrac{d_1^2}{d_1 + d_2} > 0$ and similarly $R - d_1 = \dfrac{d_2^2}{d_1 + d_2} > 0$, so the particle lands beyond both walls, as it must.

Ask yourself Where did g and u go?

They only ever appear through the single combination a, and the two walls fix a completely: $u^2 = g/(2a\cos^2\theta)$. So the walls determine the whole trajectory, and the range depends on d_1 and d_2 alone.

Checks

- Symmetry: swapping d_1 and d_2 just swaps the walls, and both answers are symmetric in d_1, d_2.
- Numbers: with $d_1 = 2$, $d_2 = 3$ we get $\tan\theta = \frac{19}{6}$, $a = \frac{5}{6}$ and $R = \frac{19}{5}$. Then $y(2) = \frac{19}{3} - \frac{10}{3} = 3$ and $y(3) = \frac{19}{2} - \frac{15}{2} = 2$, as required. Stepping the motion numerically with $g = 9.8$ also lands at $x = 3.8$.
- Since $d_1^2 + d_2^2 \geq 2d_1 d_2$, we get $\tan\theta \geq 3$, so the launch is always steep (at least about $71.6°$). That makes sense: to clear the taller wall first and then come down to the shorter one so soon, the path has to be a narrow arch.

Common slips

- Mixing up which wall has which height: the wall at distance d_1 has height d_2.
- Writing a with $\cos\theta$ instead of $\cos^2\theta$, or not treating a as a single unknown, which leaves a messy system in u and θ.
- Dividing by $d_2 - d_1$ without saying why it is non-zero.

Beyond STEP

The trick here is that a family of curves with two free parameters (here $\tan\theta$ and a) is pinned down by two points it must pass through, a small case of interpolation. The same idea of choosing the right combination of unknowns appears throughout mechanics, for example in finding the "envelope" of all trajectories with a fixed launch speed (the parabola of safety). For more projectile and particle mechanics at first-year level, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2011, Question 10

Mechanics • Elastic Collisions

STEP I 2011, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q10

The key idea

Both particles start from rest at the same point, and nothing between release and the collision (free fall, a perfectly elastic bounce) loses any energy. So at any given height each particle has the same speed, and they meet at a common height. After that, the collision is a standard two-equation problem (momentum and Newton's law of restitution with $e = 1$), and the maximum height of B follows from energy once more.

Solution

We model A and B as particles, ignore air resistance, and take upwards as positive throughout.

Why the speeds are equal

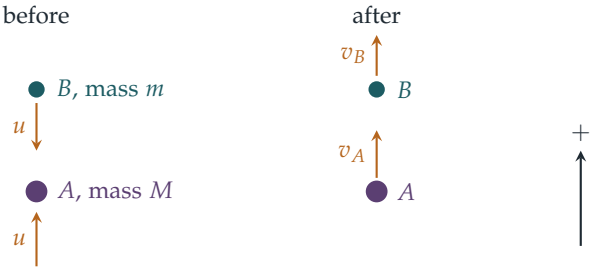
Each particle starts from rest at height h. While in free flight only gravity acts, so mechanical energy is conserved; the bounce of A on the plane is perfectly elastic, so it reverses A's velocity without changing its speed, and no energy is lost there either. Hence, per unit mass, each particle has total energy gh up to the first collision, and a particle at height y has speed v with

$$\tfrac{1}{2}v^2 + gy = gh, \qquad \text{so} \qquad v = \sqrt{2g(h-y)}.$$

The speed depends only on the height, not on the mass or on the particle's history. At the moment of the first collision the two particles are at the same height y, so their speeds are equal. Moreover A has not yet returned to P, so $y < h$ and this common speed $u = \sqrt{2g(h-y)}$ is positive. Since A has bounced it is moving up, and B (never having bounced) is moving down.

The collision

Immediately before the collision A has velocity $+u$ and B has velocity $-u$. Let their velocities immediately afterwards be v_A and v_B.



The impulsive forces in the collision are internal to the pair, and gravity gives no impulse over the instant of impact, so momentum is conserved:

$$Mu - mu = Mv_A + mv_B. \tag{1}$$

The collision is perfectly elastic ($e = 1$), so the speed of separation equals the speed of approach:

$$v_B - v_A = u - (-u) = 2u. \tag{2}$$

Substituting $v_B = v_A + 2u$ from (2) into (1):

$$(M-m)u = (M+m)v_A + 2mu \quad \Longrightarrow \quad v_A = \frac{(M-3m)u}{M+m},$$

and then

$$v_B = v_A + 2u = \frac{(M-3m)u + 2(M+m)u}{M+m} = \frac{(3M-m)u}{M+m}.$$

Since $M > 3m$ we have $M - 3m > 0$, so $v_A > 0$; and $v_B = v_A + 2u > 0$ too. So both particles move upwards after the first collision, as required.

Ask yourself How do we know B actually reaches the top of its flight before it next meets A?

After the collision both particles have acceleration $-g$, so their relative velocity stays $v_B - v_A = 2u$ while both are in flight: B pulls away from A at a steady rate, and they cannot meet before A next hits the plane. If A bounces while B is still rising, compare energies per unit mass: B has $\frac{1}{2}v_B^2 + gy$ and A has $\frac{1}{2}v_A^2 + gy$, which is smaller since $v_B > v_A > 0$ (and A's bounce keeps its energy). So whenever they are at the same height, A's speed is less than B's. To catch up with B from below while B is still rising, A would have to be rising at least as fast as B at the moment they meet, and we have just seen that it can't be. So the second collision happens only after B has passed its highest point, and that highest point is the maximum asked for.

Maximum height of B

The collision happens at height y where $u^2 = 2g(h-y)$, that is

$$y = h - \frac{u^2}{2g}.$$

After the collision B rises with initial speed v_B under gravity alone, so it climbs a further $v_B^2/(2g)$ (from $v^2 = v_B^2 - 2gs$ with $v = 0$). Its maximum height is therefore

$$\begin{aligned} y + \frac{v_B^2}{2g} &= h - \frac{u^2}{2g} + \frac{(3M-m)^2u^2}{2g(M+m)^2} \\ &= h + \frac{u^2}{2g(M+m)^2}\left[(3M-m)^2 - (M+m)^2\right] \\ &= h + \frac{u^2}{2g(M+m)^2}(2M-2m)(4M) \\ &= h + \frac{4M(M-m)u^2}{(M+m)^2 g}, \end{aligned}$$

using the difference of two squares in the third line. This is the required result.

Checks

- Energy: with v_A and v_B as above, $Mv_A^2 + mv_B^2 = (M+m)u^2$, as an elastic collision demands.
- Limiting case: if $m \to 0$, then $v_A \to u$ and $v_B \to 3u$ (a very light ball bounces off a heavy one as off a moving wall, leaving with velocity $2u$ relative to it), and the height becomes $h + 4u^2/g$, which matches $y + 9u^2/(2g)$.
- Since $M > m$, the extra height is positive: B ends up above P, as it should, having gained energy from A.
- Numerically, we released B at random times, with random h and masses satisfying $M > 3m$, stepped the motion (including A's further bounces) until the second collision, and found B's maximum height agreed with the formula to a relative error of about 10^{-10}.

Common slips

- Sign errors: taking both pre-collision velocities as $+u$ forgets that B is falling while A is rising.
- Measuring the extra climb $v_B^2/(2g)$ from the plane, or from P, rather than from the collision height $h - u^2/(2g)$.

Beyond STEP

The argument that speed depends only on height is energy conservation for a conservative force: the speed at a point depends only on the potential energy there, not on the route taken. The collision itself is often easiest in the centre-of-mass frame, where an elastic collision simply reverses each particle's velocity; transforming back gives v_A and v_B in one line. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

WORKED SOLUTION

STEP I 2011, Question 11

Mechanics • Equilibrium of a Rigid Body

STEP I 2011, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

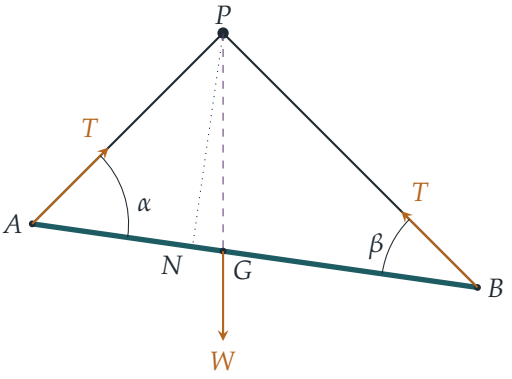
step.maths.org/questions/11-s1-q11

The key idea

Only three forces act on the bar: the two string tensions at A and B, and its weight at G. Because the peg is smooth, the two tensions are equal, so taking moments about G (where the weight has no moment) gives the relation between the angles at once. For the second part, the extra fact we need is that three forces in equilibrium that are not all parallel must pass through one point. Here that point is the peg, so PG is vertical.

Solution

We model the bar as a rigid rod, the string as light and inextensible, and the peg as smooth. Let the weight of the bar be W.



Showing that $3\sin\alpha = 4\sin\beta$

The string is light and passes over a smooth peg, so its tension T is the same throughout. The bar therefore feels a force T at A along AP and a force T at B along BP, as well as W acting vertically at G.

Take moments about G. The weight has no moment. The perpendicular distance from G to the line AP is $AG\sin\alpha = 3d\sin\alpha$, and from G to the line BP it is $GB\sin\beta = 4d\sin\beta$, since $GB = 7d - 3d = 4d$. The tension at A turns the bar one way about G and the tension at B turns it the other way, so for equilibrium

$$T\cdot 3d\sin\alpha = T\cdot 4d\sin\beta \quad\Longrightarrow\quad 3\sin\alpha = 4\sin\beta,$$

as required (dividing by $Td \neq 0$).

The length of the string

With $\cos\beta = \frac{4}{5}$ we have $\sin\beta = \frac{3}{5}$ (β is an angle of a triangle, so $\sin\beta > 0$). Then

$$\sin\alpha = \tfrac{4}{3}\sin\beta = \tfrac{4}{5}, \qquad \cos\alpha = \tfrac{3}{5},$$

taking the positive root because α is acute. Now

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta = \tfrac{3}{5}\cdot\tfrac{4}{5} - \tfrac{4}{5}\cdot\tfrac{3}{5} = 0,$$

and $0 < \alpha+\beta < \pi$, so $\alpha+\beta = \frac{\pi}{2}$. The triangle PAB therefore has a right angle at P, with hypotenuse $AB = 7d$. Hence

$$PA = 7d\cos\alpha = \tfrac{21}{5}d, \qquad PB = 7d\cos\beta = \tfrac{28}{5}d,$$

and the length of the string is

$$PA + PB = \tfrac{49}{5}d.$$

The inclination of the bar

? **Ask yourself** We know where P is relative to the bar, but where is "vertical"?

The two tensions act along lines through P, so their resultant also acts through P. For equilibrium this resultant must balance the weight, so it must be equal and opposite to W and along the same line. (Three non-parallel forces in equilibrium are concurrent.) So the vertical line through G passes through P: *PG is vertical*.

Let N be the foot of the perpendicular from P to AB. In the right-angled triangle PAN,

$$AN = PA\cos\alpha = \tfrac{21}{5}d\cdot\tfrac{3}{5} = \tfrac{63}{25}d, \qquad PN = PA\sin\alpha = \tfrac{21}{5}d\cdot\tfrac{4}{5} = \tfrac{84}{25}d.$$

Since $AN = \frac{63}{25}d < 3d = AG$, the point N lies between A and G, with

$$NG = 3d - \tfrac{63}{25}d = \tfrac{12}{25}d.$$

In the right-angled triangle PNG,

$$\tan\angle PGN = \frac{PN}{NG} = \frac{84}{12} = 7.$$

The angle PGN is the angle between the bar and the vertical GP, measured on the A side. It is acute, so the end A is above the horizontal through G and B is below it, which

matches the given configuration. The angle θ between the bar and the horizontal is $\frac{\pi}{2} - \angle PGN$, so

$$\tan\theta = \frac{1}{\tan\angle PGN} = \frac{1}{7}, \qquad \text{that is,} \qquad \theta = \arctan\tfrac{1}{7},$$

as required.

Checks

- The string, $\frac{49}{5}d = 9.8d$, is longer than the bar, $7d$, as the triangle inequality in PAB demands.
- The resultant of two equal tensions bisects the angle between them, so PG bisects angle APB. By the angle bisector theorem, $PA : PB$ should equal $AG : GB = 3 : 4$, and indeed $\frac{21}{5} : \frac{28}{5} = 3 : 4$.
- As a numerical check, we put P on the ellipse with foci A and B (the set of possible peg positions for this string length) and looked for the inclination at which the depth of G below P is stationary. It came out at 8.13°, which agrees with $\arctan\frac{1}{7} \approx 8.130^\circ$.

Common slips

- Giving the two parts of the string different tensions. A smooth peg cannot change the tension in a light string.
- Swapping the distances and getting $4\sin\alpha = 3\sin\beta$. The longer arm $GB = 4d$ goes with the angle at B.
- Forgetting that α is acute and allowing $\cos\alpha = -\frac{3}{5}$. That gives a different triangle, and in it $\alpha + \beta$ is not a right angle.

Beyond STEP

For a fixed string length, the peg must lie on an ellipse with foci A and B. The fact that PG bisects angle APB is then the reflection property of the ellipse: the normal at P bisects the angle between the focal radii. Equilibria are also exactly the positions where the potential energy is stationary, and this is the starting point of the principle of virtual work and of Lagrangian mechanics in a first-year course. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to read about equilibrium, torque and energy methods.

WORKED SOLUTION

STEP I 2011, Question 12

Probability and Statistics • Queues and Ballot Problem

STEP I 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q12

This question has a full guided treatment in *Beyond STEP*, Section 9.10, *Always Able to Give Change*: https://beyondstep.keithmcnulty.org/ch09/ballot.html

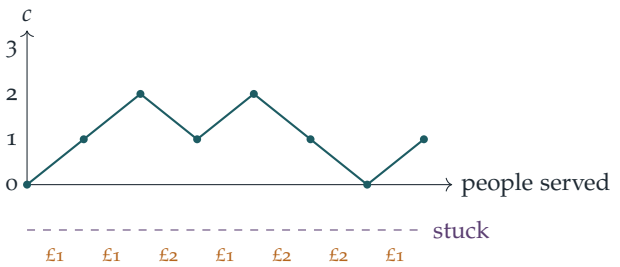
The key idea

Keep track of one number: how many £1 coins I am holding. A £1 customer adds one; a £2 customer takes one away (I keep their £2 coin, which is no use as change). I am stuck exactly when a £2 customer arrives and I am holding no £1 coins. With only a few £2 customers, the danger is all at the front of the queue, so we look at the first few people and work out what each opening leaves us with.

Solution

Only the pattern of coins in the queue matters, not who the people are. Since every arrangement of people is equally likely, the person in any given position is equally likely to be any of the people not yet placed, so we can build the queue from the front: if a people with £1 coins and b with £2 coins remain, the next person has a £1 coin with probability $\frac{a}{a+b}$ and a £2 coin with probability $\frac{b}{a+b}$.

Write c for the number of £1 coins I hold. I can serve everyone if and only if c never needs to drop below 0, that is, at every point in the queue at least as many £1 people as £2 people have been served so far.



The picture shows a successful queue: the path of c never touches the dashed line.

Part (i)

With one £2 customer, the only way to get stuck is if that person is first in the queue, when I hold no change. If anyone else is first, I hold at least one £1 coin by the time the £2 customer arrives. The £2 customer is equally likely to be in any of the $m+1$ positions, so

$$\mathrm{P}(\text{sell to everyone}) = 1 - \frac{1}{m+1} = \frac{m}{m+1}.$$

Part (ii)

Write 1 for a £1 customer and 2 for a £2 customer, and list the openings.

- If the first person is a 2, I am stuck at once.
- If the queue starts $1,2,2$, then c goes $1,0$ and I am stuck at the third person.
- If the queue starts $1,1$, then $c = 2$ after two people. There are only two £2 customers in total, and c only falls when one of them is served, so c can never fall below $2-2=0$. Success.
- If the queue starts $1,2,1$, then c goes $1,0,1$, and only one £2 customer remains, so c never falls below 0. Success.

These cases cover every possibility, so I serve everyone exactly when the queue starts $1,1$ or $1,2,1$. Building the queue from the front,

$$\mathrm{P}(1,1) = \frac{m}{m+2}\cdot\frac{m-1}{m+1}, \quad \mathrm{P}(1,2,1) = \frac{m}{m+2}\cdot\frac{2}{m+1}\cdot\frac{m-1}{m} = \frac{2(m-1)}{(m+2)(m+1)}.$$

These events are disjoint, so

$$\mathrm{P}(\text{sell to everyone}) = \frac{m(m-1)+2(m-1)}{(m+2)(m+1)} = \frac{(m-1)(m+2)}{(m+2)(m+1)} = \frac{m-1}{m+1},$$

as required. (We need $m \geq 2$ so that $m-1 \geq 1$ £1 people remain in the last factor.)

Part (iii)

Now there are three £2 customers. Again look at the first three people.

- Starts with 2, or with $1,2,2$: stuck, exactly as in (ii).
- Starts $1,1,1$: then $c = 3$, and the three £2 customers can take c down to 0 at worst. Success.
- Starts $1,1,2$ or $1,2,1$: in both cases $c = 1$ after three people, and what remains is $m-2$ people with £1 coins and 2 with £2 coins, in random order.

Ask yourself In the last case, when does the rest of the queue go wrong?

Starting the rest of the queue with $c = 1$ is like having one free £1 customer at the front of a queue with two £2 customers, which is part (ii) in disguise. With $c = 1$ and only two £2 customers left, I get stuck exactly when both of the next two people are £2 customers: c goes 0, then I cannot serve. If either of the next two people has a £1 coin, then c never falls below 0 (check $1,\ldots$ and $2,1,\ldots$ as in (ii)). Among the m people remaining after the first three,

$$\mathrm{P}(\text{next two are both 2}) = \frac{2}{m}\cdot\frac{1}{m-1},$$
$$\text{so}\quad \mathrm{P}(\text{rest succeeds}) = 1 - \frac{2}{m(m-1)} = \frac{(m-2)(m+1)}{m(m-1)},$$

using $m^2 - m - 2 = (m-2)(m+1)$.

Write $D = (m+3)(m+2)(m+1)$. The probabilities of the useful openings are

$$\mathrm{P}(1,1,1) = \frac{m(m-1)(m-2)}{D},$$
$$\mathrm{P}(1,1,2) = \frac{m\cdot(m-1)\cdot 3}{D}, \qquad \mathrm{P}(1,2,1) = \frac{m\cdot 3\cdot(m-1)}{D}.$$

Since the openings are disjoint and the rest of the queue is again in random order whatever the opening,

$$\begin{aligned}\mathrm{P}(\text{success}) &= \frac{m(m-1)(m-2)}{D} + \frac{6m(m-1)}{D}\cdot\frac{(m-2)(m+1)}{m(m-1)}\\ &= \frac{(m-2)\big[m(m-1)+6(m+1)\big]}{D} = \frac{(m-2)(m^2+5m+6)}{D}\\ &= \frac{(m-2)(m+2)(m+3)}{(m+3)(m+2)(m+1)} = \frac{m-2}{m+1},\end{aligned}$$

as required. (Here $m \geq 3$ ensures $m-1 \geq 2$, so the conditional probability makes sense.)

Checks

- Smallest cases: with $m = n$ the formulas give $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ for $n = 1,2,3$. For $m = n = 2$ the six patterns of 1s and 2s are equally likely, and only 1122 and 1212 succeed: $\frac{2}{6} = \frac{1}{3}$. ✓

- Large m: all three probabilities tend to 1, as they should when £1 coins are plentiful.
- By complete enumeration of queue orders, for $m = 5$ I get $\frac{5}{6}$, $\frac{2}{3}$, $\frac{1}{2}$ for $n = 1, 2, 3$, matching $\frac{m}{m+1}$, $\frac{m-1}{m+1}$, $\frac{m-2}{m+1}$; the formulas were also checked exactly for every m up to 14.

Common slips

- Forgetting that the £2 coins I collect are useless as change: only £1 coins count.
- Treating the people as independent draws with fixed probability $\frac{m}{m+n}$. The queue is drawn without replacement, so the probabilities change as people are placed.
- In (iii), assuming that $c = 1$ after three people is always safe, and missing the failure $1, 1, 2, 2, 2$ or $1, 2, 1, 2, 2$.

Beyond STEP

The pattern is $\frac{m+1-n}{m+1}$ for every $n \leq m$. This is the *ballot problem*, and the neatest proof uses the *reflection principle*: a path that touches the stuck level can be reflected to give a path ending somewhere else, which turns the count of bad queues into a single binomial coefficient. When $m = n$ the number of good queues is a Catalan number, which turns up all over combinatorics. For random walks and the reflection principle, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP I 2011, Question 13

Probability and Statistics • Continuous Random Variables

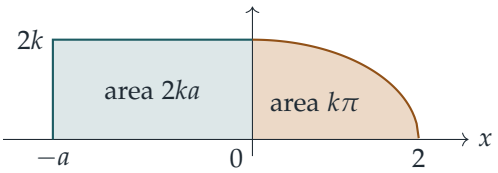
STEP I 2011, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q13

The key idea

Draw the density. It is a rectangle of height $2k$ over $[-a, 0)$ joined to a quarter of a circle of radius 2 (scaled vertically by k) over $[0, 2]$. Areas under it are probabilities, so the rectangle and the quarter-circle do most of the work without any integration. The total area must be 1, and that fixes k. After that, every question about d comes down to asking how much area lies to the right of a point.



Solution

Finding k

On $[0, 2]$ the graph of $\sqrt{4 - x^2}$ is a quarter of the circle $x^2 + y^2 = 4$, so

$$\int_0^2 \sqrt{4 - x^2}\,\mathrm{d}x = \tfrac{1}{4}\pi \cdot 2^2 = \pi.$$

(The given antiderivative agrees: $2\arcsin 1 + 0 - 0 = \pi$.) The total probability is therefore $2ka + k\pi = 1$, so

$$k = \frac{1}{2a + \pi}.$$

Part (i)

We split the integral for $\mathrm{E}(X)$ at 0:

$$\int_{-a}^0 2kx\,\mathrm{d}x = k\left[x^2\right]_{-a}^0 = -ka^2.$$

For the second piece, the derivative of $(4-x^2)^{3/2}$ is $-3x(4-x^2)^{1/2}$, so

$$\int_0^2 kx\sqrt{4-x^2}\,\mathrm{d}x = k\Big[-\tfrac{1}{3}(4-x^2)^{3/2}\Big]_0^2 = k\cdot\tfrac{1}{3}\cdot 8 = \tfrac{8}{3}k.$$

Adding the two pieces,

$$\mathrm{E}(X) = k\Big(\frac{8}{3}-a^2\Big) = \frac{8-3a^2}{3(2a+\pi)}.$$

Part (ii)

The probability to the right of 0 is the quarter-circle area:

$$\mathrm{P}(X>0) = k\pi = \frac{\pi}{2a+\pi}.$$

Ask yourself How does knowing $\mathrm{P}(X>0)$ tell us the sign of d?

The function $t \mapsto \mathrm{P}(X>t)$ is continuous and strictly decreasing for $-a \le t \le 2$, because the density is positive on $(-a,2)$. It falls from 1 to 0, so exactly one d satisfies $\mathrm{P}(X>d) = \frac{1}{10}$. Since the function is decreasing, $d<0$ exactly when $\mathrm{P}(X>0) < \frac{1}{10}$. In other words, d is negative when less than a tenth of the probability lies to the right of 0.

Now suppose $2a > 9\pi$. Then $2a+\pi > 10\pi$, so

$$\mathrm{P}(X>0) = \frac{\pi}{2a+\pi} < \frac{1}{10},$$

and hence $d<0$, as required. (The argument also runs backwards: $d<0$ only if $2a>9\pi$.)

In this case $-a \le d < 0$, so the area to the right of d is the part of the rectangle from d to 0, of width $-d$, together with the whole quarter-circle:

$$\mathrm{P}(X>d) = 2k(-d) + k\pi = \frac{\pi-2d}{2a+\pi}.$$

Setting this equal to $\frac{1}{10}$ gives $10\pi - 20d = 2a+\pi$, so

$$d = \frac{9\pi-2a}{20}.$$

This value really does lie in $[-a,0)$: it is negative because $2a>9\pi$, and $d \ge -a$ is equivalent to $9\pi+18a \ge 0$, which is true.

Part (iii)

Now $d=\sqrt{2}>0$, so d lies under the circular part and

$$\mathrm{P}(X>\sqrt{2}) = k\int_{\sqrt{2}}^2 \sqrt{4-x^2}\,\mathrm{d}x.$$

Using the given result,

$$\begin{aligned}\int_{\sqrt{2}}^{2} \sqrt{4-x^2}\,\mathrm{d}x &= \left[2\arcsin\left(\tfrac{1}{2}x\right) + \tfrac{1}{2}x\sqrt{4-x^2}\right]_{\sqrt{2}}^{2} \\ &= \left(2\cdot\tfrac{\pi}{2} + 0\right) - \left(2\cdot\tfrac{\pi}{4} + \tfrac{\sqrt{2}}{2}\cdot\sqrt{2}\right) \\ &= \frac{\pi}{2} - 1.\end{aligned}$$

(Geometrically, this is the circular segment cut off by the chord at $x = \sqrt{2}$. It equals an eighth of the disc, $\frac{\pi}{2}$, minus a right-angled triangle of area 1.) So

$$\frac{\frac{\pi}{2} - 1}{2a + \pi} = \frac{1}{10} \quad\Longrightarrow\quad 5\pi - 10 = 2a + \pi \quad\Longrightarrow\quad a = 2\pi - 5.$$

This is positive ($a \approx 1.283$), and it satisfies $2a \leq 9\pi$, which agrees with part (ii) since $d > 0$.

Checks

- In part (i), with $a = 1$ the mean is $\frac{5}{3(2+\pi)} \approx 0.3242$. A Monte Carlo simulation with 2×10^5 samples gave 0.3236.
- The formula in part (ii) gives $d = 0$ at the borderline $2a = 9\pi$, which is consistent with the condition. With $a = 20$ it gives $d = \frac{9\pi - 40}{20} \approx -0.586$, and solving $\mathrm{P}(X > d) = \frac{1}{10}$ numerically gives the same value.
- In part (iii), with $a = 2\pi - 5$, Simpson's rule gives $\mathrm{P}(X > \sqrt{2}) = 0.1000$.

Common slips

- Leaving k unknown. The normalisation $2ka + k\pi = 1$ is the hidden first step, and every answer is in terms of a alone.
- In part (ii), forgetting the quarter-circle and writing $\mathrm{P}(X > d) = -2kd$ only. All of the probability on $[0, 2]$ also lies to the right of a negative d.
- Getting the sign of $\int_{-a}^{0} 2kx\,\mathrm{d}x$ wrong. It is negative, because $x < 0$ throughout.

Beyond STEP

The value d is the upper 10% point, or 0.9-quantile, of X. Finding it means inverting the cumulative distribution function F, and that inverse is exactly what is used to simulate a random variable from a uniform one (inverse transform sampling). The mixed shape of this density, flat on one side and curved on the other, shows why quantiles often have to be found piece by piece. Grimmett and Welsh, *Probability: An Introduction*, covers distribution functions and quantiles carefully.

WORKED SOLUTION

STEP II 2011, Question 1

Pure • Equations with Square Roots

STEP II 2011, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q1

The key idea

Both parts are about square-root sums whose graphs are easy to understand once you find the domain, the endpoint values and the symmetry. In part (i) the picture gives an inequality, the curve never dips below a certain height, and the line can only reach that height at one point, so there is nothing to solve. In part (ii) the graph tells us how many roots to expect, and that lets us throw away the extra root that squaring brings in.

Solution

Part (i)

Let $f(x) = \sqrt{1-x} + \sqrt{3+x}$. Both roots are real only when $1 - x \geq 0$ and $3 + x \geq 0$, so the domain is $-3 \leq x \leq 1$. At the ends, $f(-3) = 2 + 0 = 2$ and $f(1) = 0 + 2 = 2$.

Writing $x = -1 + t$ gives $f = \sqrt{2-t} + \sqrt{2+t}$, which is unchanged when t is replaced by $-t$, so the curve is symmetric about $x = -1$. Differentiating,

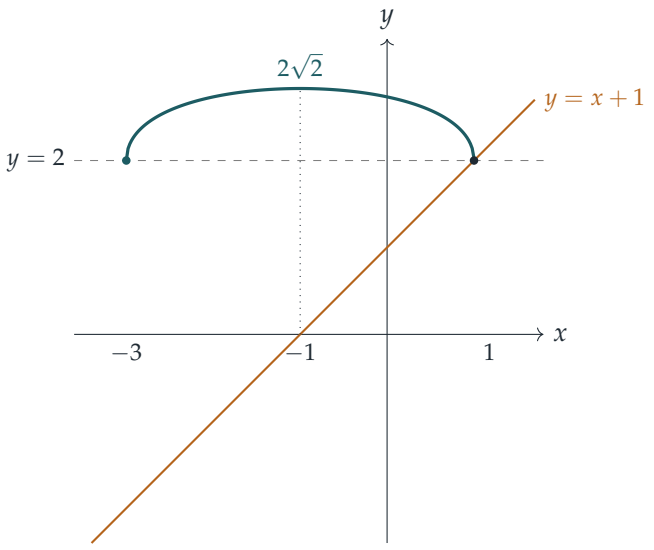
$$f'(x) = -\frac{1}{2\sqrt{1-x}} + \frac{1}{2\sqrt{3+x}},$$

which is zero only when $1 - x = 3 + x$, that is at $x = -1$, where $f(-1) = 2\sqrt{2}$. For $x < -1$ we have $3 + x < 1 - x$, so $f'(x) > 0$; for $x > -1$, $f'(x) < 0$. So f rises from 2 to a maximum of $2\sqrt{2}$ and falls back to 2. Near $x = -3$ the second term of f' blows up, so the curve leaves $(-3, 2)$ vertically, and likewise it arrives at $(1, 2)$ vertically. Each term of f is concave, so the whole curve is a smooth cap.

A cleaner way to see the lowest value is to square:

$$f(x)^2 = (1-x) + (3+x) + 2\sqrt{(1-x)(3+x)} = 4 + 2\sqrt{(1-x)(3+x)} \geq 4,$$

so $f(x) \geq 2$ on the whole domain, with equality only at $x = -3$ and $x = 1$.



Now bring in the line $y = x + 1$. On the domain $x \leq 1$, so $x + 1 \leq 2$, with equality only at $x = 1$. Meanwhile $f(x) \geq 2$. So $f(x) = x + 1$ forces both sides to equal 2, which happens only at $x = 1$. Checking: $\sqrt{0} + \sqrt{4} = 2 = 1 + 1$. So exactly one real value satisfies the equation:

$$x = 1.$$

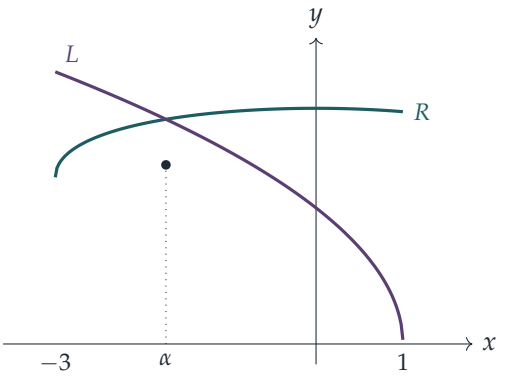
(Outside $[-3, 1]$ the left-hand side isn't defined, so there are no other candidates.)

Part (ii)

Again we need $x \leq 1$, $x \geq -3$ and $x \leq 3$, so the domain is $[-3, 1]$. Compare

$$L(x) = 2\sqrt{1 - x} \qquad \text{and} \qquad R(x) = \sqrt{3 + x} + \sqrt{3 - x}.$$

L is strictly decreasing, from $L(-3) = 4$ to $L(1) = 0$. R has the same shape as the curve in part (i): it is symmetric about $x = 0$ (swap x and $-x$), with $R(\pm 3) = \sqrt{6}$ and a maximum $R(0) = 2\sqrt{3}$, increasing on $[-3, 0]$ and decreasing on $[0, 3]$. On our domain we see only the part from $x = -3$ to $x = 1$, where $R(1) = 2 + \sqrt{2}$.



At $x = -3$, $L = 4 > \sqrt{6} = R$; at $x = 0$, $L = 2 < 2\sqrt{3} = R$. On $[-3, 0]$, L is strictly decreasing and R increasing, so $L - R$ is strictly decreasing and changes sign exactly once. On $[0, 1]$, $L \leq 2$ while $R \geq 2 + \sqrt{2}$, so there is no crossing. Hence there is **exactly one** real solution, and it lies in $-3 < x < 0$.

To find it, square. Both sides are non-negative, so squaring loses nothing:

$$\begin{aligned} & 4(1-x) = (3+x) + (3-x) + 2\sqrt{9-x^2} \\ \iff \quad & -1 - 2x = \sqrt{9-x^2}. \end{aligned}$$

Ask yourself When we square again, which solutions of the new equation are genuine?

The right-hand side is a square root, so it is ≥ 0; a genuine solution needs $-1 - 2x \geq 0$, that is $x \leq -\frac{1}{2}$. With that condition, squaring is reversible:

$$1 + 4x + 4x^2 = 9 - x^2 \iff 5x^2 + 4x - 8 = 0 \iff x = \frac{-2 \pm 2\sqrt{11}}{5}.$$

The root $\frac{-2+2\sqrt{11}}{5} \approx 0.93$ breaks $x \leq -\frac{1}{2}$, so we reject it. The other root, $\frac{-2-2\sqrt{11}}{5} \approx -1.73$, satisfies it and lies in $[-3, 0]$, as the graph predicted. So the unique solution is

$$x = -\frac{2 + 2\sqrt{11}}{5}.$$

Checks

- At $x \approx -1.72665$: $2\sqrt{1-x} \approx 3.3025$ and $\sqrt{3+x} + \sqrt{3-x} \approx 1.1284 + 2.1741 \approx 3.3025$. The two sides agree.
- The rejected root $x \approx 0.92665$ gives $L \approx 0.542$ but $R \approx 3.42$. It actually solves $\sqrt{3+x} - \sqrt{3-x} = 2\sqrt{1-x}$, which squares to the same thing.
- In part (i), $f(-1) = 2\sqrt{2} \approx 2.83 > 0 = (-1) + 1$, consistent with the line lying below the curve everywhere except at $(1, 2)$.

Common slips

- In part (i), squaring and solving algebraically, which is messy and can bring in spurious roots, when the bound $f \geq 2 \geq x + 1$ settles it at once.
- Keeping both roots of $5x^2 + 4x - 8 = 0$. Every squaring step needs its sign condition, here $-1 - 2x \geq 0$.
- Forgetting the domain: neither equation means anything for $x > 1$.

Beyond STEP

Part (i) is a small example of a common analysis technique: rather than solving $f(x) = g(x)$, show that $f \geq c \geq g$ and look at when equality holds. Part (ii) uses the intermediate value theorem together with monotonicity to count roots before finding any, which is how existence and uniqueness are proved in a first analysis course. Abbott's *Understanding Analysis* gives a careful treatment of continuity and the intermediate value theorem.

WORKED SOLUTION

STEP II 2011, Question 2

Pure • Diophantine Equations

> **STEP II 2011, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/11-s2-q2

The key idea

Whenever you see $x^3 + y^3$, factorise it: $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$. Once $x + y$ is pinned down (to k in (i), to z^2 in (ii)), the factor $x + y$ cancels against part of kz^3, leaving a quadratic condition. Completing the square in that quadratic turns it into "something $= 3\times$ a perfect square", and bounds then leave only a handful of cubes to try. That is why the question asks for the cubes up to 10^3 first.

For reference: $1, 8, 27, 64, 125, 216, 343, 512, 729, 1000$.

Solution

Part (i)

Since $x + y = k$,

$$kz^3 = x^3 + y^3 = (x + y)(x^2 - xy + y^2) = k(x^2 - xy + y^2),$$

and $k > 0$, so $z^3 = x^2 - xy + y^2$. Substituting $y = k - x$:

$$\begin{aligned} z^3 &= x^2 - x(k - x) + (k - x)^2 \\ &= x^2 - kx + x^2 + k^2 - 2kx + x^2 \\ &= k^2 - 3kx + 3x^2, \end{aligned}$$

as required.

The perfect square. Multiply by 4 and complete the square:

$$4z^3 - k^2 = 12x^2 - 12kx + 3k^2 = 3\big(4x^2 - 4kx + k^2\big) = 3(k - 2x)^2.$$

So $\dfrac{4z^3 - k^2}{3} = (k - 2x)^2$, the square of the integer $k - 2x$.

The bounds. From the line above, $4z^3 = k^2 + 3(k-2x)^2 \geq k^2$, so $\frac{1}{4}k^2 \leq z^3$. For the upper bound, note $3kx - 3x^2 = 3x(k-x) = 3xy$, so

$$z^3 = k^2 - 3xy < k^2,$$

since x and y are positive. Hence $\frac{1}{4}k^2 \leq z^3 < k^2$.

The case $k = 20$. We look for a solution with $x + y = 20$. The bounds give $100 \leq z^3 < 400$, so from the list of cubes $z^3 \in \{125, 216, 343\}$. Now test whether $(4z^3 - 400)/3$ is a perfect square:

- $z = 5$: $(500 - 400)/3 = 100/3$, not an integer;
- $z = 6$: $(864 - 400)/3 = 464/3$, not an integer;
- $z = 7$: $(1372 - 400)/3 = 324 = 18^2$. ✓

So $(k-2x)^2 = 18^2$ with $k = 20$. Since $x < y$ means $2x < x + y = k$, we have $k - 2x > 0$, so $k - 2x = 18$, giving $x = 1$ and $y = 19$. Check: $1^3 + 19^3 = 1 + 6859 = 6860 = 20 \times 343 = 20 \times 7^3$. So

$$x = 1, \quad y = 19, \quad z = 7$$

is a solution of $(*)$ with $k = 20$.

Part (ii)

Now suppose $x + y = z^2$. Then

$$kz^3 = (x+y)(x^2 - xy + y^2) = z^2(x^2 - xy + y^2),$$

and dividing by z^2 (allowed, as $z > 0$) gives $x^2 - xy + y^2 = kz$.

Ask yourself How can we get the same "perfect square" trick to work here?

Use the identity $4(x^2 - xy + y^2) = (x+y)^2 + 3(x-y)^2$, which is what we found in (i) in disguise (expand the right-hand side to check). With $x + y = z^2$ and $k = 19$ this becomes

$$76z = z^4 + 3(y-x)^2, \qquad \text{that is} \qquad 3(y-x)^2 = z\,(76 - z^3).$$

Since $y > x$, the left-hand side is positive, so $z^3 < 76$ and therefore $z \in \{1, 2, 3, 4\}$. We need $z(76 - z^3)/3$ to be a perfect square:

- $z = 1$: $75/3 = 25 = 5^2$, so $y - x = 5$; but $x + y = 1$, impossible for positive integers.
- $z = 2$: $2 \times 68/3 = 136/3$, not an integer.
- $z = 3$: $3 \times 49/3 = 49 = 7^2$, so $y - x = 7$ and $x + y = 9$, giving $x = 1$, $y = 8$.
- $z = 4$: $4 \times 12/3 = 16 = 4^2$, so $y - x = 4$ and $x + y = 16$, giving $x = 6$, $y = 10$.

Checking: $1^3 + 8^3 = 513 = 19 \times 27 = 19 \times 3^3$, and $6^3 + 10^3 = 1216 = 19 \times 64 = 19 \times 4^3$. So two solutions of $(*)$ with $k = 19$ are

$$(x, y, z) = (1, 8, 3) \quad \text{and} \quad (x, y, z) = (6, 10, 4).$$

(In fact the argument shows these are the only ones with $x + y = z^2$.)

Checks

- Test the identity of (i) on a small case: $k = 5$, $x = 2$, $y = 3$ gives $x^3 + y^3 = 35 = 5 \times 7$, and $k^2 - 3kx + 3x^2 = 25 - 30 + 12 = 7$. ✓
- Substitute every answer back into $(*)$, as done above: $6860 = 20 \times 343$, $513 = 19 \times 27$, $1216 = 19 \times 64$.
- A computer search over all $x < y$ with $x + y = 20$, and over all $x < y$ with $x + y = z^2$ for $k = 19$, finds exactly these solutions and no others.

Common slips

- Forgetting the sign: $(k - 2x)^2 = 18^2$ also allows $k - 2x = -18$, but that gives $x = 19 > y$, which the condition $x < y$ rules out.
- In (ii), keeping $z = 1$ as a solution: $y - x = 5$ is fine, but it can't coexist with $x + y = 1$.

Beyond STEP

Equations like $x^3 + y^3 = kz^3$ ask which integers k are sums of two rational cubes, a question that leads straight to elliptic curves and is still not fully solved; the famous taxicab number $1729 = 1^3 + 12^3 = 9^3 + 10^3$ lives in the same world. The tricks here (factorising, completing the square, then bounding to get a finite search) are the bread and butter of elementary number theory. Davenport's *The Higher Arithmetic* is a lovely next step.

WORKED SOLUTION

STEP II 2011, Question 3

Pure • Inequalities via Monotonicity

STEP II 2011, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q3

The key idea

Every inequality here comes from one principle: if a function starts at zero and has a non-negative derivative, it stays non-negative. The last two results add a twist. Once we know a function is increasing, we can feed it two inputs whose order we already know (such as $x \leq \arcsin x$) and read off an inequality between the outputs. Spotting *which* two inputs to compare is the whole trick.

Solution

Part (i)(a)

Differentiating, using the product rule on $x\cos x$,

$$f'(x) = \cos x - (\cos x - x\sin x) = x\sin x.$$

For $0 \leq x \leq \frac{1}{2}\pi$ both $x \geq 0$ and $\sin x \geq 0$, so $f'(x) \geq 0$ and f is increasing on this interval.

Hence for $0 \leq x \leq \frac{1}{2}\pi$ we have $x \geq 0$, so $f(x) \geq f(0) = \sin 0 - 0 = 0$, as required.

Part (i)(b)

Let $h(x) = \arcsin x - x$ for $0 \leq x < 1$. Then

$$h'(x) = \frac{\mathrm{d}}{\mathrm{d}x}(\arcsin x) - 1 \geq 1 - 1 = 0,$$

using the given fact. So h is increasing, and $h(x) \geq h(0) = 0$. That is, $\arcsin x \geq x$ for $0 \leq x < 1$.

(The given fact is just $\dfrac{1}{\sqrt{1-x^2}} \geq 1$, true because $0 < \sqrt{1-x^2} \leq 1$.)

Part (i)(c)

Write $g(x) = \dfrac{x}{\sin x}$. For $0 < x < \frac{1}{2}\pi$, the quotient rule gives

$$g'(x) = \frac{\sin x - x\cos x}{\sin^2 x} = \frac{f(x)}{\sin^2 x}.$$

By part (a), $f(x) \geq 0$ here, and $\sin^2 x > 0$, so $g'(x) \geq 0$ and g is increasing on $(0, \frac{1}{2}\pi)$.

> **Ask yourself** How does knowing that g is increasing help with an inequality involving $\arcsin x$?

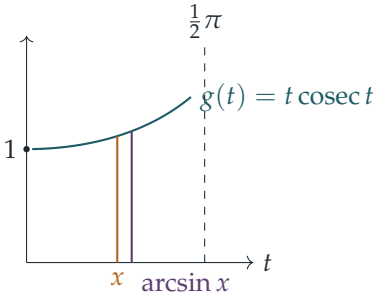
Look at the target: $\dfrac{\arcsin x}{x} \geq \dfrac{x}{\sin x}$. The right-hand side is $g(x)$. The left-hand side is also a value of g. If $y = \arcsin x$ then $\sin y = x$, so

$$g(y) = \frac{y}{\sin y} = \frac{\arcsin x}{x}.$$

So we need $g(\arcsin x) \geq g(x)$, and that follows from $\arcsin x \geq x$, which is part (b), because g is increasing.

To finish, take $0 < x < 1$ and let $y = \arcsin x$. Then $0 < y < \frac{1}{2}\pi$. Also $0 < x < 1 < \frac{1}{2}\pi$, so both x and y lie in the interval where g is increasing. By part (b), $y \geq x$, so $g(y) \geq g(x)$:

$$(\arcsin x)\, x^{-1} = \frac{y}{\sin y} \geq \frac{x}{\sin x} = x \operatorname{cosec} x \qquad (0 < x < 1).$$



The picture shows the argument for $x = 0.8$: the input $\arcsin x$ lies to the right of x, so the increasing curve is higher there.

Part (ii)

We copy the pattern of (i). First, the analogue of (b). Let $m(x) = x - \arctan x$ for $x \geq 0$. By the given fact, $m'(x) = 1 - \dfrac{\mathrm{d}}{\mathrm{d}x}(\arctan x) \geq 0$, so $m(x) \geq m(0) = 0$. Hence

$$\arctan x \leq x \qquad (x \geq 0).$$

Next, the analogue of (c). Let $k(x) = x^{-1}\tan x$ for $0 < x < \frac{1}{2}\pi$. Then

$$k'(x) = \frac{x\sec^2 x - \tan x}{x^2} = \frac{x - \sin x\cos x}{x^2\cos^2 x} = \frac{2x - \sin 2x}{2x^2\cos^2 x},$$

multiplying top and bottom by $\cos^2 x$ (which is positive) and then by 2.

? **Ask yourself** Why is $2x - \sin 2x \geq 0$?

This is the familiar $\sin t \leq t$ for $t \geq 0$, with $t = 2x$. It can be proved in the same style as everything else: $p(t) = t - \sin t$ has $p'(t) = 1 - \cos t \geq 0$ and $p(0) = 0$, so $p(t) \geq 0$ for $t \geq 0$.

So the numerator of $k'(x)$ is non-negative and the denominator is positive, giving $k'(x) \geq 0$. Thus k is increasing on $(0, \frac{1}{2}\pi)$.

Now take $0 < x < \frac{1}{2}\pi$ and let $y = \arctan x$. Then $0 < y < \frac{1}{2}\pi$, so both x and y lie in the interval where k is increasing, and $\tan y = x$. Since $y \leq x$, we get $k(y) \leq k(x)$:

$$\frac{x}{\arctan x} = \frac{\tan y}{y} \leq \frac{\tan x}{x}.$$

Both x and $\arctan x$ are positive, so multiplying through by $x\arctan x$ keeps the direction of the inequality:

$$(\tan x)(\arctan x) \geq x^2 \qquad (0 < x < \tfrac{1}{2}\pi),$$

as required.

Checks

- At $x = \frac{1}{2}$ in (c): $\arcsin(\frac{1}{2})/\frac{1}{2} = \pi/3 \approx 1.0472$, while $\frac{1}{2}\operatorname{cosec}\frac{1}{2} \approx 1.0429$. The inequality holds, and only just, which is what we expect: both sides tend to 1 as $x \to 0$.
- At $x = 1$ in (ii): $\tan 1 \cdot \arctan 1 = \frac{1}{4}\pi\tan 1 \approx 1.2232 \geq 1$.
- For small x, $\tan x \approx x + \frac{1}{3}x^3$ and $\arctan x \approx x - \frac{1}{3}x^3$, so the product is $x^2 + O(x^6)$: the inequality in (ii) is extremely tight near 0. A numerical check at 20,000 random points confirmed every inequality and every derivative formula above.

Common slips

- In (c), comparing g at the wrong pair of points, or forgetting to check that $\arcsin x$ is inside $(0, \frac{1}{2}\pi)$, where g is known to be increasing.
- In (ii), the inequality between the inputs points the other way ($\arctan x \leq x$), so the inequality between the outputs does too. Getting this backwards gives the wrong inequality.
- Multiplying an inequality through without saying that the multiplier ($x\arctan x$, or $\cos^2 x$) is positive.

Beyond STEP

The principle used throughout, that a non-negative derivative forces a function to increase, is a consequence of the mean value theorem, which is proved in any first course in analysis. The trick of comparing $g(x)$ with $g(\arcsin x)$, where sin and arcsin are inverses, is a cousin of ideas you'll meet with inverse functions and convexity. Abbott's *Understanding Analysis* gives a careful, readable proof of the mean value theorem and its consequences.

WORKED SOLUTION

STEP II 2011, Question 4

Pure • Trigonometric Equations

STEP II 2011, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q4

The key idea

Write $\cos\theta$ as the sine (or cosine) of a related angle, so that "$\cos\theta = \sin 4\theta$" becomes an equation between two cosines, which we can solve exactly. Expanding $\sin 4\theta$ in powers of $\sin\theta$ gives a cubic whose roots are the sines of those angles, and that is how a surd like $\sin 18^\circ$ appears. Part (ii) is a quadratic in $\sin^2 x$, and part (iii) is part (ii) in disguise once you spot an angle whose sine squared is $\frac{1}{4}$.

Solution

Part (i)

To compare like with like, write $\sin 4\theta = \cos(90^\circ - 4\theta)$. Two angles have the same cosine exactly when they are equal or opposite, up to multiples of 360°, so

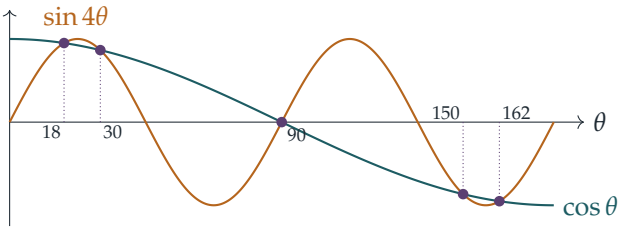
$$\cos\theta = \cos(90^\circ - 4\theta) \iff \theta = \pm(90^\circ - 4\theta) + 360^\circ k \quad \text{for some integer } k.$$

- With the $+$ sign: $5\theta = 90^\circ + 360^\circ k$, so $\theta = 18^\circ + 72^\circ k$. In $0^\circ < \theta < 180^\circ$ this gives $\theta = 18^\circ,\ 90^\circ,\ 162^\circ$.
- With the $-$ sign: $3\theta = 90^\circ - 360^\circ k$, so $\theta = 30^\circ - 120^\circ k$, giving $\theta = 30^\circ,\ 150^\circ$.

So the solutions are

$$\theta = 18^\circ,\ 30^\circ,\ 90^\circ,\ 150^\circ,\ 162^\circ.$$

The graphs below confirm there are exactly five crossings.



Now for the "hence". Write $s = \sin\theta$. Using the double-angle formulae twice,

$$\sin 4\theta = 2\sin 2\theta\cos 2\theta = 4\sin\theta\cos\theta\,(1 - 2\sin^2\theta),$$

so the equation $\cos\theta = \sin 4\theta$ becomes

$$\cos\theta\big(1 - 4s(1 - 2s^2)\big) = 0.$$

Ask yourself Why can we cancel the $\cos\theta$?

Of our five solutions, only $\theta = 90^\circ$ has $\cos\theta = 0$. The other four, $18^\circ, 30^\circ, 150^\circ, 162^\circ$, all have $\cos\theta \neq 0$, so for them we may divide by $\cos\theta$ and get

$$8s^3 - 4s + 1 = 0.$$

We know $s = \sin 30^\circ = \frac{1}{2}$ is one root, and factorising,

$$8s^3 - 4s + 1 = (2s - 1)(4s^2 + 2s - 1).$$

Now $\sin 18^\circ$ is a root of the cubic, and $0 < \sin 18^\circ < \sin 30^\circ = \frac{1}{2}$ because sine is increasing on $[0^\circ, 90^\circ]$. So $\sin 18^\circ \neq \frac{1}{2}$, and it must be a root of $4s^2 + 2s - 1 = 0$, whose roots are

$$s = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-1 \pm \sqrt{5}}{4}.$$

Only the $+$ root is positive, so

$$\sin 18^\circ = \tfrac{1}{4}(\sqrt{5} - 1),$$

as required.

Part (ii)

Let $s = \sin x$. Since $\sin^2 2x = 4\sin^2 x\cos^2 x = 4s^2(1 - s^2)$, the equation depends only on s:

$$4s^2 + 1 = 16s^2(1 - s^2) \iff 16s^4 - 12s^2 + 1 = 0.$$

This is a quadratic in s^2:

$$s^2 = \frac{12 \pm \sqrt{144 - 64}}{32} = \frac{12 \pm 4\sqrt{5}}{32} = \frac{3 \pm \sqrt{5}}{8}.$$

To take square roots, notice that

$$\left(\frac{1 + \sqrt{5}}{4}\right)^2 = \frac{6 + 2\sqrt{5}}{16} = \frac{3 + \sqrt{5}}{8}, \qquad \left(\frac{\sqrt{5} - 1}{4}\right)^2 = \frac{6 - 2\sqrt{5}}{16} = \frac{3 - \sqrt{5}}{8}.$$

Both values of s^2 lie between 0 and 1, so every one of the four square roots is a genuine value of $\sin x$, and each satisfies the original equation because the steps above were all reversible. The possible values are

$$\sin x = \tfrac{1}{4} + \tfrac{1}{4}\sqrt{5}, \qquad -\tfrac{1}{4} - \tfrac{1}{4}\sqrt{5}, \qquad -\tfrac{1}{4} + \tfrac{1}{4}\sqrt{5}, \qquad \tfrac{1}{4} - \tfrac{1}{4}\sqrt{5}.$$

By part (i), $\frac{1}{4}(\sqrt{5} - 1) = \sin 18^\circ$; and $\frac{1}{4}(1 + \sqrt{5}) = \sin 54^\circ$ (it is $\cos 36^\circ = 1 - 2\sin^2 18^\circ$, which you can check comes to the same thing).

Part (iii)

Divide the equation of part (ii) by 4:

$$\sin^2 x + \tfrac{1}{4} = \sin^2 2x.$$

Compare this with $\sin^2 3\alpha + \sin^2 5\alpha = \sin^2 6\alpha$. If we put $x = 3\alpha$, then $2x = 6\alpha$ automatically, and the two equations match provided

$$\sin^2 5\alpha = \tfrac{1}{4} \qquad \text{and} \qquad \sin 3\alpha \text{ is one of the four values in part (ii).}$$

Ask yourself Is it enough for $\sin 3\alpha$ to be one of those values?

Yes: in part (ii) we saw that the equation $\sin^2 x + \frac{1}{4} = \sin^2 2x$ involves x only through $s = \sin x$, and those four values of s satisfy it. So any x with one of those sines works.

Take $\alpha = 6°$. Then $5\alpha = 30°$, so $\sin^2 5\alpha = \frac{1}{4}$, and $3\alpha = 18°$, so $\sin 3\alpha = \frac{1}{4}(\sqrt{5} - 1)$, one of our values. Hence

$$\sin^2 18° + \sin^2 30° = \sin^2 36°,$$

and $\alpha = 6°$ is a solution.

Take $\alpha = 42°$. Then $5\alpha = 210°$, so $\sin 5\alpha = -\frac{1}{2}$ and $\sin^2 5\alpha = \frac{1}{4}$; and $3\alpha = 126°$, so $\sin 3\alpha = \sin 54° = \frac{1}{4}(1 + \sqrt{5})$, again one of our values. So $\alpha = 42°$ is a solution.

So two suitable values are $\alpha = 6°$ and $\alpha = 42°$.

The same method also gives $\alpha = 66°$ ($5\alpha = 330°$, $\sin 3\alpha = \sin 198° = -\sin 18°$) and $\alpha = 78°$ ($5\alpha = 390°$, $\sin 3\alpha = \sin 234° = -\sin 54°$). The other angle in range with $\sin^2 5\alpha = \frac{1}{4}$ is $\alpha = 30°$, and it fails, since $\sin 90° = 1$ is not one of our values. (Indeed $1 + \frac{1}{4} \neq \sin^2 180° = 0$.)

Checks

- $\sin 18° \approx 0.30902$ and $\frac{1}{4}(\sqrt{5} - 1) \approx 0.30902$.
- For $\alpha = 6°$: $\sin^2 18° + \sin^2 30° \approx 0.09549 + 0.25 = 0.34549$, and $\sin^2 36° \approx 0.34549$.
- A numerical scan of $\sin^2 3\alpha + \sin^2 5\alpha - \sin^2 6\alpha$ on $(0°, 90°)$ finds zeros at exactly $6°, 42°, 45°, 66°, 78°$. The extra one, $45°$, does not come from part (ii): there $\frac{1}{2} + \frac{1}{2} = 1$.

Common slips

- Forgetting the second family of solutions of $\cos A = \cos B$ (namely $A = -B + 360°k$), and so missing $30°$ and $150°$ in part (i).
- Cancelling $\cos\theta$ without comment. It is fine here, but only because the solution with $\cos\theta = 0$ is $90°$, not $18°$.
- Giving only the positive values of $\sin x$ in part (ii). All four are possible.

Beyond STEP

The exact value of $\sin 18^\circ$ comes from the fact that 18° is a fifth of a right angle, and so it is tied to the regular pentagon and the golden ratio: $\frac{1}{4}(1+\sqrt{5}) = \frac{1}{2}\varphi$. More generally, $\cos(360^\circ/n)$ can be written using square roots alone exactly when a regular n-gon can be constructed with ruler and compasses, which Gauss showed depends on the Fermat primes dividing n. This is a classic first application of field theory and Galois theory. Allenby's *Rings, Fields and Groups* builds up the algebra, and Needham's *Visual Complex Analysis* shows why roots of unity and polygons belong together.

WORKED SOLUTION

STEP II 2011, Question 5

Pure • Vectors and Reflections

STEP II 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q5

The key idea

To reflect a point in a line through the origin, split its position vector into a part along the line and a part perpendicular to it: the reflection keeps the first and reverses the second. The component of $\mathbf{b}$ along $\mathbf{a}$ is the familiar projection $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$, and everything else is bookkeeping with dot products. For the collinearity, remember that a point lies on line AB exactly when its position vector is $s\mathbf{a}+t\mathbf{b}$ with $s+t=1$.

Solution

Throughout, $\mathbf{a}$ and $\mathbf{b}$ are non-zero and not parallel, because O, A, B are not collinear. In particular $\mathbf{a}\cdot\mathbf{a}\neq 0$ and $\mathbf{b}\cdot\mathbf{b}\neq 0$, so we may divide by them. Write $\theta=\angle AOB$.

The reflection C

Let P be the foot of the perpendicular from B to the line OA. Its position vector is a multiple $k\mathbf{a}$, and $\mathbf{b}-k\mathbf{a}$ is perpendicular to $\mathbf{a}$:

$$(\mathbf{b}-k\mathbf{a})\cdot\mathbf{a}=0 \quad\Longrightarrow\quad k=\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}.$$

Reflecting in the line OA fixes P and sends B to the point C on the other side with P as the midpoint of BC. So $\mathbf{c}=2\overrightarrow{OP}-\mathbf{b}$, that is,

$$\mathbf{c}=\frac{2\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}-\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}, \qquad \lambda=\frac{2\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}},$$

as required.

The reflection D

The same argument with the line OB in place of OA gives

$$\mathbf{d}=\frac{2\,\mathbf{b}\cdot\mathbf{c}}{\mathbf{b}\cdot\mathbf{b}}\,\mathbf{b}-\mathbf{c}.$$

Now $\mathbf{b}\cdot\mathbf{c} = \lambda\,\mathbf{a}\cdot\mathbf{b} - \mathbf{b}\cdot\mathbf{b}$, so

$$\begin{aligned}\mathbf{d} &= \left(\frac{2\lambda\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}} - 2\right)\mathbf{b} - (\lambda\mathbf{a} - \mathbf{b}) \\ &= \left(\frac{2\lambda\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}} - 1\right)\mathbf{b} - \lambda\mathbf{a}.\end{aligned}$$

This is $\mathbf{d} = \mu\mathbf{b} - \lambda\mathbf{a}$ with

$$\mu = \frac{2\lambda\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}} - 1 = \frac{4(\mathbf{a}\cdot\mathbf{b})^2}{(\mathbf{a}\cdot\mathbf{a})(\mathbf{b}\cdot\mathbf{b})} - 1 = 4\cos^2\theta - 1,$$

using $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ in the last step. The form $4\cos^2\theta - 1$ is the one we shall need below.

A, B, D collinear

The point D lies on the line AB if and only if $\mathbf{d} = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) = (1-t)\mathbf{a} + t\mathbf{b}$ for some scalar t. We also have $\mathbf{d} = -\lambda\mathbf{a} + \mu\mathbf{b}$.

Ask yourself Why can we compare the coefficients of $\mathbf{a}$ and $\mathbf{b}$?

Because $\mathbf{a}$ and $\mathbf{b}$ are not parallel. If $s\mathbf{a} + t\mathbf{b} = s'\mathbf{a} + t'\mathbf{b}$ then $(s - s')\mathbf{a} = (t' - t)\mathbf{b}$, and unless both brackets are zero this would make $\mathbf{a}$ and $\mathbf{b}$ parallel. This is where the non-collinearity of O, A, B is used.

So $-\lambda = 1 - t$ and $\mu = t$. Eliminating t, A, B and D are collinear exactly when

$$\mu = \lambda + 1.$$

The case $\lambda = -\frac{1}{2}$

Collinearity now gives $\mu = \frac{1}{2}$, so $4\cos^2\theta - 1 = \frac{1}{2}$ and $\cos^2\theta = \frac{3}{8}$. Since $\lambda < 0$ we have $\mathbf{a}\cdot\mathbf{b} < 0$, so $\cos\theta < 0$ and

$$\cos\angle AOB = -\sqrt{\tfrac{3}{8}} = -\frac{\sqrt{6}}{4}.$$

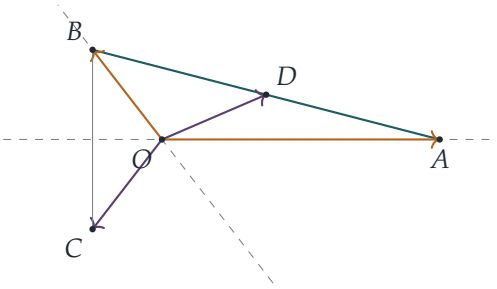
Then

$$\mathbf{d} = \tfrac{1}{2}\mathbf{b} + \tfrac{1}{2}\mathbf{a} = \tfrac{1}{2}(\mathbf{a} + \mathbf{b}),$$

so D **is the midpoint of** AB. We can say more about the shape. From $\lambda = \dfrac{2|\mathbf{b}|\cos\theta}{|\mathbf{a}|} = -\frac{1}{2}$,

$$\frac{|\mathbf{a}|}{|\mathbf{b}|} = -4\cos\theta = \sqrt{6},$$

so $OA = \sqrt{6}\,OB$, and the angle AOB is obtuse (about 127.8°). Also, reflections preserve distance from O, so $OD = OC = OB$: the midpoint of AB is exactly as far from O as B is.



Checks

- Reflecting twice, first in OA and then in OB, is a rotation about O through 2θ. So D should be B rotated through 2θ, with $\cos\angle BOD = \cos 2\theta = 2\cos^2\theta - 1$. In the final case this is $2\cdot\frac{3}{8} - 1 = -\frac{1}{4}$. Directly, $\mathbf{b}\cdot\frac{1}{2}(\mathbf{a}+\mathbf{b}) = \frac{1}{2}(-\frac{3}{2}+1)|\mathbf{b}|^2 = -\frac{1}{4}|\mathbf{b}|^2$, using $\mathbf{a}\cdot\mathbf{b} = -\frac{1}{4}\,\mathbf{a}\cdot\mathbf{a} = -\frac{3}{2}\,\mathbf{b}\cdot\mathbf{b}$, which agrees.
- If $\theta = 90°$ then $\lambda = 0$, C is $-\mathbf{b}$ and D is $-\mathbf{b}$ again. The formula gives $\mu = 4\cdot 0 - 1 = -1$, as it should.
- Numerically, with $\mathbf{a} = (\sqrt{6}, 0)$ and $\mathbf{b} = (-0.6124, 0.7906)$, reflecting twice gives $\mathbf{d} = (0.9186, 0.3953)$, which is the midpoint of AB, and $|\mathbf{d}| = 1 = |\mathbf{b}|$.

Common slips

- Writing the reflection as $\mathbf{b} - 2\times$ (projection). That reflects in the line *perpendicular* to OA. Check by testing $\mathbf{b} = \mathbf{a}$, which should be fixed.
- Taking $\cos\theta = +\sqrt{6}/4$. The sign of λ fixes the sign of $\mathbf{a}\cdot\mathbf{b}$.
- Comparing coefficients without saying that $\mathbf{a}$ and $\mathbf{b}$ are independent.

Beyond STEP

The map $\mathbf{x} \mapsto \frac{2\,\mathbf{a}\cdot\mathbf{x}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a} - \mathbf{x}$ is a linear map, and its matrix is a standard first-year example of an orthogonal matrix with determinant -1. Composing two reflections gives a rotation through twice the angle between the mirrors. This is the start of the study of isometries and of groups generated by reflections, such as the symmetry groups of regular polygons. Strang's *Introduction to Linear Algebra* covers projections and reflection matrices well.

WORKED SOLUTION

STEP II 2011, Question 6

Pure • Integration by Parts

STEP II 2011, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q6

The key idea

The integrand $[f'(x)]^2[f(x)]^n$ is one factor of f' short of being something we can integrate on sight, because $f'(x)[f(x)]^n$ is the derivative of $[f(x)]^{n+1}/(n+1)$. So we integrate by parts, splitting off that piece. The leftover integral contains f'', and the condition $f'' = kff'$ turns it back into the same easy shape, $f'f^m$. The applications then come down to choosing f and n so that the given integrand is $[f']^2 f^n$, and finding k.

Solution

The general result

Write f for $f(x)$, and so on. Integrate by parts with $u = f'$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = f'f^n$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}x} = f''$ and $v = \dfrac{f^{n+1}}{n+1}$ (fine, since $n \geq 1$ means $n+1 \neq 0$):

$$\begin{aligned} I = \int f' \cdot f'f^n \,\mathrm{d}x &= \frac{f'f^{n+1}}{n+1} - \frac{1}{n+1}\int f''f^{n+1}\,\mathrm{d}x \\ &= \frac{f'f^{n+1}}{n+1} - \frac{k}{n+1}\int f'f^{n+2}\,\mathrm{d}x \qquad \text{using } f'' = kff' \\ &= \frac{f'f^{n+1}}{n+1} - \frac{kf^{n+3}}{(n+1)(n+3)} + C. \end{aligned}$$

So

$$\boxed{I = \frac{f'(x)\,[f(x)]^{n+1}}{n+1} - \frac{k\,[f(x)]^{n+3}}{(n+1)(n+3)} + C}$$

which is an expression in terms of $f(x)$, $f'(x)$, k and n, as required.

Part (i)

For $f(x) = \tan x$ we have $f'(x) = \sec^2 x$ and

$$f''(x) = 2\sec x \cdot \sec x \tan x = 2\tan x \sec^2 x = 2f(x)f'(x),$$

so the condition holds with $k = 2$. The formula then gives

$$\int \sec^4 x \tan^n x \, \mathrm{d}x = \frac{\sec^2 x \tan^{n+1} x}{n+1} - \frac{2\tan^{n+3} x}{(n+1)(n+3)} + C. \tag{1}$$

Verification. Directly, using $\sec^2 x = 1 + \tan^2 x$ and the substitution $t = \tan x$, $\mathrm{d}t = \sec^2 x \, \mathrm{d}x$:

$$\int \sec^4 x \tan^n x \, \mathrm{d}x = \int (1+t^2) t^n \, \mathrm{d}t = \frac{t^{n+1}}{n+1} + \frac{t^{n+3}}{n+3} + C.$$

Rewriting the right-hand side of (1) with $\sec^2 x = 1 + t^2$ gives

$$\frac{(1+t^2)t^{n+1}}{n+1} - \frac{2t^{n+3}}{(n+1)(n+3)} = \frac{t^{n+1}}{n+1} + \frac{(n+3)-2}{(n+1)(n+3)} t^{n+3} = \frac{t^{n+1}}{n+1} + \frac{t^{n+3}}{n+3},$$

the same answer. So the result is verified for $f = \tan$.

The integral. Since

$$\frac{\sin^4 x}{\cos^8 x} = \frac{\sin^4 x}{\cos^4 x} \cdot \frac{1}{\cos^4 x} = \tan^4 x \sec^4 x = [f'(x)]^2 [f(x)]^4,$$

this is the case $n = 4$ of (1). Using the simplified form just found,

$$\int \frac{\sin^4 x}{\cos^8 x} \, \mathrm{d}x = \frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + C.$$

Part (ii)

? **Ask yourself** Which f should we take, and where does the $\sec^2 x$ come from?

The bracket $(\sec x + \tan x)$ is the obvious candidate for f, so try $f(x) = \sec x + \tan x$. Its derivative is

$$f'(x) = \sec x \tan x + \sec^2 x = \sec x(\tan x + \sec x) = \sec x \, f(x).$$

This is the key fact: differentiating f just multiplies it by $\sec x$. Hence

$$\begin{aligned} f''(x) &= \sec x \tan x \, f(x) + \sec x \, f'(x) = \sec x \tan x \, f + \sec^2 x \, f \\ &= \sec x \, f \, (\tan x + \sec x) = \sec x \, f^2, \end{aligned}$$

and since $f f' = f \cdot \sec x \, f = \sec x \, f^2$ too, we have $f'' = f f'$, so $k = 1$.

Also $[f'(x)]^2 = \sec^2 x \, [f(x)]^2$, so

$$\sec^2 x \, (\sec x + \tan x)^6 = \sec^2 x \, f^2 \cdot f^4 = [f'(x)]^2 [f(x)]^4,$$

which is the case $n = 4$, $k = 1$ of the general result:

$$\begin{aligned}\int \sec^2 x\,(\sec x + \tan x)^6\,\mathrm{d}x &= \frac{f'f^5}{5} - \frac{f^7}{35} + C = \frac{\sec x\, f^6}{5} - \frac{f^7}{35} + C \\ &= \frac{f^6}{35}\big(7\sec x - (\sec x + \tan x)\big) + C \\ &= \frac{(\sec x + \tan x)^6\,(6\sec x - \tan x)}{35} + C.\end{aligned}$$

Checks

- Differentiate the answer to (i): $\sec^2 x \tan^4 x + \sec^2 x \tan^6 x = \sec^2 x \tan^4 x(1 + \tan^2 x) = \tan^4 x \sec^4 x$.
- Numerically, Simpson's rule gives $\int_{-0.7}^{1.1} \sin^4 x / \cos^8 x\,\mathrm{d}x \approx 22.12954$, matching $\left[\frac{1}{5}\tan^5 x + \frac{1}{7}\tan^7 x\right]_{-0.7}^{1.1}$; and $\int_{-1}^{0.9} \sec^2 x(\sec x + \tan x)^6\,\mathrm{d}x \approx 133.68566$, matching the answer to (ii).
- The general formula agrees with quadrature for $f = \tan$ and every n from 1 to 6.

Common slips

- Choosing $u = f^n$ in the integration by parts: this produces $f^{n-1}[f']^3$, which is worse. Pair one f' with f^n so that it integrates exactly.
- Taking $k = 1$ for $\tan x$: $f'' = 2\sec^2 x \tan x$, so $k = 2$.
- In (ii), forgetting that $\sec^2 x$ must be absorbed as $[f']^2/f^2$, which is why the power drops from 6 to $n = 4$.

Beyond STEP

The condition $f'' = kff'$ is itself a differential equation: it integrates once to $f' = \frac{1}{2}kf^2 + c$, a first-order separable equation whose solutions include $\tan$ (with $k = 2$, $c = 1$) and $\sec x + \tan x$ (with $k = 1$, $c = \frac{1}{2}$). Recognising that an integrand becomes tractable because the function satisfies an ODE is a theme that recurs in first-year work on special functions and reduction formulae. Boyce and DiPrima, *Elementary Differential Equations*, covers separable and autonomous equations like this one.

WORKED SOLUTION

STEP II 2011, Question 7

Pure • Sequences and Series

STEP II 2011, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q7

The key idea

Never work with the sequences term by term: split every sum into a sum of powers of λ and a sum of powers of μ, and use the geometric series formula on each. The two numbers satisfy $\lambda + \mu = 2$ and $\lambda\mu = -1$, so they are the roots of $x^2 = 2x + 1$. That single quadratic makes all the awkward denominators collapse. The other tool is squaring: b_k^2 and a_k^2 both turn into a_{2k} plus a constant.

Solution

First, some facts we will use throughout:

$$\lambda - 1 = \sqrt{2}, \qquad \mu - 1 = -\sqrt{2}, \qquad \lambda\mu = 1 - 2 = -1, \qquad \lambda - \mu = 2\sqrt{2}.$$

Also $a_0 = 2$ and $b_0 = 0$.

Part (i)

Since $\lambda \neq 1$ and $\mu \neq 1$, we can use the geometric series formula on each sum:

$$\sum_{r=0}^{n} \lambda^r = \frac{\lambda^{n+1} - 1}{\lambda - 1} = \frac{\lambda^{n+1} - 1}{\sqrt{2}}, \qquad \sum_{r=0}^{n} \mu^r = \frac{\mu^{n+1} - 1}{\mu - 1} = -\frac{\mu^{n+1} - 1}{\sqrt{2}}.$$

Subtracting the second sum from the first,

$$\sum_{r=0}^{n} b_r = \frac{\lambda^{n+1} - 1}{\sqrt{2}} + \frac{\mu^{n+1} - 1}{\sqrt{2}} = \frac{a_{n+1} - 2}{\sqrt{2}} = -\sqrt{2} + \frac{1}{\sqrt{2}}\, a_{n+1},$$

as required. Adding the two sums instead, the -1's cancel:

$$\boxed{\sum_{r=0}^{n} a_r = \frac{\lambda^{n+1} - 1}{\sqrt{2}} - \frac{\mu^{n+1} - 1}{\sqrt{2}} = \frac{1}{\sqrt{2}}\, b_{n+1}.}$$

(Quick test: with $n = 0$ this says $a_0 = b_1/\sqrt{2} = 2\sqrt{2}/\sqrt{2} = 2$, which is correct.)

Part (ii)

By part (i), the inner sum is $b_{m+1}/\sqrt{2}$. So

$$\sum_{m=0}^{2n}\left(\sum_{r=0}^{m} a_r\right) = \frac{1}{\sqrt{2}}\sum_{m=0}^{2n} b_{m+1} = \frac{1}{\sqrt{2}}\sum_{s=1}^{2n+1} b_s = \frac{1}{\sqrt{2}}\sum_{s=0}^{2n+1} b_s,$$

where the last step is allowed because $b_0 = 0$. Now part (i), with $2n+1$ in place of n, gives

$$\sum_{m=0}^{2n}\left(\sum_{r=0}^{m} a_r\right) = \frac{1}{\sqrt{2}}\left(-\sqrt{2} + \frac{a_{2n+2}}{\sqrt{2}}\right) = \tfrac{1}{2}a_{2n+2} - 1. \qquad (*)$$

Ask yourself How do we connect a_{2n+2} to b_{n+1}^2?

We square, and use $\lambda\mu = -1$ for the cross term:

$$b_{n+1}^2 = \lambda^{2n+2} + \mu^{2n+2} - 2(\lambda\mu)^{n+1} = a_{2n+2} - 2(-1)^{n+1}, \qquad a_{n+1}^2 = a_{2n+2} + 2(-1)^{n+1}.$$

If n is odd, then $n+1$ is even, so $b_{n+1}^2 = a_{2n+2} - 2$. Then $(*)$ equals $\frac{1}{2}(a_{2n+2} - 2) = \frac{1}{2}b_{n+1}^2$, as required.

If n is even, then $(-1)^{n+1} = -1$, so it is now a_{n+1}^2 that equals $a_{2n+2} - 2$. Hence

$$\boxed{n \text{ even}: \quad \sum_{m=0}^{2n}\left(\sum_{r=0}^{m} a_r\right) = \tfrac{1}{2}a_{n+1}^2}$$

or, equivalently, $\frac{1}{2}b_{n+1}^2 - 2$.

Part (iii)

By part (i), and then the squaring identity from part (ii),

$$\left(\sum_{r=0}^{n} a_r\right)^2 = \tfrac{1}{2}b_{n+1}^2 = \tfrac{1}{2}a_{2n+2} - (-1)^{n+1}.$$

For the second sum we again split into two geometric series, this time with ratios λ^2 and μ^2:

$$\sum_{r=0}^{n}\lambda^{2r+1} = \lambda\,\frac{\lambda^{2n+2} - 1}{\lambda^2 - 1}.$$

Ask yourself Can $\lambda^2 - 1$ be simplified?

Yes. Since $\lambda^2 = 2\lambda + 1$, we have $\lambda^2 - 1 = 2\lambda$, so the factor of λ cancels and the sum is $\frac{1}{2}(\lambda^{2n+2} - 1)$. Because μ satisfies the same quadratic, $\mu^2 - 1 = 2\mu$ and the μ-sum is $\frac{1}{2}(\mu^{2n+2} - 1)$ (note $\mu^2 \neq 1$, so the formula is valid). Adding the two,

$$\sum_{r=0}^{n} a_{2r+1} = \tfrac{1}{2}\left(a_{2n+2} - 2\right) = \tfrac{1}{2}a_{2n+2} - 1.$$

Therefore

$$\left(\sum_{r=0}^{n} a_r\right)^2 - \sum_{r=0}^{n} a_{2r+1} = \left(\tfrac{1}{2}a_{2n+2} - (-1)^{n+1}\right) - \left(\tfrac{1}{2}a_{2n+2} - 1\right) = 1 + (-1)^n.$$

If n is even this equals 2, as required. If n is odd it is 0, so

$$\boxed{n \text{ odd}: \quad \left(\sum_{r=0}^{n} a_r\right)^2 = \sum_{r=0}^{n} a_{2r+1}.}$$

Checks

The a_n are integers satisfying $a_{n+1} = 2a_n + a_{n-1}$: $2, 2, 6, 14, 34, 82, 198, 478, \ldots$. For part (ii) with $n = 3$, the partial sums $2, 4, 10, 24, 58, 140, 338$ add up to 576. Since $b_4 = 24\sqrt{2}$, we get $\frac{1}{2}b_4^2 = 576$, which matches. For part (iii) with $n = 2$: $(2 + 2 + 6)^2 - (2 + 14 + 82) = 100 - 98 = 2$. With $n = 1$: $(2 + 2)^2 - (2 + 14) = 0$. A computer checked all three parts, including both parities, for every n up to 24.

Common slips

- Forgetting that $\mu - 1 = -\sqrt{2}$ is negative, which swaps the roles of a and b in part (i).
- Writing $(\lambda\mu)^{n+1} = 1$ instead of $(-1)^{n+1}$; the sign is the whole point of the odd/even split.
- In part (ii), shifting the index of the sum of b_{m+1} without noticing that $b_0 = 0$ is what lets you start the sum at 0.

Beyond STEP

Both sequences satisfy the linear recurrence $x_{n+1} = 2x_n + x_{n-1}$, whose characteristic equation is $x^2 = 2x + 1$. So λ and μ are its roots, and a_n and b_n are particular solutions. Since $\lambda^n = \frac{1}{2}(a_n + b_n)$, the identity $\lambda^n\mu^n = (-1)^n$ becomes $a_n^2 - b_n^2 = 4(-1)^n$, a Pell-type equation in the ring $\mathbb{Z}[\sqrt{2}]$. Davenport's *The Higher Arithmetic* explains how such units and continued fractions for $\sqrt{2}$ fit together.

WORKED SOLUTION

STEP II 2011, Question 8

Pure • Involute of a Circle

STEP II 2011, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q8

The key idea

At every moment the string is made of two pieces: an arc of the circle, whose length we can read off from the angle, and a straight piece that leaves the circle along a tangent. Since the total length is fixed, the length of the straight piece is whatever is left over, and a tangent is perpendicular to the radius, so B is easy to locate with vectors. For the final area, notice that the whole region swept out is "area under the outer branch, minus area under the inner branch, minus the semicircle".

Solution

The parametric equations

Let $P = (\cos t, \sin t)$ be the point where the string leaves the circle, so t is the angle from the positive x-axis to OP. At the start $P = A = (-1, 0)$, so $t = \pi$; as the string wraps round over the top of the circle, P moves clockwise and t decreases to 0.

The part of the string in contact with the circle is the arc from angle π to angle t. The radius is 1, so this arc has length $\pi - t$. The string has length π, so the straight piece PB has length

$$\pi - (\pi - t) = t.$$

Because the string is taut, PB is tangent to the circle at P, so it is perpendicular to $OP = (\cos t, \sin t)$. The two unit vectors perpendicular to OP are $\pm(\sin t, -\cos t)$. At the start ($t = \pi$) the string points straight up from A, in the direction $(0, 1) = (\sin \pi, -\cos \pi)$, and by continuity the sign stays the same throughout. Hence

$$\overrightarrow{OB} = \overrightarrow{OP} + t\,(\sin t, -\cos t),$$

that is,

$$x = \cos t + t \sin t, \qquad y = \sin t - t \cos t,$$

as required. (Check: $t = \pi$ gives $(-1, \pi)$, the starting position of B, and $t = 0$ gives $(1, 0)$, where B meets the circle.)

The maximum of x

Differentiating, the terms $\pm\sin t$ and $\pm\cos t$ cancel:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t + \sin t + t\cos t = t\cos t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \cos t - \cos t + t\sin t = t\sin t.$$

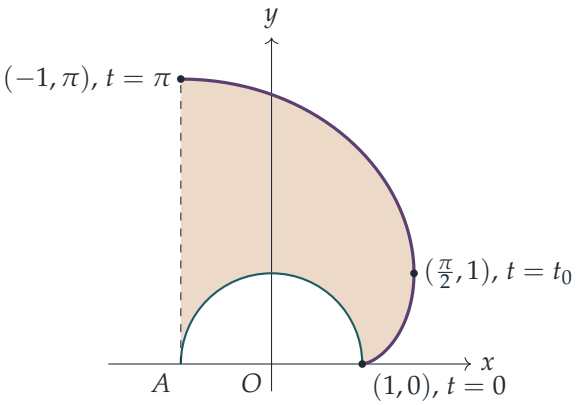
For $0 < t < \pi$, $\frac{\mathrm{d}x}{\mathrm{d}t} = t\cos t$ is positive when $t < \frac{\pi}{2}$ and negative when $t > \frac{\pi}{2}$. So as t runs over $[0, \pi]$, x increases and then decreases, and the maximum is at

$$t_0 = \frac{\pi}{2}, \qquad \text{where } (x, y) = \left(\frac{\pi}{2}, 1\right).$$

The sketch

Two facts help. First, $y > 0$ for $0 < t \leq \pi$: if $t \leq \frac{\pi}{2}$ then $y = \cos t\,(\tan t - t) > 0$ since $\tan t > t$ (for $t = \frac{\pi}{2}$, $y = 1$ directly), and if $t > \frac{\pi}{2}$ then $\cos t < 0$, so both terms of y are non-negative and $\sin t - t\cos t > 0$. Second, $x^2 + y^2 = 1 + t^2$, so B is always outside the circle, by an amount that grows with t. Also $\frac{\mathrm{d}y}{\mathrm{d}t} = t\sin t \geq 0$, so y increases steadily with t.

So the curve starts at $(1, 0)$ (where $\frac{\mathrm{d}y}{\mathrm{d}x} = \tan t = 0$, so it leaves the x-axis horizontally), climbs to its rightmost point $(\frac{\pi}{2}, 1)$, where it is vertical, then sweeps back over the top of the circle to $(-1, \pi)$, where it is horizontal again. This curve is an *involute* of the circle.



The shaded region, between the curve, the initial position of the string (dashed) and the semicircle, is the area swept out by the string.

Area for $\pi \geq t \geq t_0$

On this part of the curve, x runs from $\frac{\pi}{2}$ (at $t = \frac{\pi}{2}$) down to -1 (at $t = \pi$), and $y > 0$. So the area between this arc and the x-axis is

$$\int_{-1}^{\pi/2} y\,\mathrm{d}x = \int_{\pi}^{\pi/2} y\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \int_{\pi/2}^{\pi} \left(t^2\cos^2 t - t\sin t\cos t\right)\mathrm{d}t,$$

where we swapped the limits to absorb the minus sign. We need two integrals, both by parts after a double-angle formula. Write $\sin t\cos t = \frac{1}{2}\sin 2t$ and $\cos^2 t = \frac{1}{2}(1+\cos 2t)$. By parts,

$$\int t\sin 2t\,\mathrm{d}t = -\tfrac{1}{2}t\cos 2t + \tfrac{1}{4}\sin 2t,$$
$$\int t^2\cos 2t\,\mathrm{d}t = \tfrac{1}{2}t^2\sin 2t - \int t\sin 2t\,\mathrm{d}t = \tfrac{1}{2}t^2\sin 2t + \tfrac{1}{2}t\cos 2t - \tfrac{1}{4}\sin 2t.$$

At $t=\pi$ we have $\sin 2t = 0$, $\cos 2t = 1$; at $t = \frac{\pi}{2}$, $\sin 2t = 0$, $\cos 2t = -1$. Hence

$$\int_{\pi/2}^{\pi} t\sin t\cos t\,\mathrm{d}t = \tfrac{1}{2}\Big[-\tfrac{1}{2}t\cos 2t\Big]_{\pi/2}^{\pi} = \tfrac{1}{2}\Big(-\frac{\pi}{2} - \frac{\pi}{4}\Big) = -\frac{3\pi}{8},$$
$$\int_{\pi/2}^{\pi} t^2\cos^2 t\,\mathrm{d}t = \tfrac{1}{2}\Big[\tfrac{1}{3}t^3\Big]_{\pi/2}^{\pi} + \tfrac{1}{2}\Big[\tfrac{1}{2}t\cos 2t\Big]_{\pi/2}^{\pi} = \frac{7\pi^3}{48} + \tfrac{1}{2}\Big(\frac{\pi}{2} + \frac{\pi}{4}\Big) = \frac{7\pi^3}{48} + \frac{3\pi}{8}.$$

So the required area is

$$\frac{7\pi^3}{48} + \frac{3\pi}{8} - \Big(-\frac{3\pi}{8}\Big) = \frac{7\pi^3}{48} + \frac{3\pi}{4}.$$

The area swept out by the string

Ask yourself The region under the arc for $t \geq t_0$ includes more than the swept region. What must be removed?

The area just found is everything between the outer arc ($t_0 \leq t \leq \pi$) and the x-axis for $-1 \leq x \leq \frac{\pi}{2}$. The swept region is this minus two pieces lying underneath it: the upper semicircle, of area $\frac{\pi}{2}$, and the sliver between the inner arc ($0 \leq t \leq t_0$) and the x-axis for $1 \leq x \leq \frac{\pi}{2}$. The inner arc lies outside the circle (since $x^2 + y^2 = 1 + t^2 > 1$ for $t > 0$) and below the outer arc, so these pieces do not overlap and the subtraction is legitimate.

On the inner arc x increases with t, so the sliver has area

$$\int_0^{\pi/2} y\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \int_0^{\pi/2}\big(t\sin t\cos t - t^2\cos^2 t\big)\,\mathrm{d}t.$$

Using the same antiderivatives, now with $t=0$ as the lower limit (where $\sin 2t = 0$, $\cos 2t = 1$),

$$\int_0^{\pi/2} t\sin t\cos t\,\mathrm{d}t = \tfrac{1}{2}\Big[-\tfrac{1}{2}t\cos 2t\Big]_0^{\pi/2} = \frac{\pi}{8},$$
$$\int_0^{\pi/2} t^2\cos^2 t\,\mathrm{d}t = \tfrac{1}{2}\cdot\frac{\pi^3}{24} + \tfrac{1}{2}\Big[\tfrac{1}{2}t\cos 2t\Big]_0^{\pi/2} = \frac{\pi^3}{48} - \frac{\pi}{8},$$

so the sliver has area $\dfrac{\pi}{4} - \dfrac{\pi^3}{48}$ (about 0.14, which is reassuringly small and positive). Therefore the area swept out by the string is

$$\Big(\frac{7\pi^3}{48} + \frac{3\pi}{4}\Big) - \Big(\frac{\pi}{4} - \frac{\pi^3}{48}\Big) - \frac{\pi}{2} = \frac{8\pi^3}{48} = \frac{\pi^3}{6}.$$

Checks

- There is a slicker route to the last answer. When t decreases by δt, the straight piece of string (length t) turns through angle δt about a point that is essentially fixed, sweeping a thin sector of area about $\frac{1}{2}t^2\,\delta t$. Adding these up gives $\int_0^{\pi} \frac{1}{2}t^2\,\mathrm{d}t = \frac{\pi^3}{6}$, in agreement.
- Numerically: Simpson's rule gives 6.87794 for the area under the outer arc, matching $\frac{7\pi^3}{48} + \frac{3\pi}{4}$, and the shoelace formula applied to the boundary of the shaded region gives $5.16771 \approx \frac{\pi^3}{6}$.

Common slips

- Taking the straight piece to have length $\pi - t$ (that is the arc), or choosing the wrong one of the two tangent directions; checking $t = \pi$ catches both.
- Leaving the limits as t_0 to π without noticing that x decreases along that arc, which gives a negative "area".
- Forgetting the small region under the inner arc, which gives $\frac{7\pi^3}{48} + \frac{\pi}{4}$ for the swept area.

Beyond STEP

The curve traced by the end of a string unwinding from a curve is called an involute, and the curve it unwinds from is its evolute: the centre of curvature of the involute at B is exactly the point P where the string leaves the circle. Involutes of circles give the profile of most gear teeth, because two such teeth roll against each other with a constant speed ratio. The "thin sector" argument in the Checks is the polar-area formula $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ in disguise. For more on curvature and curves of this kind, see Siklos, *Advanced Problems in Mathematics*, and for the careful calculus behind it, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2011, Question 9

Mechanics • Collisions and Restitution

STEP II 2011, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

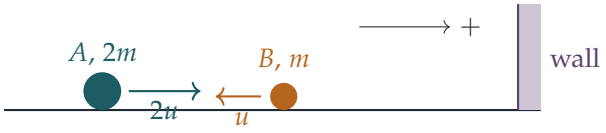
step.maths.org/questions/11-s2-q9

The key idea

Every collision here is direct, so each one is settled by the same two equations: conservation of momentum and Newton's law of restitution. The only real danger is signs. Fix one positive direction (towards the wall) at the start, write every velocity as a signed number in that direction, and the algebra takes care of itself. The final claim then comes down to showing that one expression stays positive, and since it is linear in f we only need to look at the ends of the range of f.

Solution

The model. We treat A and B as particles on a smooth plane, so no horizontal forces act except at impacts, and the wall is smooth, so it only reverses B's velocity (scaled by f). After the first collision B goes on to hit the wall, so the wall lies on B's side, beyond B in the direction A was originally moving. We take that direction as positive. The order along the line is then always A, B, wall.



The first collision

Before impact A has velocity $2u$ and B has velocity $-u$. Let their velocities afterwards be v_A and v_B. Momentum is conserved:

$$2m(2u) + m(-u) = 2m\,v_A + m\,v_B \quad \Longrightarrow \quad 2v_A + v_B = 3u.$$

The speed of approach is $2u - (-u) = 3u$. Since B is in front of A, the speed of separation is $v_B - v_A$, and Newton's law of restitution gives

$$v_B - v_A = 3eu.$$

Subtracting the second equation from the first gives $3v_A = 3u - 3eu$, so

$$v_A = (1-e)u, \qquad v_B = (1+2e)u.$$

Both are ≥ 0, so after the collision A moves towards the wall with speed $(1-e)u$ (and is at rest if $e = 1$), and B moves towards the wall with speed $(1+2e)u$. This is consistent with B then reaching the wall.

The wall

The wall is smooth and the impact is direct, so B rebounds along the line with speed $f(1+2e)u$. Its velocity becomes $-f(1+2e)u$. Meanwhile A is still moving with velocity $(1-e)u \geq 0$, because nothing acts on it. B is now moving towards A, so the two will certainly meet again.

The second collision

Just before the second impact the velocities are

$$A: \ (1-e)u, \qquad B: \ -f(1+2e)u.$$

Let the velocities afterwards be w_A and w_B. Momentum gives

$$2w_A + w_B = 2(1-e)u - f(1+2e)u.$$

Ask yourself Which way round does the restitution equation go this time?

B is still the particle nearer the wall, so the separation velocity is still $w_B - w_A$. The speed of approach is A's velocity minus B's, namely $(1-e)u + f(1+2e)u$. Hence

$$w_B - w_A = e\big[(1-e)u + f(1+2e)u\big].$$

To isolate w_B, add twice this equation to the momentum equation:

$$\begin{aligned} 3w_B &= 2(1-e)u - f(1+2e)u + 2e(1-e)u + 2ef(1+2e)u \\ &= 2(1-e)(1+e)u + f(1+2e)(2e-1)u \\ &= 2(1-e^2)u - (1-4e^2)fu. \end{aligned}$$

Therefore

$$w_B = \tfrac{2}{3}(1-e^2)u - \tfrac{1}{3}(1-4e^2)fu,$$

measured in the positive direction, that is, towards the wall. This is the required result.

B always moves towards the wall

We must show that $w_B > 0$ for all $0 < e \leq 1$ and $0 < f \leq 1$. Since $u > 0$, this is the same as showing

$$g(f) = 2(1-e^2) + (4e^2 - 1)f > 0.$$

The term $1-4e^2$ can have either sign, so we can't simply drop it. Instead, for fixed e, note that g is linear in f, and

$$g(0) = 2(1-e^2) \geq 0, \qquad g(1) = 2 - 2e^2 + 4e^2 - 1 = 1 + 2e^2 > 0.$$

A linear function on $[0,1]$ is the weighted average of its end values: $g(f) = (1-f)\,g(0) + f\,g(1)$. For $0 < f \leq 1$ the first term is ≥ 0 and the second is strictly positive (as $f > 0$ and $g(1) > 0$), so $g(f) > 0$. Hence $w_B > 0$, and B moves towards the wall after the second collision, whatever the values of e and f.

Checks

- With $e = f = 1$ (perfectly elastic throughout), A stops dead at the first impact, B goes out at $3u$, comes back at $3u$, and hits a stationary particle of twice its mass. The formula gives $w_B = 0 + u = u$, which matches the elastic result $\frac{m-2m}{3m}(-3u) = u$.
- With $e = f = \frac{1}{2}$, the formula gives $w_B = \frac{2}{3} \cdot \frac{3}{4}u = \frac{1}{2}u$. A step-by-step simulation of positions and impacts gives the same value, $0.5u$.
- The worst case for the sign is $e = 1$ with f small. Then $w_B = fu$, which is small but still positive.

Common slips

- Using speeds instead of signed velocities in the second collision, and so getting B's rebound velocity with the wrong sign.
- Writing the restitution equation as $w_A - w_B$ because B "came from the wall". The order of the particles along the line hasn't changed, so separation is still $w_B - w_A$.
- Claiming $w_B > 0$ by assuming $1 - 4e^2 > 0$. It fails for $e > \frac{1}{2}$, which is why the argument has to use $f \leq 1$.

Beyond STEP

Restitution is an empirical model. In a first-year course you meet it alongside the cleaner idea of the centre-of-mass frame: in that frame the total momentum is zero, and a collision with coefficient e simply multiplies both velocities by $-e$. This makes results like the first collision almost immediate. Repeated collisions with a wall turn up again in billiard problems and in counting arguments, such as the famous one where the number of collisions between two blocks and a wall gives the digits of π. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats momentum and collisions in this spirit.

WORKED SOLUTION

STEP II 2011, Question 10

Mechanics • Projectiles and Impulse

STEP II 2011, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q10

The key idea

Because R is always vertically below P, the two particles share the same horizontal velocity throughout, so the string is vertical and only the vertical motion matters. The string goes taut when P is exactly $\frac{1}{2}H$ above the plane; the jerk is then a vertical impulse that shares P's upward momentum with R. After that the two particles have identical velocities, so they fly together as one projectile until R lands.

Solution

Maximum height

Model the particle as a point under gravity alone. The vertical component of velocity is $u\sin\theta - gt$, which is zero at the top. Using $v^2 = u^2 + 2as$ vertically, with final vertical speed 0,

$$0 = u^2\sin^2\theta - 2gH \qquad \Longrightarrow \qquad H = \frac{u^2\sin^2\theta}{2g}.$$

It is worth noting for later that this gives $u\sin\theta = \sqrt{2gH}$.

When the string becomes taut

Take O as origin, x horizontal and y vertically up. The particle P has horizontal velocity $u\cos\theta$ for all time. For R to stay vertically below P, it must be projected with speed $u\cos\theta$; the plane is smooth, so R then slides at this constant speed. Hence the string is always vertical, and the distance PR is simply the height of P:

$$y_P = u\sin\theta\, t - \tfrac{1}{2}gt^2 = \sqrt{2gH}\, t - \tfrac{1}{2}gt^2.$$

The string becomes taut when y_P first reaches $\frac{1}{2}H$ (which it does, since P rises to $H > \frac{1}{2}H$):

$$\tfrac{1}{2}gt^2 - \sqrt{2gH}\,t + \tfrac{1}{2}H = 0 \quad \Longrightarrow \quad t = \frac{\sqrt{2gH} \pm \sqrt{2gH - gH}}{g} = (\sqrt{2} \pm 1)\sqrt{\frac{H}{g}}.$$

The smaller root is the first time, so the string becomes taut after

$$t_1 = (\sqrt{2} - 1)\sqrt{H/g},$$

as required. (The larger root is when P would come back down through height $\frac{1}{2}H$.)

The jerk

At t_1 the vertical velocity of P is

$$u\sin\theta - gt_1 = \sqrt{2gH} - (\sqrt{2} - 1)\sqrt{gH} = \sqrt{gH},$$

upwards, while R has no vertical velocity. Both have horizontal velocity $u\cos\theta$.

Ask yourself Which impulses act at the instant the string jerks, and in which direction?

The string is vertical, so the tension impulse J is vertical: downwards on P, upwards on R. The plane is smooth and can only push up on R; since R is pulled upwards off the plane, the plane exerts no impulse. So horizontal velocities are unchanged, and vertically the momentum of the pair is conserved. The string is inextensible and stays taut, so just afterwards P and R have the same vertical velocity w. With equal masses m,

$$m\sqrt{gH} + m \cdot 0 = 2mw \qquad \Longrightarrow \qquad w = \tfrac{1}{2}\sqrt{gH}.$$

The distance D

After the jerk, P and R have the same velocity and the same acceleration g downwards if the tension is zero. So they move as two identical projectiles a fixed distance $\frac{1}{2}H$ apart: the string stays exactly taut with zero tension, which is consistent. R starts at height 0 with upward velocity w, so it returns to the plane after

$$t_2 = \frac{2w}{g} = \sqrt{\frac{H}{g}}.$$

The whole time of R's motion is therefore

$$t_1 + t_2 = \sqrt{2}\sqrt{\frac{H}{g}} = \sqrt{\frac{2H}{g}} = \sqrt{\frac{u^2\sin^2\theta}{g^2}} = \frac{u\sin\theta}{g},$$

and throughout it R moves horizontally at $u\cos\theta$. Hence

$$D = u\cos\theta \cdot \frac{u\sin\theta}{g} = \frac{u^2\sin\theta\cos\theta}{g} = \frac{u^2\sin 2\theta}{2g}.$$

This is exactly half the range of P on its own: R lands at the moment P, without a string, would have been at the top of its path.

The case $D = H$

Setting

$$\frac{u^2 \sin\theta \cos\theta}{g} = \frac{u^2 \sin^2\theta}{2g}$$

and dividing by $u^2 \sin\theta / g$ (non-zero, as $0 < \theta < \frac{1}{2}\pi$ for the particle to rise) gives $\cos\theta = \frac{1}{2}\sin\theta$, so

$$\tan\theta = 2.$$

Checks

- The two roots for t multiply to H/g and sum to $2\sqrt{2H/g}$, the time for P to reach the top and come back to the plane, as they should by symmetry.
- A numerical simulation (stepping P's flight until it is $\frac{1}{2}H$ up, applying the jerk, then stepping again until R lands) for random u and θ matched t_1 and D; for example $u = 18.11$, $\theta = 0.711$, $g = 9.8$ gives $D \approx 16.543$ both ways.
- With $\tan\theta = 2$, $u = 10$, $g = 9.8$: $D = H \approx 4.082$.

Common slips

- Taking the larger root $(\sqrt{2}+1)\sqrt{H/g}$, which is when P comes back down through height $\frac{1}{2}H$.
- Forgetting the time t_1 that R spends sliding before the jerk, and computing D from the flight after the jerk alone.
- Using energy conservation at the jerk: kinetic energy is lost when an inextensible string jerks taut; only momentum is conserved.

Beyond STEP

After the jerk the pair behaves exactly like a single particle at its centre of mass: internal forces (here the tension) never change the motion of the centre of mass of a system. This is the starting point for the mechanics of systems of particles and rigid bodies in a first-year course, where impulsive constraints like this one are handled the same way. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

WORKED SOLUTION

STEP II 2011, Question 11

Mechanics • Equilibrium in Three Dimensions

STEP II 2011, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q11

The key idea

Everything rests on writing a unit vector along each string. Each one splits into a vertical part, fixed by the string's angle with the vertical, and a horizontal part, which points from O towards the string's end in the ceiling and is fixed by the angle that direction makes with OA. Once the three unit vectors are known, equilibrium is just three scalar equations, one for each of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.

Solution

We model P as a particle and the strings as light and inextensible, so each string pulls P towards its fixed end with a force along the string. Take O as the origin, with $\mathbf{j}$ along OA and $\mathbf{k}$ vertically upwards; $\mathbf{i}$ is horizontal and perpendicular to OA.

First the trigonometry. Since $\tan\theta = \sqrt{2}$, a right-angled triangle with sides $1, \sqrt{2}, \sqrt{3}$ gives

$$\sin\theta = \frac{\sqrt{2}}{\sqrt{3}}, \qquad \cos\theta = \frac{1}{\sqrt{3}}.$$

Since $\tan\phi = \frac{1}{4}\sqrt{2}$, the sides are $4, \sqrt{2}$ and hypotenuse $\sqrt{18} = 3\sqrt{2}$, so

$$\sin\phi = \frac{1}{3}, \qquad \cos\phi = \frac{2\sqrt{2}}{3}.$$

Part (i)

A string making angle α with the vertical has unit vector (from P towards the ceiling) with vertical component $\cos\alpha$ and horizontal part of length $\sin\alpha$.

For PB, $\alpha = 90^\circ - \theta$, so the vertical component is $\cos(90^\circ - \theta) = \sin\theta = \frac{\sqrt{2}}{\sqrt{3}}$ and the horizontal part has length $\sin(90^\circ - \theta) = \cos\theta = \frac{1}{\sqrt{3}}$.

The horizontal part points along OB (since O is directly above P), which makes angle $90^\circ + \theta$ with $\mathbf{j}$. A horizontal unit vector at angle $90^\circ + \theta$ to $\mathbf{j}$ is $\pm\sin(90^\circ + \theta)\,\mathbf{i} + \cos(90^\circ + \theta)\,\mathbf{j} = \pm\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}$. We choose the sense of $\mathbf{i}$ so that B lies on the

negative-**i** side of OA; the required form fixes this choice. The horizontal part of the unit vector along PB is then

$$\tfrac{1}{\sqrt{3}}(-\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}) = \tfrac{1}{\sqrt{3}}\left(-\tfrac{1}{\sqrt{3}}\,\mathbf{i} - \tfrac{\sqrt{2}}{\sqrt{3}}\,\mathbf{j}\right) = -\tfrac{1}{3}\,\mathbf{i} - \tfrac{\sqrt{2}}{3}\,\mathbf{j}.$$

Adding the vertical component, the unit vector in the direction PB is

$$-\tfrac{1}{3}\,\mathbf{i} - \tfrac{\sqrt{2}}{3}\,\mathbf{j} + \tfrac{\sqrt{2}}{\sqrt{3}}\,\mathbf{k},$$

as required. (Check: $\frac{1}{9} + \frac{2}{9} + \frac{2}{3} = 1$.)

Part (ii)

The same recipe gives the other two strings.

String PA: angle 30° with the vertical, horizontal part along **j**:

$$\mathbf{e}_A = \tfrac{1}{2}\,\mathbf{j} + \tfrac{\sqrt{3}}{2}\,\mathbf{k}.$$

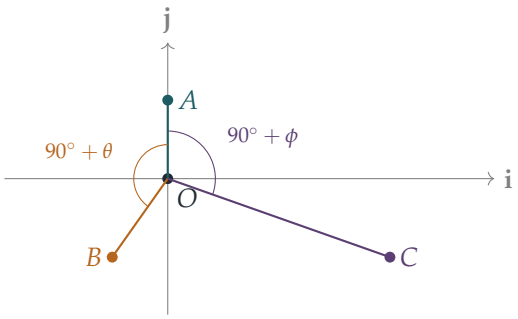
String PC: angle 60° with the vertical, so vertical component $\frac{1}{2}$ and horizontal length $\frac{\sqrt{3}}{2}$. The direction OC makes angle $90° + \phi$ with **j**, so its horizontal unit vector is $\pm\cos\phi\,\mathbf{i} - \sin\phi\,\mathbf{j}$.

Ask yourself Is C on the same side of OA as B, or the opposite side?

The diagram shows the opposite side, and equilibrium demands it anyway. The tension in PA has no **i**-component and that in PB has a negative one. If PC also had a negative **i**-component, nothing could balance them. So the sign is $+$, and

$$\mathbf{e}_C = \tfrac{\sqrt{3}}{2}\left(\tfrac{2\sqrt{2}}{3}\,\mathbf{i} - \tfrac{1}{3}\,\mathbf{j}\right) + \tfrac{1}{2}\,\mathbf{k} = \tfrac{\sqrt{6}}{3}\,\mathbf{i} - \tfrac{\sqrt{3}}{6}\,\mathbf{j} + \tfrac{1}{2}\,\mathbf{k}.$$

The plan view of the ceiling below (taking $OP = 1$, so $OA = \tan 30°$, $OB = \cot\theta$, $OC = \tan 60°$) shows the layout.



The forces on P are therefore

$$\begin{aligned}\text{tension in } PA: &\quad T\left(\tfrac{1}{2}\,\mathbf{j}+\tfrac{\sqrt{3}}{2}\,\mathbf{k}\right),\\ \text{tension in } PB: &\quad U\left(-\tfrac{1}{3}\,\mathbf{i}-\tfrac{\sqrt{2}}{3}\,\mathbf{j}+\tfrac{\sqrt{2}}{\sqrt{3}}\,\mathbf{k}\right),\\ \text{tension in } PC: &\quad V\left(\tfrac{\sqrt{6}}{3}\,\mathbf{i}-\tfrac{\sqrt{3}}{6}\,\mathbf{j}+\tfrac{1}{2}\,\mathbf{k}\right),\\ \text{weight}: &\quad -W\,\mathbf{k}.\end{aligned}$$

Part (iii)

P is in equilibrium, so the four forces sum to zero. We take components.

i:

$$-\tfrac{1}{3}U+\tfrac{\sqrt{6}}{3}V=0 \quad\Longrightarrow\quad U=\sqrt{6}\,V,$$

as required.

j:

$$\tfrac{1}{2}T-\tfrac{\sqrt{2}}{3}U-\tfrac{\sqrt{3}}{6}V=0.$$

Substituting $U=\sqrt{6}V$ and using $\sqrt{2}\sqrt{6}=2\sqrt{3}$:

$$\tfrac{1}{2}T=\tfrac{2\sqrt{3}}{3}V+\tfrac{\sqrt{3}}{6}V=\tfrac{5\sqrt{3}}{6}V \quad\Longrightarrow\quad T=\tfrac{5\sqrt{3}}{3}V.$$

k:

$$\tfrac{\sqrt{3}}{2}T+\tfrac{\sqrt{2}}{\sqrt{3}}U+\tfrac{1}{2}V=W.$$

Substituting both results,

$$\tfrac{\sqrt{3}}{2}\cdot\tfrac{5\sqrt{3}}{3}V+\tfrac{\sqrt{2}\sqrt{6}}{\sqrt{3}}V+\tfrac{1}{2}V=\tfrac{5}{2}V+2V+\tfrac{1}{2}V=5V=W.$$

Hence

$$V=\frac{W}{5},\qquad U=\frac{\sqrt{6}\,W}{5},\qquad T=\frac{5\sqrt{3}}{3}\cdot\frac{W}{5}=\frac{W}{\sqrt{3}}=\frac{\sqrt{3}\,W}{3}.$$

All three are positive, so every string is taut, which is consistent with the model.

Checks

- Each unit vector has length 1: for $\mathbf{e}_C$, $\frac{6}{9}+\frac{3}{36}+\frac{1}{4}=1$.
- The angle between the horizontal parts of $\mathbf{e}_A$ and $\mathbf{e}_C$ has cosine $-\frac{1}{3}=-\sin\phi$, so it is $90^\circ+\phi\approx 109.47^\circ$, as it should be.
- With $W=1$: $T\approx 0.5774$, $U\approx 0.4899$, $V=0.2$. Then the vertical components $0.5774\times 0.8660+0.4899\times 0.8165+0.2\times 0.5=0.5+0.4+0.1=1$ balance the weight. Note the neat split: PA, PB, PC carry $\frac{1}{2}$, $\frac{2}{5}$, $\frac{1}{10}$ of the weight.

Common slips

- Mixing up sin and cos for the string PB: its angle with the vertical is $90^\circ - \theta$, so its vertical component is $\sin\theta$, not $\cos\theta$.
- Putting C on the same side of OA as B. The **i**-equation then forces a negative tension (that is, $V = -\frac{1}{3}W$), which a string cannot have.
- Forgetting that $\cos(90^\circ + \phi) = -\sin\phi$, which gives the wrong sign for the **j**-components of $\mathbf{e}_B$ and $\mathbf{e}_C$.

Beyond STEP

Three strings in three independent directions make the system *statically determinate*: the equilibrium equations form a 3×3 linear system with a unique solution. With a fourth string the tensions are no longer fixed by statics alone, and engineers have to bring in the elasticity of the strings. That is the start of structural analysis and of the linear algebra of rank and null spaces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats forces as vectors in this way, and Strang, *Introduction to Linear Algebra*, explains when such systems have unique solutions.

WORKED SOLUTION

STEP II 2011, Question 12

Probability and Statistics • Geometric Series of Games

STEP II 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q12

The key idea

Each game is only three points long, so we can list every way a game can go (there are just eight sequences) and sort them into "Xavier wins", "Younis wins" and "nobody wins, play another game". Because each new game starts afresh, the match is a geometric process: the probability Younis wins the match is his chance of winning a game divided by the chance that a game is decisive.

Solution

Write each game as a string of point-winners, so XYY means Xavier won the first point and Younis the second and third. The first point goes to X with probability p. After that, each point goes to the *same* player as the previous point with probability p and *switches* with probability $1-p$.

A game is decided as soon as the same player wins two points in a row, so the possible games are:

Sequence	Probability	Outcome
XX	$p \cdot p$	Xavier wins
YXX	$(1-p)(1-p)p$	Xavier wins
YY	$(1-p)p$	Younis wins
XYY	$p(1-p)p$	Younis wins
XYX	$p(1-p)(1-p)$	another game
YXY	$(1-p)(1-p)(1-p)$	another game

(Once a game reaches XX or YY the third point isn't needed, and it makes no difference whether we imagine it being played.) The undecided games are exactly those where the points alternate, i.e. both later points "switch".

Part (i)

Let A be the event that Younis wins a given game and C the event that the game is undecided. From the table,

$$\mathrm{P}(A) = (1-p)p + p^2(1-p) = p(1-p)(1+p) = p(1-p^2),$$
$$\mathrm{P}(C) = p(1-p)^2 + (1-p)^3 = (1-p)^2\big(p + (1-p)\big) = (1-p)^2.$$

Games are played independently with the same rules, so Younis wins the match if the first $n-1$ games are undecided and he wins game n, for some $n \geq 1$. Hence

$$w = \sum_{n=1}^{\infty} \mathrm{P}(C)^{\,n-1}\,\mathrm{P}(A) = \frac{\mathrm{P}(A)}{1-\mathrm{P}(C)},$$

where the geometric series converges because $\mathrm{P}(C) = (1-p)^2 < 1$ when $p \neq 0$. (Equivalently: if the first game is undecided we are back where we started, so $w = \mathrm{P}(A) + \mathrm{P}(C)\,w$.) Now $1-(1-p)^2 = 2p - p^2 = p(2-p)$, so

$$w = \frac{p(1-p^2)}{p(2-p)} = \frac{1-p^2}{2-p},$$

where cancelling p uses $p \neq 0$. This is the required result.

Comparing with $\frac{1}{2}$. We have

$$w - \tfrac{1}{2} = \frac{2(1-p^2) - (2-p)}{2(2-p)} = \frac{p - 2p^2}{2(2-p)} = \frac{p(1-2p)}{2(2-p)}.$$

For $0 < p \leq 1$ the factors p and $2-p$ are positive, so $w - \frac{1}{2}$ has the sign of $1-2p$. Hence $w > \frac{1}{2}$ when $0 < p < \frac{1}{2}$ and $w < \frac{1}{2}$ when $p > \frac{1}{2}$ (and $w = \frac{1}{2}$ at $p = \frac{1}{2}$, as symmetry demands).

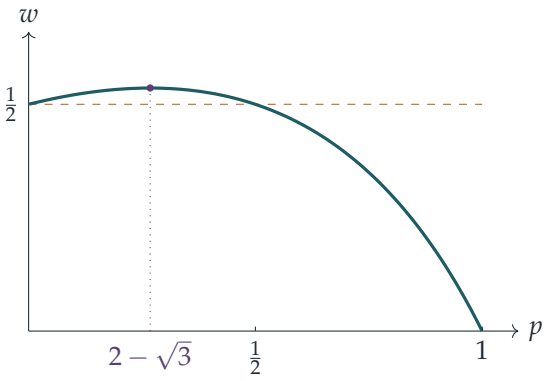
> **Ask yourself** Lowering p makes Younis more likely to win the first point. Must it always help him?

Is w decreasing in p? No. Differentiating with the quotient rule,

$$\frac{\mathrm{d}w}{\mathrm{d}p} = \frac{-2p(2-p) + (1-p^2)}{(2-p)^2} = \frac{p^2 - 4p + 1}{(2-p)^2}.$$

The numerator has roots $p = 2 \pm \sqrt{3}$, and only $2 - \sqrt{3} \approx 0.268$ lies in $[0,1]$. So $\mathrm{d}w/\mathrm{d}p > 0$ for $0 < p < 2-\sqrt{3}$ and $\mathrm{d}w/\mathrm{d}p < 0$ for $2-\sqrt{3} < p \leq 1$. On the interval $0 < p < 2-\sqrt{3}$, decreasing p actually *decreases* w. For example, $w(0.2) = \frac{8}{15} \approx 0.533$ but $w(0.1) = \frac{99}{190} \approx 0.521$.

The reason is that p plays two roles. Small p gives Younis the first point, but it also makes winning streaks unlikely, since a winner of a point keeps winning with probability only p. When p is tiny the points tend to alternate, and the advantage of the first point gets washed out. The largest value is $w(2-\sqrt{3}) = 4 - 2\sqrt{3} \approx 0.536$, and $w \to \frac{1}{2}$ as $p \to 0$.



Part (ii)

With $p = \frac{2}{3}$,

$$w = \frac{1 - \frac{4}{9}}{2 - \frac{2}{3}} = \frac{5/9}{4/3} = \frac{5}{12}.$$

Since $\mathrm{P}(C) = \frac{1}{9} < 1$, the match ends with probability 1, so Xavier wins it with probability $1 - \frac{5}{12} = \frac{7}{12}$. Xavier's expected gain is

$$\tfrac{7}{12} \times 1 - \tfrac{5}{12} \times k,$$

and the game is fair when this is zero, i.e. when

$$k = \frac{7}{5}.$$

So Xavier should pay Younis £1.40 when Younis wins.

Part (iii)

When $p = 0$, Younis wins the first point of every game for certain, and every later point is won for certain by the player who *lost* the previous point. So every game goes YXY with probability 1. Nobody ever wins two consecutive points, and the match never ends: neither player wins.

This is why part (i) excludes $p = 0$. There $\mathrm{P}(A) = 0$ and $1 - \mathrm{P}(C) = 0$, so the formula $w = \mathrm{P}(A)/(1 - \mathrm{P}(C))$ becomes $0/0$. The fact that $\frac{1-p^2}{2-p} \to \frac{1}{2}$ as $p \to 0$ describes nearby matches that do eventually end; it is not a probability for the case $p = 0$ itself.

Checks

- At $p = \frac{1}{2}$ every point is a fair coin toss, so w should be $\frac{1}{2}$. The formula gives $\frac{3/4}{3/2} = \frac{1}{2}$.
- At $p = 1$ Xavier wins the first two points for certain, so $w = 0$. The formula gives 0.
- The six game probabilities in the table add to 1.

- An exact enumeration of all eight point sequences (for p from 0.01 to 0.99) reproduced w, and a simulation of 200 000 matches at $p = \frac{2}{3}$ gave Younis winning 41.6% of the time, against $\frac{5}{12} \approx 41.7\%$.

Common slips

- Getting the transition rule the wrong way round. The probability p belongs to whoever won the *previous point*, not to Xavier.
- Answering $k = \frac{5}{7}$ by equating the wrong products. Fairness means $\mathrm{P}(\text{X wins}) \times 1 = \mathrm{P}(\text{Y wins}) \times k$, so the less likely winner gets the bigger payout.
- Plugging $p = 0$ into the formula in part (iii) and saying each player has probability $\frac{1}{2}$, when in fact the match never ends.

Beyond STEP

The points here form a *Markov chain*: each point depends only on the one before it. Splitting the match into identical, independent games is an example of a *renewal* argument, and the equation $w = \mathrm{P}(A) + \mathrm{P}(C)w$ is the simplest example of "first-step analysis", the standard method for absorption probabilities in Markov chains. Part (iii) shows what happens when absorption fails: the chain cycles for ever. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains and these methods at first-year level.

WORKED SOLUTION

STEP II 2011, Question 13

Probability and Statistics • Measures of Skewness

STEP II 2011, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q13

The key idea

Skewness is about lopsidedness, and the three measures here detect it in three different ways: through the third central moment, through where the median sits between two deciles, and through the gap between mean and median. For the density $2x$ on $[0,1]$ everything is explicit: the moments are one-line integrals, and the cumulative distribution function is x^2, so its inverse is just a square root. The only real work is careful arithmetic with surds, and a clean way to compare them at the end.

Solution

What skewness measures. Skewness measures the *asymmetry* of a distribution about its centre. A symmetric distribution has zero skewness. If the longer tail is on the right, the skewness is positive; if the longer tail is on the left, it is negative.

Part (i)

Expand the cube and use linearity of expectation, remembering that μ is a constant:

$$\begin{aligned}\mathrm{E}\big((X-\mu)^3\big) &= \mathrm{E}(X^3) - 3\mu\,\mathrm{E}(X^2) + 3\mu^2\,\mathrm{E}(X) - \mu^3 \\ &= \mathrm{E}(X^3) - 3\mu(\sigma^2+\mu^2) + 3\mu^3 - \mu^3 \\ &= \mathrm{E}(X^3) - 3\mu\sigma^2 - \mu^3,\end{aligned}$$

using $\mathrm{E}(X) = \mu$ and $\mathrm{E}(X^2) = \sigma^2 + \mu^2$ (the usual formula for the variance, rearranged). Dividing by σ^3 gives the required expression for γ.

The distribution with density $2x$. For $k \geq 0$,

$$\mathrm{E}(X^k) = \int_0^1 x^k \cdot 2x \,\mathrm{d}x = \frac{2}{k+2}.$$

So $\mu = \frac{2}{3}$, $\mathrm{E}(X^2) = \frac{1}{2}$ and $\mathrm{E}(X^3) = \frac{2}{5}$. Hence

$$\sigma^2 = \tfrac{1}{2} - \tfrac{4}{9} = \tfrac{1}{18}, \qquad \sigma = \frac{1}{3\sqrt{2}}, \qquad \sigma^3 = \frac{1}{27 \cdot 2\sqrt{2}} = \frac{1}{54\sqrt{2}}.$$

The numerator is

$$\mathrm{E}(X^3) - 3\mu\sigma^2 - \mu^3 = \frac{2}{5} - 3 \cdot \frac{2}{3} \cdot \frac{1}{18} - \frac{8}{27} = \frac{2}{5} - \frac{1}{9} - \frac{8}{27} = \frac{54 - 15 - 40}{135} = -\frac{1}{135}.$$

Therefore

$$\gamma = -\frac{1}{135} \cdot 54\sqrt{2} = -\frac{54\sqrt{2}}{135} = -\frac{2\sqrt{2}}{5},$$

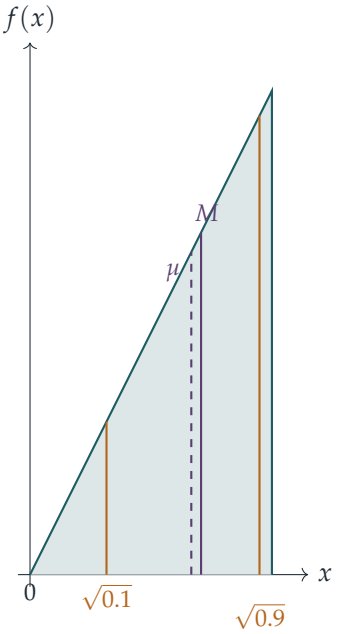
as required. The negative sign makes sense: the density piles up near $x = 1$, leaving a long tail towards 0 on the left.

Part (ii)

For $0 \le x \le 1$ the cumulative distribution function is

$$\mathrm{F}(x) = \int_0^x 2t \, \mathrm{d}t = x^2,$$

which is increasing on $[0, 1]$, so for $0 \le p \le 1$ its inverse is $\mathrm{F}^{-1}(p) = \sqrt{p}$.



Decile skewness. Multiplying the numerator and denominator of D by $\sqrt{10}$ clears all the surds except one:

$$D = \frac{\sqrt{\frac{9}{10}} - 2\sqrt{\frac{1}{2}} + \sqrt{\frac{1}{10}}}{\sqrt{\frac{9}{10}} - \sqrt{\frac{1}{10}}} = \frac{3 - 2\sqrt{5} + 1}{3 - 1} = \frac{4 - 2\sqrt{5}}{2} = 2 - \sqrt{5},$$

since $\sqrt{10}\cdot\sqrt{\frac{1}{2}}=\sqrt{5}$. This is what we had to show.

Pearson skewness. The median satisfies $\mathrm{F}(M)=\frac{1}{2}$, so $M=\frac{1}{\sqrt{2}}$. Using $\frac{1}{\sigma}=3\sqrt{2}$,

$$P=3\Big(\frac{2}{3}-\frac{1}{\sqrt{2}}\Big)\cdot 3\sqrt{2}=9\sqrt{2}\cdot\frac{2}{3}-9=6\sqrt{2}-9.$$

Numerically, $D\approx -0.236$, $P\approx -0.515$ and $\gamma\approx -0.566$, so the ordering looks right; now we prove it.

> **Ask yourself** How do we compare numbers like $2-\sqrt{5}$ and $6\sqrt{2}-9$ without a calculator?

Rearrange so that each side is positive, then square: for positive numbers, $a<b$ if and only if $a^2<b^2$. Each step below is reversible, so a true final statement proves the first one.

$D>P$**.** We need $2-\sqrt{5}>6\sqrt{2}-9$, that is, $\sqrt{5}+6\sqrt{2}<11$. Both sides are positive, so this is equivalent to

$$\begin{aligned}(\sqrt{5}+6\sqrt{2})^2=77+12\sqrt{10}<121 &\iff 12\sqrt{10}<44\\ &\iff 3\sqrt{10}<11\\ &\iff 90<121,\end{aligned}$$

where the last step squares two positive numbers again. Since $90<121$ is true, $D>P$.

$P>\gamma$**.** We need $6\sqrt{2}-9>-\frac{2\sqrt{2}}{5}$. Multiplying by 5 and rearranging, this is $32\sqrt{2}>45$. Both sides are positive, so it is equivalent to $32^2\cdot 2>45^2$, that is, $2048>2025$, which is true. So $P>\gamma$.

Hence $D>P>\gamma$. All three measures are negative and agree that the distribution is skewed to the left, but they disagree about how much.

Checks

- The identity in (i) gives 0 for any symmetric distribution: for example, if X is ± 1 with equal probability then $\mu=0$, $\sigma=1$ and $\mathrm{E}(X^3)=0$.
- Numerically, a midpoint-rule evaluation of $\mathrm{E}\big((X-\mu)^3\big)/\sigma^3$ gives -0.56569, matching $-2\sqrt{2}/5$.
- Simulating $X=\sqrt{U}$ with U uniform on $[0,1]$ (which has $\mathrm{F}(x)=x^2$) gives a sample median close to 0.707 and deciles close to 0.316 and 0.949, as in the sketch.
- The mean $\frac{2}{3}$ lies to the left of the median $\frac{1}{\sqrt{2}}\approx 0.707$, which is typical of a left-skewed distribution and is why $P<0$.

Common slips

- Writing $\sigma^3 = (\frac{1}{18})^3$ instead of $(\frac{1}{18})^{3/2}$: it is the standard deviation, not the variance, that is cubed.
- Squaring an inequality when one side may be negative. Rearrange first so that both sides are positive.
- Using $\mathrm{E}(X^2) = \frac{1}{2}$ where $\sigma^2 = \frac{1}{18}$ is needed, in either the formula for γ or the denominator of P.

Beyond STEP

The quantity $\mathrm{E}\big((X-\mu)^3\big)$ is the third *central moment*, and the expansion in (i) is the start of the relationship between moments and *cumulants*, which add up nicely for sums of independent random variables. Quantile-based measures such as D are prized in statistics because they are *robust*: a single wild observation barely moves a decile but can change a third moment dramatically. Grimmett and Welsh, *Probability: An Introduction*, develops moments and moment generating functions at the right level.

WORKED SOLUTION

STEP III 2011, Question 1

Pure • Reduction of Order

STEP III 2011, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q1

The key idea

Part (i) is a separable equation, and it is there for a reason: it is exactly the equation that appears when we substitute $y = ze^{-x}$ into $(*)$. The term in z itself cancels, leaving an equation involving only z' and z'', which is first order in $u = z'$. This is the method of *reduction of order*: once we know one solution (e^{-x}) we can find the other. Part (iii) then runs the same substitution with a right-hand side, which turns into a linear first-order equation solved by an integrating factor.

Solution

Throughout we work on an interval where $x \neq -1$, so we may divide by $x+1$.

Part (i)

Since

$$\frac{x+2}{x+1} = 1 + \frac{1}{x+1},$$

separating the variables (for $u \neq 0$) gives

$$\int \frac{\mathrm{d}u}{u} = \int \left(1 + \frac{1}{x+1}\right) \mathrm{d}x \quad \Longrightarrow \quad \ln|u| = x + \ln|x+1| + c.$$

Exponentiating, $|u| = e^c\,|x+1|\,e^x$. The sign and the factor e^c combine into a single arbitrary constant, and $u = 0$ is also a solution, so the general solution is

$$u = C(x+1)e^x, \qquad C \text{ an arbitrary constant.}$$

Part (ii)

With $y = ze^{-x}$ the product rule gives

$$y' = (z' - z)e^{-x},$$
$$y'' = (z'' - z')e^{-x} - (z' - z)e^{-x} = (z'' - 2z' + z)e^{-x}.$$

Substituting into $(*)$ and cancelling the non-zero factor e^{-x}:

$$\begin{aligned} 0 &= (x+1)(z''-2z'+z)+x(z'-z)-z \\ &= (x+1)z''+\big(-2(x+1)+x\big)z'+\big((x+1)-x-1\big)z \\ &= (x+1)z''-(x+2)z'. \end{aligned}$$

The z terms have cancelled completely. Writing $u=\dfrac{\mathrm{d}z}{\mathrm{d}x}$ and dividing by $x+1$,

$$\frac{\mathrm{d}u}{\mathrm{d}x}-\left(\frac{x+2}{x+1}\right)u=0,$$

which is a first-order equation for $\dfrac{\mathrm{d}z}{\mathrm{d}x}$, as required. It is precisely the equation of part (i).

Ask yourself Why did the z term vanish? Could we have predicted it?

The coefficient of z is what we get by putting $z=1$, that is, by testing $y=\mathrm{e}^{-x}$ in $(*)$. Since $(x+1)\mathrm{e}^{-x}-x\mathrm{e}^{-x}-\mathrm{e}^{-x}=0$, $y=\mathrm{e}^{-x}$ is already a solution, so constant z must work and no bare z term can survive. That is why this substitution was chosen.

By part (i), $z'=C(x+1)\mathrm{e}^{x}$. Integrating by parts,

$$\int (x+1)\mathrm{e}^{x}\,\mathrm{d}x=(x+1)\mathrm{e}^{x}-\int \mathrm{e}^{x}\,\mathrm{d}x=x\mathrm{e}^{x}+\text{const},$$

so

$$z=Cx\mathrm{e}^{x}+B.$$

Hence $y=z\mathrm{e}^{-x}=Cx+B\mathrm{e}^{-x}$, and renaming C as A,

$$y=Ax+B\mathrm{e}^{-x},$$

with A and B arbitrary constants. Every solution of $(*)$ arises this way, since any y can be written as $z\mathrm{e}^{-x}$ with $z=y\mathrm{e}^{x}$; so this is the general solution, as required.

Part (iii)

The same substitution changes only the right-hand side, which becomes $(x+1)^2$ divided by e^{-x}:

$$(x+1)z''-(x+2)z'=(x+1)^2\mathrm{e}^{x}.$$

With $u=z'$ and dividing by $x+1$,

$$\frac{\mathrm{d}u}{\mathrm{d}x}-\left(\frac{x+2}{x+1}\right)u=(x+1)\mathrm{e}^{x}.$$

This is linear. From part (i), $(x+1)\mathrm{e}^{x}$ solves the homogeneous equation, so its reciprocal $\dfrac{\mathrm{e}^{-x}}{x+1}$ is an integrating factor. (It is $\exp\!\big(-\int \frac{x+2}{x+1}\,\mathrm{d}x\big)$ up to sign.) Multiplying through,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u\,\mathrm{e}^{-x}}{x+1}\right)=1 \quad\Longrightarrow\quad u=(x+1)\mathrm{e}^{x}(x+k)$$

for a constant k. So

$$z' = (x^2 + x)\mathrm{e}^x + k(x+1)\mathrm{e}^x.$$

The second term integrates to $kx\mathrm{e}^x$ as in part (ii). For the first, integrating by parts twice,

$$\begin{aligned}\int (x^2 + x)\mathrm{e}^x \,\mathrm{d}x &= (x^2 + x)\mathrm{e}^x - \int (2x+1)\mathrm{e}^x \,\mathrm{d}x \\ &= (x^2 + x)\mathrm{e}^x - (2x+1)\mathrm{e}^x + 2\mathrm{e}^x + \text{const} = (x^2 - x + 1)\mathrm{e}^x + \text{const}.\end{aligned}$$

Therefore $z = (x^2 - x + 1)\mathrm{e}^x + kx\mathrm{e}^x + B$, and

$$y = z\mathrm{e}^{-x} = x^2 - x + 1 + kx + B\mathrm{e}^{-x}.$$

The $-x$ merges with the arbitrary kx, so writing $A = k - 1$ the general solution is

$$y = x^2 + 1 + Ax + B\mathrm{e}^{-x},$$

with A and B arbitrary constants.

Alternatively, since the right-hand side is a quadratic, we could try a particular integral $y = px^2 + qx + r$. Substituting gives $(x+1)2p + x(2px + q) - (px^2 + qx + r) = px^2 + 2px + 2p - r$, and matching with $x^2 + 2x + 1$ gives $p = 1$, $r = 1$, with q free (because x is a complementary function). Adding the complementary function from part (ii) gives the same answer.

Checks

Substitute $y = x^2 + 1$ directly: $y' = 2x$, $y'' = 2$, so the left-hand side is $2(x+1) + 2x^2 - x^2 - 1 = x^2 + 2x + 1 = (x+1)^2$, as needed. Numerically, with random A, B and random $x \in (-0.9, 3)$, finite-difference derivatives of $y = x^2 + 1 + Ax + B\mathrm{e}^{-x}$ satisfy the equation to within about 3×10^{-6}; the same test confirms parts (i) and (ii).

Common slips

- Getting y'' wrong: it is $(z'' - 2z' + z)\mathrm{e}^{-x}$, with a *plus* on the z; a sign error here leaves a stray z term.
- Forgetting to multiply the right-hand side by e^x in part (iii) when cancelling the factor e^{-x}.
- Losing a constant: the final answer needs two arbitrary constants, one from each integration.

Beyond STEP

Reduction of order is a general technique: if y_1 solves a linear second-order equation, putting $y = y_1 v$ always removes the v term and leaves a first-order equation for v'. It is

also the idea behind *variation of parameters*, which produces particular integrals for any right-hand side. Both are treated clearly in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 2011, Question 2

Pure • Rational Roots of Polynomials

STEP III 2011, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q2

The key idea

Multiplying $\mathrm{f}(p/q) = 0$ by q^{n-1} clears every denominator except one: the leading term becomes p^n/q while everything else is an integer. So p^n/q must be an integer, and since p and q share no factor, that forces $q = 1$. After that, each part is a question about *integer* roots, and these can be ruled out by size, by divisibility or, neatly, by parity.

Solution

The opening result

Suppose $\mathrm{f}(p/q) = 0$ with p, q coprime integers and $q > 0$. Then

$$0 = q^{n-1}\,\mathrm{f}\left(\frac{p}{q}\right) = \frac{p^n}{q} + a_{n-1}p^{n-1} + a_{n-2}p^{n-2}q + \cdots + a_1 p\,q^{n-2} + a_0 q^{n-1},$$

because the term $a_k(p/q)^k$ becomes $a_k p^k q^{n-1-k}$, and $n - 1 - k \geq 0$ for every $k \leq n - 1$. Every term after the first is an integer, so

$$\frac{p^n}{q} = -\left(a_{n-1}p^{n-1} + \cdots + a_0 q^{n-1}\right)$$

is an integer; that is, q divides p^n.

? **Ask yourself** Why does $q \mid p^n$ force $q = 1$?

If $q > 1$, it has some prime factor r. Then r divides p^n, and since r is prime it must divide p itself. But then r is a common factor of p and q greater than 1, which is impossible. So q has no prime factor, and as $q > 0$ we get

$$q = 1.$$

Hence every rational root p/q of $\mathrm{f}(x) = 0$ equals p, an integer. (Notice that this is where we use that f is *monic*: a leading coefficient a_n would leave $a_n p^n/q$, and we could only conclude $q \mid a_n$.)

Two quick facts will help below. If m is an integer root, then

$$m\left(m^{n-1}+a_{n-1}m^{n-2}+\cdots+a_1\right)=-a_0,$$

so m divides a_0; and $m\neq 0$ since $a_0\neq 0$.

Part (i)

The number $2^{1/n}$ is a root of $x^n-2=0$, which is monic with integer coefficients and $a_0=-2\neq 0$. If $2^{1/n}$ were rational, it would therefore be an integer m with $m^n=2$. But

$$1^n=1<2<4\leq 2^n,$$

so the positive real number $2^{1/n}$ lies strictly between the consecutive integers 1 and 2, and it cannot be an integer. Hence $2^{1/n}$ is irrational for every $n\geq 2$.

Part (ii)

Here $n=3$ and $a_0=1$. Any rational root is an integer m dividing 1, so $m=\pm 1$. But

$$1^3-1+1=1\neq 0,\qquad (-1)^3-(-1)+1=1\neq 0.$$

So $x^3-x+1=0$ has no rational roots.

(Alternatively: $m^3-m=(m-1)m(m+1)$ is a product including two consecutive integers, so it is even, and m^3-m+1 is odd, never zero. This parity idea is exactly what part (iii) needs.)

Part (iii)

The polynomial x^n-5x+7 is monic with integer coefficients and constant term $7\neq 0$, so any rational root is an integer m. We show that m^n-5m+7 is always odd.

- If m is even, then m^n and $5m$ are even, so m^n-5m+7 is even $-$ even $+$ odd, which is odd.
- If m is odd, then m^n and $5m$ are both odd, so m^n-5m is even and m^n-5m+7 is odd.

Either way m^n-5m+7 is odd, so in particular it is not 0. Hence $x^n-5x+7=0$ has no integer roots, and therefore no rational roots, for any $n\geq 2$.

(Checking the divisors of 7 instead would mean showing $m^n-5m+7\neq 0$ for $m=\pm 1,\pm 7$ and *every* n, which works but is messier; parity handles all n at once.)

Checks

- Try a polynomial that *does* have a rational root, such as $x^2-x-6=(x-3)(x+2)$: the roots 3 and -2 are integers, as promised, and they divide $a_0=-6$.

- Non-monic polynomials can have non-integer rational roots, e.g. $2x - 1$; this is why the leading coefficient 1 matters.
- Numerically, $2^{1/2} \approx 1.414$ and $2^{1/3} \approx 1.260$ sit between 1 and 2; and $x^3 - x + 1$ takes the value 1 at $x = \pm 1$. A computer search over 3000 random monic integer polynomials found only integer rational roots, and confirmed $m^n - 5m + 7$ is odd for $|m| \leq 200$, $2 \leq n \leq 14$.

Common slips

- Multiplying by q^n instead of q^{n-1}: you get $p^n = -q(\cdots)$, which also works, but you must then say that $q \mid p^n$; with q^{n-1} the fraction p^n/q makes this explicit.
- Asserting "$q \mid p^n$ so $q \mid p$" without saying why; the argument goes through a prime factor of q.
- In (iii), checking only a few values of n. The argument must work for all $n \geq 2$ simultaneously.

Beyond STEP

This is the *rational root theorem* in its monic form: a rational number that is a root of a monic integer polynomial is an integer. In the language of algebraic number theory, the integers are *integrally closed*, and numbers that are roots of monic integer polynomials are called algebraic integers. The parity trick in (iii) is the simplest case of reducing a polynomial modulo a prime, a powerful tool for proving that polynomials have no roots or are irreducible. Davenport's *The Higher Arithmetic* is a lovely next step, and Allenby's *Rings, Fields and Groups* develops the algebraic side.

WORKED SOLUTION

STEP III 2011, Question 3

Pure • Solving Cubic Equations

STEP III 2011, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q3

The key idea

Two cubes $(x-\alpha)^3$ and $(x-\beta)^3$ give us two free constants a and b, and a monic cubic with no x^2 term imposes four conditions. So we use two of the conditions to find a and b, and the other two must then be exactly the facts about $\alpha+\beta$ and $\alpha\beta$ that the quadratic supplies. Once the cubic is a sum of two cubes, setting it to zero becomes "a ratio cubed equals a constant", which we solve by taking cube roots, all three of them.

Solution

Writing the cubic as a sum of two cubes

Since $p \neq 0$, the quadratic $pt^2 - qt + p^2 = 0$ has

$$\alpha+\beta = \frac{q}{p}, \qquad \alpha\beta = \frac{p^2}{p} = p,$$

and its discriminant is $q^2 - 4p^3$. The proviso $q^2 \neq 4p^3$ therefore says precisely that $\alpha \neq \beta$. Also $\alpha\beta = p \neq 0$, so neither root is zero.

Expanding,

$$a(x-\alpha)^3 + b(x-\beta)^3 = (a+b)x^3 - 3(a\alpha+b\beta)x^2 + 3(a\alpha^2+b\beta^2)x - (a\alpha^3+b\beta^3).$$

To match $x^3 - 3px + q$ we need

$$a+b=1, \qquad a\alpha+b\beta=0, \qquad a\alpha^2+b\beta^2=-p, \qquad a\alpha^3+b\beta^3=-q.$$

The first two are linear in a and b. Substituting $b = 1-a$ into the second gives $a(\alpha-\beta) = -\beta$, and since $\alpha \neq \beta$ we may divide:

$$a = \frac{\beta}{\beta-\alpha}, \qquad b = \frac{\alpha}{\alpha-\beta}.$$

Now we check that the remaining two conditions hold automatically:

$$a\alpha^2 + b\beta^2 = \frac{\beta\alpha^2 - \alpha\beta^2}{\beta - \alpha} = \frac{\alpha\beta(\alpha - \beta)}{\beta - \alpha} = -\alpha\beta = -p,$$
$$a\alpha^3 + b\beta^3 = \frac{\beta\alpha^3 - \alpha\beta^3}{\beta - \alpha} = \frac{\alpha\beta(\alpha - \beta)(\alpha + \beta)}{\beta - \alpha} = -\alpha\beta(\alpha + \beta) = -p \cdot \frac{q}{p} = -q.$$

So, with these values of a and b,

$$x^3 - 3px + q = \frac{\beta}{\beta - \alpha}(x - \alpha)^3 + \frac{\alpha}{\alpha - \beta}(x - \beta)^3,$$

as required.

Solving $x^3 - 24x + 48 = 0$

Here $p = 8$ and $q = 48$, and $q^2 = 2304 \neq 2048 = 4p^3$, so the result applies. The quadratic is $8t^2 - 48t + 64 = 0$, that is $t^2 - 6t + 8 = 0$, with roots 2 and 4. Take $\alpha = 2$, $\beta = 4$; then $a = 2$ and $b = -1$, so

$$x^3 - 24x + 48 = 2(x - 2)^3 - (x - 4)^3.$$

(A quick expansion confirms this: $2x^3 - 12x^2 + 24x - 16 - x^3 + 12x^2 - 48x + 64$.)

Ask yourself How do we turn $2(x - 2)^3 = (x - 4)^3$ into something we can solve?

Divide by $(x - 4)^3$. This is allowed, because $x = 4$ is not a root ($64 - 96 + 48 = 16$). The equation becomes

$$\left(\frac{x - 2}{x - 4}\right)^3 = \frac{1}{2}.$$

Write $c = 2^{1/3}$. The complex numbers whose cube is $\frac{1}{2}$ are $k = c^{-1}$, $c^{-1}\omega$ and $c^{-1}\omega^2$. For each, solve $\dfrac{x - 2}{x - 4} = k$:

$$x - 2 = kx - 4k \quad \Longrightarrow \quad x = \frac{2 - 4k}{1 - k},$$

which is valid since $k \neq 1$. The map $k \mapsto x$ is one-to-one, so the three values of k give three distinct roots, which must be all the roots of the cubic.

For $k = c^{-1}$, multiply top and bottom by c:

$$x = \frac{2c - 4}{c - 1} = \frac{2(2 - 2^{1/3})}{1 - 2^{1/3}},$$

which is the stated solution. For $k = c^{-1}\omega$, multiply top and bottom by $c\,\omega^{-1} = c\,\omega^2$ (using $\omega^3 = 1$):

$$x = \frac{2c\omega^2 - 4}{c\omega^2 - 1} = \frac{2(2 - 2^{1/3}\omega^2)}{1 - 2^{1/3}\omega^2},$$

and in the same way $k = c^{-1}\omega^2$ gives

$$x = \frac{2(2 - 2^{1/3}\omega)}{1 - 2^{1/3}\omega}.$$

So the three solutions are $x = \dfrac{2(2 - 2^{1/3}\omega^j)}{1 - 2^{1/3}\omega^j}$ for $j = 0, 1, 2$.

The case $p = r^2$, $q = 2r^3$

Now $q^2 = 4r^6 = 4p^3$, so this is exactly the case the method excludes: the quadratic $r^2t^2 - 2r^3t + r^4 = r^2(t - r)^2$ has the repeated root $t = r$, and we cannot find a and b. But a repeated root of the quadratic suggests a repeated root of the cubic too. Indeed $x = r$ gives $r^3 - 3r^3 + 2r^3 = 0$, and the derivative $3x^2 - 3r^2$ also vanishes at $x = r$, so $(x - r)^2$ is a factor. Comparing the x^3 and constant terms,

$$x^3 - 3r^2x + 2r^3 = (x - r)^2(x + 2r).$$

The roots are $x = r$ (twice) and $x = -2r$.

? **Ask yourself** Could we have found these from the earlier formula?

No. With $\alpha = \beta$ the two cubes collapse into one, so $a(x - \alpha)^3 + b(x - \alpha)^3$ could only ever be $(x - \alpha)^3$, which has no x-term to match $-3px$. That is why we needed a different approach. Writing the cubic in powers of $(x - r)$ shows what happens instead: $x^3 - 3r^2x + 2r^3 = (x - r)^3 + 3r(x - r)^2$, so the double root sits at the repeated root $t = r$ of the quadratic.

Checks

- The real root is $2(2 - 1.259921)/(1 - 1.259921) \approx -5.694644$, and substituting gives $x^3 - 24x + 48 \approx 0$ to about 10^{-13}. The other two are $2.847322 \pm 0.567212\,\mathrm{i}$, also with zero residual.
- The three roots sum to 0 (no x^2 term) and multiply to -48: $-5.694644 \times (2.847322^2 + 0.567212^2) \approx -48.0$.
- One real root and two complex ones fits the theory: $x^3 - 3px + q$ has three real roots only when $q^2 < 4p^3$, and here $q^2 > 4p^3$.
- The identity can be tested numerically: we checked it at random complex x for 2000 random pairs (p, q), and the factorisation $(x - r)^2(x + 2r)$ for several values of r.

Common slips

- Taking only the real cube root of $\frac{1}{2}$ and so finding only one solution. A cubic has three roots, and they come from the three cube roots.

- Mixing up which of a, b goes with which root: a (the coefficient of $(x-\alpha)^3$) has β on top.
- Swapping ω and ω^2 when clearing fractions. Since the two answers form a conjugate pair, the final set of roots is the same either way.

Beyond STEP

This is a version of the 16th-century method for solving cubics, often credited to Cardano and Tartaglia: reduce to a "depressed" cubic with no x^2 term, then solve an auxiliary quadratic (the *resolvent*). Galois theory explains why cubics and quartics have such resolvents and why general quintics do not. Allenby, *Rings, Fields and Groups*, gives a readable route towards these ideas.

WORKED SOLUTION

STEP III 2011, Question 4

Pure • Young's Inequality

STEP III 2011, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 3.4, *Areas That Prove Inequalities*: https://beyondstep.keithmcnulty.org/ch03/areas-inequalities.html

The key idea

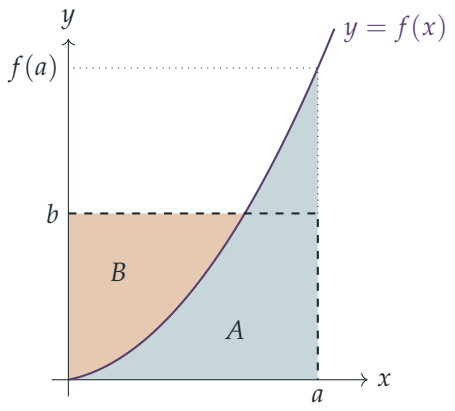
Both integrals in $(*)$ are areas on the same picture: $\int_0^a f(x)\,\mathrm{d}x$ is the area between the curve and the x-axis, and $\int_0^b f^{-1}(y)\,\mathrm{d}y$ is the area between the curve and the y-axis. Together they always cover the rectangle with corners $(0,0)$ and (a,b), with some to spare unless the corner (a,b) sits on the curve. Parts (ii) and (iii) are then just a matter of choosing f and doing two integrals.

Solution

Part (i)

Since f is continuous with positive gradient, it is strictly increasing, so f^{-1} exists and is also increasing, and $f(0) = 0$ means the curve $y = f(x)$ starts at the origin.

Write A for the region between the curve and the x-axis for $0 \leq x \leq a$; its area is $\int_0^a f(x)\,\mathrm{d}x$. Write B for the region between the curve and the y-axis for $0 \leq y \leq b$; integrating with respect to y, its area is $\int_0^b f^{-1}(y)\,\mathrm{d}y$. The two regions lie on opposite sides of the curve, so they don't overlap.



Suppose first that $b \leq f(a)$, as in the picture. Every point of the rectangle $0 \leq x \leq a$, $0 \leq y \leq b$ lies either on or below the curve (so in A) or on or to the left of it (so in B). So A and B together cover the rectangle, of area ab, and $(*)$ holds. The surplus is the part of A above the line $y = b$, the sliver between the line and the curve for $f^{-1}(b) \leq x \leq a$.

If instead $b \geq f(a)$, the same argument works with the roles swapped: the rectangle is covered, and now the surplus is the part of B to the right of the line $x = a$.

So $(*)$ holds in every case, and equality needs the surplus to have zero area. When $a, b > 0$ and $b \neq f(a)$, the surplus is bounded by the curve and a straight line that meet at only one end, so it has positive area. Hence, for $a > 0$ and $b > 0$,

$$\text{equality holds if and only if } b = f(a),$$

that is, when the corner (a, b) of the rectangle lies on the curve.

Part (ii)

For $p > 1$, $f(x) = x^{p-1}$ is continuous on $x \geq 0$, has $f(0) = 0$, and is strictly increasing there, so $(*)$ applies. (Its gradient $(p-1)x^{p-2}$ is positive for $x > 0$; if $p < 2$ it blows up at 0, which doesn't affect the area argument.) Its inverse is $f^{-1}(y) = y^{1/(p-1)}$.

The condition $\frac{1}{p} + \frac{1}{q} = 1$ gives $\frac{1}{q} = \frac{p-1}{p}$, so

$$q = \frac{p}{p-1}, \qquad \frac{1}{p-1} + 1 = \frac{p}{p-1} = q.$$

So $f^{-1}(y) = y^{q-1}$, and the two integrals are

$$\int_0^a x^{p-1}\,\mathrm{d}x = \frac{a^p}{p}, \qquad \int_0^b y^{q-1}\,\mathrm{d}y = \frac{b^q}{q}.$$

Substituting into $(*)$ gives

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

as required.

Ask yourself What does the equality condition $b = f(a)$ say here, and how do we simplify b^q?

The condition is $b = a^{p-1}$. Then $b^q = a^{(p-1)q} = a^p$, because $(p-1)q = p$. So

$$\frac{a^p}{p} + \frac{b^q}{q} = a^p\left(\frac{1}{p} + \frac{1}{q}\right) = a^p = a \cdot a^{p-1} = ab,$$

and equality does hold.

Part (iii)

Take $f(x) = \sin x$ on $0 \le x \le \frac{1}{2}\pi$. It is continuous, $f(0) = 0$, and its gradient $\cos x$ is positive for $0 \le x < \frac{1}{2}\pi$, so it is strictly increasing (the gradient vanishes only at the endpoint, which does not affect the argument). It maps $[0, \frac{1}{2}\pi]$ onto $[0, 1]$, with inverse arcsin, and these are exactly the ranges allowed for a and b.

The first integral is

$$\int_0^a \sin x \, \mathrm{d}x = 1 - \cos a.$$

For the second, integrate by parts with $u = \arcsin y$ and $\mathrm{d}v = \mathrm{d}y$:

$$\begin{aligned}\int_0^b \arcsin y \, \mathrm{d}y &= \Big[y \arcsin y\Big]_0^b - \int_0^b \frac{y}{\sqrt{1-y^2}} \, \mathrm{d}y \\ &= b \arcsin b + \Big[\sqrt{1-y^2}\Big]_0^b \\ &= b \arcsin b + \sqrt{1-b^2} - 1.\end{aligned}$$

Adding, the 1s cancel and $(*)$ becomes

$$ab \le b \arcsin b + \sqrt{1-b^2} - \cos a,$$

which is the required inequality, with equality when $b = \sin a$.

Ask yourself We have a free choice of a. Which one gives the inequality we want?

For $t \ge 1$, put $b = t^{-1}$, which lies in $(0, 1]$ as needed. Multiplying the inequality by $t > 0$ and using $t\sqrt{1-t^{-2}} = \sqrt{t^2-1}$ (valid because $t > 0$) gives

$$a \le \arcsin(t^{-1}) + \sqrt{t^2-1} - t\cos a \quad \text{for every } a \in [0, \tfrac{1}{2}\pi].$$

The target has no a in it and has a t on the right, which is what $t\cos a$ becomes at $a = 0$. Taking $a = 0$:

$$0 \le \arcsin(t^{-1}) + \sqrt{t^2-1} - t, \qquad \text{that is,} \qquad \arcsin(t^{-1}) \ge t - \sqrt{t^2-1} \quad (t \ge 1).$$

Checks

- At $t = 1$ the final inequality reads $\frac{1}{2}\pi \geq 1$, and at $t = 2$ it reads $\frac{1}{6}\pi \approx 0.5236 \geq 2 - \sqrt{3} \approx 0.2679$.
- As $t \to \infty$, both sides are of order $\frac{1}{t}$: $\arcsin(t^{-1}) \approx t^{-1} + \frac{1}{6}t^{-3}$, while $t - \sqrt{t^2-1} = \frac{1}{t+\sqrt{t^2-1}} \approx \frac{1}{2t}$. So the bound is about half the true value.
- We tested the part (iii) inequality at 2×10^5 random points (a, b); it never failed, and the gap shrank to zero only near the curve $b = \sin a$. We also checked Young's inequality for random p, a and b, and checked both integrals by Simpson's rule.

Common slips

- Only drawing the case $b \leq f(a)$ in part (i). The case $b \geq f(a)$ is symmetric, but you should say so.
- Writing $f^{-1}(y) = y^{1/(p-1)}$ and then not spotting that $1/(p-1) = q - 1$, so the integral b^q/q doesn't appear.
- In the last step, looking for the a that optimises the bound. That a is $\arcsin(t^{-1})$, which only gives the trivial identity. The simplest choice, $a = 0$, is the one that works.

Beyond STEP

Part (ii) is Young's inequality, the standard route to Hölder's inequality $\sum |x_i y_i| \leq \left(\sum |x_i|^p\right)^{1/p} \left(\sum |y_i|^q\right)^{1/q}$, which in turn underlies the L^p spaces of analysis. The pairing of $\int f$ with $\int f^{-1}$ is the germ of the Legendre transform and convex duality, which reappear in mechanics (moving from the Lagrangian to the Hamiltonian) and in optimisation. For the analysis background, see Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP III 2011, Question 5

Pure • Areas by Integration

STEP III 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 4.9, *Holditch's Theorem*: https://beyondstep.keithmcnulty.org/ch04/holditch.html

The key idea

The cartesian formula $(*)$ is useful because it is *bilinear*: if A is P shifted by a vector that depends only on t, the area swept by OA splits into the area swept by OP, an area from the shift alone, and a cross term. The rod gives exactly such shifts, $\mp$ a multiple of the unit vector $(\cos t, \sin t)$. The cross term f looks impossible to evaluate, but we never need to: A and B both go once round the same curve $\mathcal{C}$, so $[A] = [B]$, and that equation pins down f.

Solution

The area formula

With $x = r\cos\theta$ and $y = r\sin\theta$, where r and θ are functions of t, the product rule gives

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \dot{r}\cos\theta - r\sin\theta\,\dot{\theta}, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \dot{r}\sin\theta + r\cos\theta\,\dot{\theta},$$

where dots denote $\mathrm{d}/\mathrm{d}t$. So

$$\begin{aligned} x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t} &= r\cos\theta\,(\dot{r}\sin\theta + r\cos\theta\,\dot{\theta}) - r\sin\theta\,(\dot{r}\cos\theta - r\sin\theta\,\dot{\theta}) \\ &= r^2(\cos^2\theta + \sin^2\theta)\,\dot{\theta} = r^2\frac{\mathrm{d}\theta}{\mathrm{d}t}, \end{aligned}$$

because the $r\dot{r}\sin\theta\cos\theta$ terms cancel. Changing the variable of integration from θ to t,

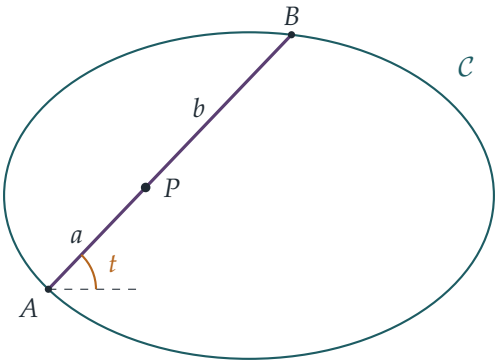
$$\tfrac{1}{2}\int r^2\,\mathrm{d}\theta = \tfrac{1}{2}\int r^2\frac{\mathrm{d}\theta}{\mathrm{d}t}\,\mathrm{d}t = \tfrac{1}{2}\int\left(x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}t,$$

which is $(*)$.

Coordinates of A and B

The rod points in the direction of the unit vector $\mathbf{u} = (\cos t, \sin t)$, going from A to B, and P lies between them with $AP = a$ and $PB = b$. So A is $P - a\mathbf{u}$ and B is $P + b\mathbf{u}$:

$$A = (x - a\cos t,\ y - a\sin t), \qquad B = (x + b\cos t,\ y + b\sin t).$$



The area swept by OA

Write dots for $\mathrm{d}/\mathrm{d}t$ again. For A we have

$$x_A = x - a\cos t, \quad \dot{x}_A = \dot{x} + a\sin t, \qquad y_A = y - a\sin t, \quad \dot{y}_A = \dot{y} - a\cos t.$$

Expanding,

$$x_A\dot{y}_A = x\dot{y} - a\cos t\,(x + \dot{y}) + a^2\cos^2 t,$$
$$y_A\dot{x}_A = y\dot{x} + a\sin t\,(y - \dot{x}) - a^2\sin^2 t,$$

so

$$x_A\dot{y}_A - y_A\dot{x}_A = (x\dot{y} - y\dot{x}) + a^2 - a\Big[(x + \dot{y})\cos t + (y - \dot{x})\sin t\Big].$$

Now apply $(*)$ with t running from 0 to 2π. The first bracket gives $[P]$, the constant a^2 gives $\frac{1}{2}\cdot 2\pi a^2 = \pi a^2$, and the last term gives $-af$ by the definition of f. Hence

$$[A] = [P] + \pi a^2 - af,$$

as required.

For B, the only change is that $-a$ becomes $+b$ throughout (replace a by $-b$ in the working above). So

$$[B] = [P] + \pi b^2 + bf.$$

The area between $\mathcal{C}$ and $\mathcal{D}$

> **Ask yourself** What are $[A]$, $[B]$ and $[P]$ actually equal to?

As t goes from 0 to 2π the rod turns through one full revolution. Both ends travel anticlockwise once round $\mathcal{C}$ and return to their starting points: after half a turn, A and B have swapped places, and after a full turn they are back where they began. The origin lies inside $\mathcal{D}$, and so inside $\mathcal{C}$. So OA and OB each sweep out the region enclosed by $\mathcal{C}$ exactly once, and OP sweeps out the region enclosed by $\mathcal{D}$ once. Writing $|\mathcal{C}|$ and $|\mathcal{D}|$ for these areas,

$$[A] = [B] = |\mathcal{C}|, \qquad [P] = |\mathcal{D}|.$$

Setting $[A] = [B]$ eliminates $[P]$:

$$\pi a^2 - af = \pi b^2 + bf \quad \Longrightarrow \quad (a+b)f = \pi(a^2 - b^2) = \pi(a-b)(a+b).$$

Since $a + b > 0$ (it is the length of the rod), we may divide to get $f = \pi(a - b)$. Therefore

$$|\mathcal{C}| - |\mathcal{D}| = [A] - [P] = \pi a^2 - a\pi(a-b) = \pi ab.$$

So the area between the curves is πab. Using $[B]$ instead gives $\pi b^2 + b\pi(a-b) = \pi ab$ as well, which is a reassuring consistency check.

Checks

- *A circle.* If $\mathcal{C}$ is a circle of radius R, the chord of length $a + b$ stays at distance $d = \sqrt{R^2 - \frac{1}{4}(a+b)^2}$ from the centre. By Pythagoras, P is at distance $\sqrt{d^2 + \frac{1}{4}(a-b)^2}$ from the centre, so $\mathcal{D}$ is a concentric circle with radius squared $R^2 - ab$. The area between them is $\pi R^2 - \pi(R^2 - ab) = \pi ab$.
- *Symmetry.* The answer is symmetric in a and b, as it must be, because reversing the rod swaps their roles.
- *Numerically.* We moved a rod with $a = 1$, $b = 1.5$ round the ellipse $x^2/16 + y^2/9 = 1$ and computed the area of the traced curve with the shoelace formula. The area between the curves came out as 4.712390, matching $\pi ab = 1.5\pi \approx 4.712389$. We also checked the identities for $[A]$ and $[B]$ by quadrature for randomly chosen curves $P(t)$.

Common slips

- Getting the sign of $\dot{x}_A = \dot{x} + a\sin t$ wrong. Differentiating $-a\cos t$ gives $+a\sin t$.
- Trying to evaluate f directly. It depends on the shape of $\mathcal{D}$ and can't be computed without more information. The point is that $[A] = [B]$ determines it.

Beyond STEP

This is Holditch's theorem (1858): the area between $\mathcal{C}$ and $\mathcal{D}$ depends only on a and b, not on the shape of $\mathcal{C}$. The formula $(*)$ is a special case of Green's theorem, which turns an area integral over a region into a line integral round its boundary. It is also the principle behind the planimeter, a mechanical device that measures area by tracing a boundary. Green's theorem is a centrepiece of first-year vector calculus, and Needham's *Visual Complex Analysis* gives a beautifully geometric view of such boundary integrals.

WORKED SOLUTION

STEP III 2011, Question 6

Pure • Integral Substitutions

STEP III 2011, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q6

The key idea

We are not allowed to evaluate anything, so the only tools are the ones that turn one integral into another: substitution and integration by parts. The limits are the clue. The numbers $\frac{1}{3}, \frac{1}{2}$ and $\ln 2, \ln 3$ are linked by $v = e^{-u}$, and $\frac{1}{2}\ln 2, \frac{1}{2}\ln 3$ are linked to $\frac{1}{2}, \frac{1}{3}$ by $t = e^{-2x}$. The integrals T and V have the same limits, and $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{artanh} t = \frac{1}{1-t^2}$ suggests integration by parts between them. Three links are enough to make all four integrals equal.



Solution

Throughout we use

$$\operatorname{artanh} t = \tfrac{1}{2}\ln\frac{1+t}{1-t} \quad (|t|<1), \qquad \frac{\mathrm{d}}{\mathrm{d}t}\operatorname{artanh} t = \frac{1}{1-t^2},$$

and in particular $\operatorname{artanh}\frac{1}{2} = \frac{1}{2}\ln 3$ and $\operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln 2$. All the integrands are continuous on their (closed, bounded) ranges, so every integral exists and the manipulations below are legitimate.

$T = V$: integration by parts

In T we integrate $\frac{1}{t}$ and differentiate $\operatorname{artanh} t$, taking $u = \operatorname{artanh} t$ and $\mathrm{d}w = \frac{\mathrm{d}t}{t}$, so $w = \ln t$ (fine, as $t > 0$ on the range):

$$\begin{aligned} T &= \Big[\ln t \operatorname{artanh} t\Big]_{1/3}^{1/2} - \int_{1/3}^{1/2} \frac{\ln t}{1-t^2}\,\mathrm{d}t \\ &= \Big[\ln t \operatorname{artanh} t\Big]_{1/3}^{1/2} + V, \end{aligned}$$

since the remaining integral is exactly V (the name of the dummy variable doesn't matter).

Ask yourself Doesn't the boundary term spoil the equality?

It vanishes, and this is the neat part of the question. At the upper limit $\ln\frac{1}{2}\cdot\operatorname{artanh}\frac{1}{2} = (-\ln 2)(\frac{1}{2}\ln 3)$; at the lower limit $\ln\frac{1}{3}\cdot\operatorname{artanh}\frac{1}{3} = (-\ln 3)(\frac{1}{2}\ln 2)$. Both are $-\frac{1}{2}\ln 2\ln 3$, so

$$\Big[\ln t\,\operatorname{artanh} t\Big]_{1/3}^{1/2} = -\tfrac{1}{2}\ln 2\ln 3 + \tfrac{1}{2}\ln 2\ln 3 = 0,$$

and therefore $T = V$.

$V = U$: the substitution $v = e^{-u}$

Put $v = e^{-u}$, so $\ln v = -u$ and $\mathrm{d}v = -e^{-u}\,\mathrm{d}u$. When $v = \frac{1}{3}$, $u = \ln 3$; when $v = \frac{1}{2}$, $u = \ln 2$. Then

$$\begin{aligned} V &= -\int_{\ln 3}^{\ln 2} \frac{-u}{1-e^{-2u}}\left(-e^{-u}\right)\mathrm{d}u = -\int_{\ln 3}^{\ln 2} \frac{u\,e^{-u}}{1-e^{-2u}}\,\mathrm{d}u \\ &= \int_{\ln 2}^{\ln 3} \frac{u\,e^{-u}}{1-e^{-2u}}\,\mathrm{d}u \qquad \text{(swapping the limits)} \\ &= \int_{\ln 2}^{\ln 3} \frac{u}{e^{u}-e^{-u}}\,\mathrm{d}u \qquad \text{(multiplying top and bottom by } e^{u}\text{)} \\ &= \int_{\ln 2}^{\ln 3} \frac{u}{2\sinh u}\,\mathrm{d}u = U. \end{aligned}$$

$X = T$: the substitution $t = e^{-2x}$

First rewrite the integrand of X. For $x > 0$,

$$\coth x = \frac{e^{x}+e^{-x}}{e^{x}-e^{-x}} = \frac{1+e^{-2x}}{1-e^{-2x}}, \qquad \text{so} \qquad \ln(\coth x) = \ln\frac{1+e^{-2x}}{1-e^{-2x}} = 2\operatorname{artanh}\left(e^{-2x}\right),$$

which is valid because $0 < e^{-2x} < 1$. Now put $t = e^{-2x}$, so $\mathrm{d}t = -2t\,\mathrm{d}x$, that is $\mathrm{d}x = -\frac{\mathrm{d}t}{2t}$. When $x = \frac{1}{2}\ln 2$, $t = \frac{1}{2}$; when $x = \frac{1}{2}\ln 3$, $t = \frac{1}{3}$. Hence

$$X = \int_{1/2}^{1/3} 2\operatorname{artanh} t\left(-\frac{1}{2t}\right)\mathrm{d}t = \int_{1/3}^{1/2} \frac{\operatorname{artanh} t}{t}\,\mathrm{d}t = T.$$

Conclusion

Putting the three links together, $X = T = V = U$, so all four integrals are equal, as required. We never had to find a value for any of them.

Checks

Simpson's rule on each integral separately gives $T \approx U \approx V \approx X \approx 0.177704$, agreeing to about 12 decimal places. A quick check of the vanishing boundary term on

a calculator is also reassuring: $\ln\frac{1}{2}\cdot\operatorname{artanh}\frac{1}{2}$ and $\ln\frac{1}{3}\cdot\operatorname{artanh}\frac{1}{3}$ both come out as $-0.3807\ldots$.

Common slips

- Losing a minus sign in $V = U$: there are three of them (the one in front of V, $\ln v = -u$ and $\mathrm{d}v = -e^{-u}\,\mathrm{d}u$), and swapping the limits supplies a fourth.
- Abandoning the integration by parts because the boundary term looks non-zero, without evaluating it at both ends.
- Writing $\ln\coth x = \operatorname{artanh}(e^{-2x})$ and dropping the factor 2; it is cancelled later by the $\frac{1}{2}$ in $\mathrm{d}x = -\frac{\mathrm{d}t}{2t}$.

Beyond STEP

Expanding $\operatorname{artanh} t/t$ as a power series shows that the common value is $\chi_2(\frac{1}{2}) - \chi_2(\frac{1}{3})$, where $\chi_2(z) = \sum_{k\geq 0} z^{2k+1}/(2k+1)^2$ is Legendre's chi function, a close relative of the dilogarithm $\mathrm{Li}_2(z) = \sum_{n\geq 1} z^n/n^2$. Many identities for these functions come from exactly this kind of change of variable. Showing that two integrals agree without computing either is a standard technique in analysis, and the theorems that justify substitution and integration by parts (continuity, the fundamental theorem of calculus) are made precise in a first analysis course. Abbott's *Understanding Analysis* is a good place to see them proved carefully.

WORKED SOLUTION

STEP III 2011, Question 7

Pure • Proof by Induction

STEP III 2011, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q7

The key idea

Squaring $\sqrt{a+1}+\sqrt{a}$ gets rid of the separate square roots and leaves a number of the shape $p+q\sqrt{a(a+1)}$, so for even n we can step from T_n to T_{n+2} by multiplying by T_2. The quantity $A^2-a(a+1)B^2$ then turns out to be unchanged by that multiplication, which is exactly what the induction needs. One more factor of $\sqrt{a+1}+\sqrt{a}$ handles odd n, and the identities in (i) and (ii) say that the two terms are square roots of consecutive integers.

Solution

Throughout, write $N=a(a+1)$, so that $\sqrt{a+1}\,\sqrt{a}=\sqrt{N}$.

Part (i)

Base case, $n=2$. Expanding,

$$T_2=(a+1)+2\sqrt{a+1}\,\sqrt{a}+a=(2a+1)+2\sqrt{N},$$

so $A_2=2a+1$ and $B_2=2$ are integers, and

$$A_2^2-NB_2^2=4a^2+4a+1-4a(a+1)=1.$$

Inductive step. Suppose that for some even $n\geq 2$ we have $T_n=A_n+B_n\sqrt{N}$ with integers A_n, B_n satisfying $A_n^2-NB_n^2=1$. Write $p=2a+1$ for short, and note that $p^2=4a^2+4a+1=4N+1$. Since $T_{n+2}=T_nT_2$,

$$\begin{aligned}T_{n+2}&=\left(A_n+B_n\sqrt{N}\right)\left(p+2\sqrt{N}\right)\\&=\left(pA_n+2NB_n\right)+\left(2A_n+pB_n\right)\sqrt{N}.\end{aligned}$$

So we may take

$$A_{n+2}=pA_n+2NB_n,\qquad B_{n+2}=2A_n+pB_n,$$

which are integers because p, N, A_n and B_n are. For the identity, the cross terms $\pm 4pNA_nB_n$ cancel:

$$\begin{aligned} A_{n+2}^2 - NB_{n+2}^2 &= \left(p^2A_n^2 + 4pNA_nB_n + 4N^2B_n^2\right) - N\left(4A_n^2 + 4pA_nB_n + p^2B_n^2\right) \\ &= (p^2 - 4N)A_n^2 - N(p^2 - 4N)B_n^2 \\ &= A_n^2 - NB_n^2 = 1, \end{aligned}$$

using $p^2 - 4N = 1$ and then the inductive hypothesis. So the statement holds for $n+2$, and by induction it holds for every even positive integer n.

Note also that $A_2, B_2 > 0$ and the recurrences have positive coefficients, so $A_n > 0$ and $B_n > 0$ for every even n. We shall need this in part (iii).

Ask yourself Why does $A^2 - NB^2$ survive multiplication by T_2?

Because $A^2 - NB^2 = (A + B\sqrt{N})(A - B\sqrt{N})$, and the same calculation shows that the "conjugate" $A - B\sqrt{N}$ is multiplied by $p - 2\sqrt{N}$ at each step. So the product gets multiplied by $(p + 2\sqrt{N})(p - 2\sqrt{N}) = p^2 - 4N = 1$. The algebra above is just this fact written out.

Part (ii)

The case $n = 1$. $T_1 = 1 \cdot \sqrt{a+1} + 1 \cdot \sqrt{a}$, so $C_1 = D_1 = 1$, and $(a+1)C_1^2 = a + 1 = aD_1^2 + 1$.

The case $n \geq 3$ odd. Then $m = n - 1$ is even and positive, so by part (i) $T_m = A_m + B_m\sqrt{N}$ with $A_m^2 - NB_m^2 = 1$. Using $\sqrt{N}\sqrt{a+1} = (a+1)\sqrt{a}$ and $\sqrt{N}\sqrt{a} = a\sqrt{a+1}$,

$$\begin{aligned} T_n = \left(\sqrt{a+1} + \sqrt{a}\right)\left(A_m + B_m\sqrt{N}\right) &= A_m\sqrt{a+1} + A_m\sqrt{a} + (a+1)B_m\sqrt{a} + aB_m\sqrt{a+1} \\ &= \left(A_m + aB_m\right)\sqrt{a+1} + \left(A_m + (a+1)B_m\right)\sqrt{a}. \end{aligned}$$

So $C_n = A_m + aB_m$ and $D_n = A_m + (a+1)B_m$ are integers (and positive). Now

$$\begin{aligned} (a+1)C_n^2 - aD_n^2 &= (a+1)\left(A_m^2 + 2aA_mB_m + a^2B_m^2\right) \\ &\quad - a\left(A_m^2 + 2(a+1)A_mB_m + (a+1)^2B_m^2\right) \\ &= A_m^2 + a(a+1)\left[a - (a+1)\right]B_m^2 \\ &= A_m^2 - NB_m^2 = 1, \end{aligned}$$

since the A_mB_m terms cancel. Hence $(a+1)C_n^2 = aD_n^2 + 1$, as required. (The case $n = 1$ is really the same calculation with $T_0 = 1$, that is $A_0 = 1$, $B_0 = 0$.)

Part (iii)

n even. Since $A_n > 0$ and $B_n > 0$,

$$T_n = A_n + B_n\sqrt{N} = \sqrt{A_n^2} + \sqrt{NB_n^2},$$

and by part (i) the integers NB_n^2 and $A_n^2 = NB_n^2 + 1$ are consecutive.

Ask yourself Why do we need the signs of A_n and B_n?

Because $\sqrt{A^2} = |A|$, not A. Writing $A_n = \sqrt{A_n^2}$ is only valid once we know $A_n \geq 0$, which is why we recorded positivity in part (i).

n odd. Since $C_n, D_n > 0$,

$$T_n = C_n\sqrt{a+1} + D_n\sqrt{a} = \sqrt{(a+1)C_n^2} + \sqrt{aD_n^2},$$

and by part (ii) the integers aD_n^2 and $(a+1)C_n^2 = aD_n^2 + 1$ are consecutive.

So for every positive integer n, $T_n = \sqrt{k+1} + \sqrt{k}$ for some positive integer k: namely $k = a(a+1)B_n^2$ for even n and $k = aD_n^2$ for odd n.

Checks

Take $a = 3$, so $T_n = (2+\sqrt{3})^n$. Then $T_2 = 7 + 4\sqrt{3} = 7 + 2\sqrt{12} = \sqrt{49} + \sqrt{48}$, and $49 = 12 \cdot 4 + 1$. Next, $A_2 = 7$, $B_2 = 2$ give $C_3 = 13$, $D_3 = 15$, and $T_3 = 26 + 15\sqrt{3} = \sqrt{676} + \sqrt{675} \approx 51.9808$, which agrees with $(2+\sqrt{3})^3$ computed directly. A computer ran the recurrences for $1 \leq a \leq 39$ and $n \leq 23$, confirming both identities exactly in integer arithmetic and the expansions of T_n numerically.

Common slips

- Forgetting the base case, or starting the induction at $n = 1$ when part (i) is only about even n.
- Writing $A_n = \sqrt{A_n^2}$ in part (iii) without noting that A_n (and B_n, C_n, D_n) are positive.

Beyond STEP

The equation $x^2 - Ny^2 = 1$ is Pell's equation, and part (i) shows that powers of $(2a+1) + 2\sqrt{a(a+1)}$ generate infinitely many solutions for $N = a(a+1)$. The key fact is that the "norm" $x^2 - Ny^2$ of $x + y\sqrt{N}$ is multiplicative, which is the start of the theory of quadratic number fields and their units. Davenport's *The Higher Arithmetic* has a very readable account of Pell's equation and continued fractions.

WORKED SOLUTION

STEP III 2011, Question 8

Pure • Complex Loci

STEP III 2011, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q8

The key idea

Multiply top and bottom by the conjugate of the denominator to get u and v; after that, each part is a matter of substituting the given line and spotting a familiar curve. The half-angle substitution turns $\frac{2x}{1+x^2}$ and $\frac{x^2-1}{x^2+1}$ into a sine and a cosine, which is exactly why the question suggests it. Throughout, keep an eye on *which* points of the curve are actually reached: the endpoints and missing points are half the answer.

Solution

Write $z = x + \mathrm{i}y$. Then $1 + \mathrm{i}z = (1-y) + \mathrm{i}x$ and $\mathrm{i} + z = x + \mathrm{i}(1+y)$, so, multiplying numerator and denominator by $x - \mathrm{i}(1+y)$,

$$\begin{aligned} w &= \frac{\big((1-y)+\mathrm{i}x\big)\big(x-\mathrm{i}(1+y)\big)}{x^2+(1+y)^2} \\ &= \frac{(1-y)x + x(1+y) + \mathrm{i}\big(x^2-(1-y)(1+y)\big)}{x^2+(1+y)^2}. \end{aligned}$$

Hence

$$u = \frac{2x}{x^2+(1+y)^2}, \qquad v = \frac{x^2+y^2-1}{x^2+(1+y)^2},$$

valid for $z \neq -\mathrm{i}$ (where w is undefined).

It also helps to have the inverse map. From $w(\mathrm{i}+z) = 1 + \mathrm{i}z$ we get $z(w-\mathrm{i}) = 1 - \mathrm{i}w$, so

$$z = \frac{1-\mathrm{i}w}{w-\mathrm{i}}, \qquad w \neq \mathrm{i}.$$

So every $w \neq \mathrm{i}$ comes from exactly one z, and $w = \mathrm{i}$ is never attained. That is the omitted point we shall keep meeting.

Part (i)

With $y = 0$,

$$u = \frac{2x}{1+x^2}, \qquad v = \frac{x^2-1}{x^2+1}.$$

Put $x = \tan(\theta/2)$. As x runs over all real numbers, θ runs over $-\pi < \theta < \pi$, and each x gives exactly one such θ. By the half-angle formulae,

$$u = \frac{2\tan(\theta/2)}{1+\tan^2(\theta/2)} = \sin\theta, \qquad v = -\frac{1-\tan^2(\theta/2)}{1+\tan^2(\theta/2)} = -\cos\theta.$$

So $u^2+v^2 = 1$. Moreover $(\sin\theta, -\cos\theta)$ for $-\pi < \theta < \pi$ traces the whole unit circle except the point given by $\theta = \pm\pi$, namely $(u,v) = (0,1)$. So the locus of w is the circle $u^2+v^2 = 1$ with the single point $w = \mathrm{i}$ omitted, as required. (The omitted point is the limit of w as $x \to \pm\infty$.)

Part (ii)

Now $-1 < x < 1$, which corresponds to $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. On this range $v = -\cos\theta < 0$, and $u = \sin\theta$ increases from -1 to 1 (not inclusive). So the locus of w is

$$u^2+v^2 = 1, \quad v < 0,$$

the lower half of the unit circle, *not* including its endpoints $w = \pm 1$ (which come from $x = \pm 1$).

Part (iii)

With $x = 0$ we get $u = 0$ and

$$v = \frac{y^2-1}{(1+y)^2} = \frac{(y-1)(y+1)}{(y+1)^2} = \frac{y-1}{y+1} = 1 - \frac{2}{y+1},$$

where cancelling $y+1$ is fine because $y > -1$.

? **Ask yourself** Is every negative value of v reached, or only some of them?

As y increases from -1 to 1, $y+1$ increases through $(0,2)$, so $\frac{2}{y+1}$ decreases through $(1,\infty)$, and v increases continuously and strictly through $(-\infty, 0)$. As $y \to -1^+$, $v \to -\infty$, and as $y \to 1^-$, $v \to 0^-$. So every negative v is reached exactly once. The locus of w is the negative imaginary axis

$$u = 0, \quad v < 0,$$

with the origin excluded (it comes from $y = 1$).

Part (iv)

With $y = 1$,

$$u = \frac{2x}{x^2+4}, \qquad v = \frac{x^2}{x^2+4}.$$

Then

$$u^2+v^2=\frac{4x^2+x^4}{(x^2+4)^2}=\frac{x^2(x^2+4)}{(x^2+4)^2}=\frac{x^2}{x^2+4}=v,$$

so w lies on $u^2+v^2=v$, that is,

$$u^2+\left(v-\tfrac{1}{2}\right)^2=\tfrac{1}{4},$$

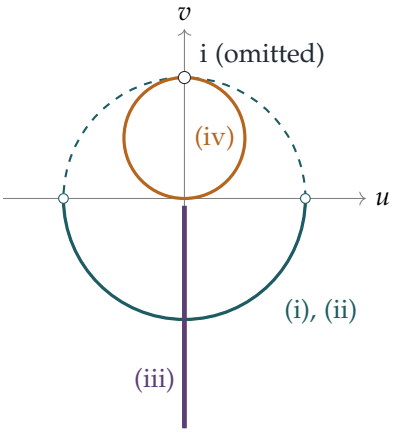
the circle with centre $\frac{1}{2}\mathrm{i}$ and radius $\frac{1}{2}$.

Ask yourself Which points of this circle are actually reached?

Since $v=1-\frac{4}{x^2+4}$, we have $0\le v<1$, so the top point $w=\mathrm{i}$ is never reached. Every other point is: setting $x=2\tan(\phi/2)$ with $-\pi<\phi<\pi$ gives

$$u=\frac{\tan(\phi/2)}{1+\tan^2(\phi/2)}=\tfrac{1}{2}\sin\phi,\qquad v=\frac{\tan^2(\phi/2)}{1+\tan^2(\phi/2)}=\tfrac{1}{2}(1-\cos\phi),$$

which runs round the whole circle except $\phi=\pm\pi$, the point $(0,1)$. (Alternatively, the inverse map shows that each $w\neq\mathrm{i}$ comes from exactly one z.) So the locus is the circle $u^2+v^2=v$ with the point $w=\mathrm{i}$ omitted. It touches the real axis at $w=0$, the image of $z=\mathrm{i}$.



The dashed and solid teal circle is the image of the real axis, part (i); the solid lower half is part (ii). The plum ray is part (iii) and the amber circle is part (iv).

Checks

- $z=0$ gives $w=1/\mathrm{i}=-\mathrm{i}$: on the lower semicircle (ii) and on the ray (iii), and indeed $z=0$ lies on both segments.
- $z=1+\mathrm{i}$ gives $w=\frac{\mathrm{i}}{1+2\mathrm{i}}=0.4+0.2\mathrm{i}$, and $0.4^2+0.2^2=0.2=v$, so it lies on the circle in (iv).

- Large real z: $w(10^8) \approx 2 \times 10^{-8} + \mathrm{i}$, approaching the omitted point i.
- We also checked numerically, for thousands of random points, both the formulae for u, v and that every point of each claimed locus has a preimage on the right line or segment.

Common slips

- Giving the circle or semicircle but not saying which points are omitted or excluded: here these are $w = \mathrm{i}$ in (i) and (iv), and the endpoints ± 1 in (ii).
- Claiming the locus in (iii) is the whole imaginary axis: $v = \frac{y-1}{y+1}$ is negative for every $-1 < y < 1$.

Beyond STEP

The map is a Möbius transformation; in fact $w = \mathrm{i}\,\dfrac{z-\mathrm{i}}{z+\mathrm{i}}$, so $|w| = |z-\mathrm{i}|/|z+\mathrm{i}|$, which equals 1 exactly when z is equidistant from $\pm\mathrm{i}$, i.e. real. Möbius maps send lines and circles to lines and circles (a line being a "circle through ∞"), and the omitted point i is the image of ∞. This map (a Cayley transform) sends the upper half-plane onto the unit disc, a key tool in complex analysis and hyperbolic geometry. Needham's *Visual Complex Analysis* has a beautiful, picture-led treatment of Möbius transformations.

WORKED SOLUTION

STEP III 2011, Question 9

Mechanics • Motion on a Cylinder

STEP III 2011, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

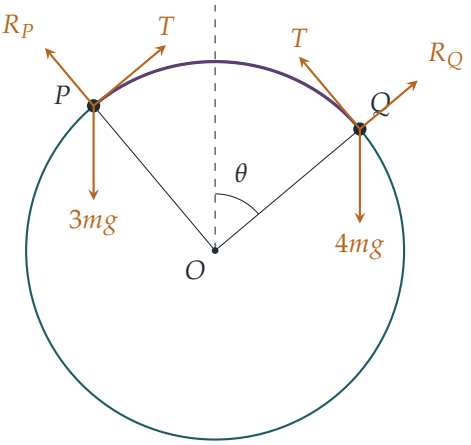
step.maths.org/questions/11-s3-q9

The key idea

A string of length $\frac{1}{2}\pi a$ lying on the cylinder subtends a right angle at O, so OP and OQ are always perpendicular and the whole system has a single coordinate, θ. Energy conservation then gives $\dot{\theta}^2$ directly, and Newton's second law towards the centre gives each normal reaction in terms of $\dot{\theta}^2$. Losing contact means the normal reaction falls to zero. For the tension, use the tangential equations: adding them eliminates T and gives $\ddot{\theta}$, and either one alone then gives T.

Solution

Model. P and Q are particles, the string is light and inextensible, and the cylinder is smooth. A light string over a smooth surface has the same tension T throughout. The arc between the particles has length $\frac{1}{2}\pi a$, so $\angle POQ = \frac{1}{2}\pi$. Measure θ as the angle OQ makes with the upward vertical, with Q on one side of the top. Then OP makes angle $\frac{1}{2}\pi - \theta$ with the upward vertical on the other side. While the particles stay on the surface, both move on the circle of radius a with angular speed $\dot{\theta}$, so each has speed $a\dot{\theta}$.



Equilibrium. Resolving along the surface at each particle (the normal reaction has no tangential component, and the string pulls tangentially towards the top), for Q we get $T = 4mg\sin\theta$ and for P we get $T = 3mg\sin(\frac{1}{2}\pi - \theta) = 3mg\cos\theta$. So initially $\tan\theta = \frac{3}{4}$, that is, $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$.

Energy. Take O as the level of zero potential energy. Q is at height $a\cos\theta$ and P at height $a\cos(\frac{1}{2}\pi - \theta) = a\sin\theta$. The normal reactions do no work (they're perpendicular to the motion), and the tension does no net work on the system (the string is inextensible), so energy is conserved. The system starts from rest, as the disturbance is slight, so

$$\tfrac{1}{2}(3m + 4m)a^2\dot\theta^2 + 4mga\cos\theta + 3mga\sin\theta = 4mga\cdot\tfrac{4}{5} + 3mga\cdot\tfrac{3}{5} = 5mga.$$

Multiplying by $2/(ma)$ gives

$$7a\dot\theta^2 + 8g\cos\theta + 6g\sin\theta = 10g,$$

as required. As Q moves down, θ increases from $\tan^{-1}\frac{3}{4}$.

Part (i)

Q moves on a circle of radius a with speed $a\dot\theta$, so it has acceleration $a\dot\theta^2$ towards O. The component of its weight towards O is $4mg\cos\theta$, and R_Q acts outwards. The string is tangential, so it contributes nothing radially. Hence

$$4mg\cos\theta - R_Q = 4ma\dot\theta^2, \qquad \text{so} \qquad R_Q = 4mg\cos\theta - 4ma\dot\theta^2.$$

Q leaves the surface when R_Q falls to zero, that is, when $a\dot\theta^2 = g\cos\theta$. Substituting this into the energy equation at $\theta = \beta$:

$$7g\cos\beta + 8g\cos\beta + 6g\sin\beta = 10g \quad\Longrightarrow\quad 15\cos\beta + 6\sin\beta = 10,$$

as required.

? **Ask yourself** Why does the energy equation still hold right up to $\theta = \beta$?

It holds as long as both particles are on the cylinder, and the question tells us Q leaves first. You can also check this. The same argument for P gives $R_P = 3mg\sin\theta - 3ma\dot\theta^2$, which vanishes when $13\sin\theta + 8\cos\theta = 10$. At $\beta \approx 73.6^\circ$ the left-hand side is about 14.7, so R_P is still positive.

Part (ii)

Now resolve tangentially, taking the direction of motion as positive. Q moves away from the top with tangential acceleration $a\ddot\theta$, and P moves towards the top with the same tangential acceleration $a\ddot\theta$ (the string keeps their speeds equal). For Q, gravity has component $4mg\sin\theta$ forwards and the tension pulls back:

$$4ma\ddot\theta = 4mg\sin\theta - T. \tag{1}$$

For P, the tension pulls it forwards and gravity has component $3mg\cos\theta$ backwards:

$$3ma\ddot{\theta} = T - 3mg\cos\theta. \tag{2}$$

Adding (1) and (2) eliminates T:

$$7ma\ddot{\theta} = 4mg\sin\theta - 3mg\cos\theta.$$

(This agrees with differentiating the energy equation with respect to time and dividing by $2\dot{\theta}$.) Substituting into (1):

$$\begin{aligned} T &= 4mg\sin\theta - 4ma\ddot{\theta} \\ &= 4mg\sin\theta - \tfrac{4}{7}m(4g\sin\theta - 3g\cos\theta) \\ &= \tfrac{12}{7}mg\sin\theta + \tfrac{12}{7}mg\cos\theta = \tfrac{12}{7}mg(\sin\theta + \cos\theta), \end{aligned}$$

as required. This is positive for $0 < \theta < \frac{1}{2}\pi$, so the string stays taut throughout.

Checks

- At equilibrium, $\sin\theta + \cos\theta = \frac{7}{5}$, so $T = \frac{12}{5}mg$. This matches $4mg \cdot \frac{3}{5}$ and $3mg \cdot \frac{4}{5}$.
- Writing $15\cos\beta + 6\sin\beta = \sqrt{261}\cos(\beta - \alpha)$ with $\tan\alpha = \frac{2}{5}$ gives $\beta \approx 1.2839$ rad $\approx 73.56°$. Stepping the motion numerically from just past equilibrium, we found R_Q first reaches zero at this same angle, while R_P stays positive. The energy relation and the tension formula held along the whole path to within rounding error.

Common slips

- Taking P's height as $a\cos\theta$ rather than $a\sin\theta$. The right angle $\angle POQ$ is the whole point of the string's length.
- Giving each particle a different tension, or putting T into the radial equation. The string touches the particle tangentially, so T appears only in the tangential equations.

Beyond STEP

With one coordinate θ describing everything, this is a one-degree-of-freedom system. Lagrangian mechanics gets the equation of motion straight from the kinetic and potential energies, with no forces of constraint at all. The normal reactions and tension then come out as the "Lagrange multipliers" of the constraints. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen your feel for constrained motion and energy methods before meeting Lagrange.

WORKED SOLUTION

STEP III 2011, Question 10

Mechanics • Elastic Strings and SHM

STEP III 2011, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q10

The key idea

No external horizontal force acts on the pair, so their centre of mass moves at a constant speed fixed by the impulse. Relative to the centre of mass, the extension of the string obeys simple harmonic motion, and the string is taut for exactly half an oscillation. Splitting the motion into "centre of mass" plus "extension" makes every answer drop out with almost no work.

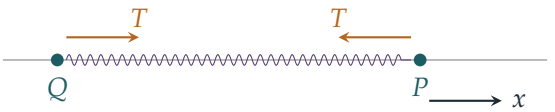
Solution

Model and set-up

The particles move on a smooth horizontal plane, so the only horizontal forces on them are the tensions in the light string, and these are equal and opposite. Take the line QP as the x-axis, pointing from Q to P, with Q starting at $x = 0$ and P at $x = a$. Let x_P and x_Q be their positions at time t after the impulse, and let

$$e = (x_P - x_Q) - a$$

be the extension of the string. Straight after the impulse, $\dot{x}_P = u$, $\dot{x}_Q = 0$ and $e = 0$, with $\dot{e} = u > 0$, so the string at once becomes taut.



While the string is taut, Hooke's law gives tension

$$T = \frac{\lambda e}{a} = \frac{\frac{1}{2}ma\omega^2 e}{a} = \tfrac{1}{2}m\omega^2 e.$$

The tension pulls P back towards Q and pulls Q forward towards P:

$$m\ddot{x}_P = -\tfrac{1}{2}m\omega^2 e, \qquad m\ddot{x}_Q = +\tfrac{1}{2}m\omega^2 e.$$

The centre of mass

Adding the two equations, $\ddot{x}_P + \ddot{x}_Q = 0$. So the centre of mass G, at $\bar{x} = \frac{1}{2}(x_P + x_Q)$, moves with constant velocity equal to its initial value,

$$\dot{\bar{x}} = \tfrac{1}{2}(u + 0) = \tfrac{1}{2}u.$$

(This is just conservation of momentum: the total momentum mu is shared by mass $2m$.)

The extension

Subtracting the equations,

$$\ddot{e} = \ddot{x}_P - \ddot{x}_Q = -\omega^2 e.$$

This is simple harmonic motion with angular frequency ω. With $e(0) = 0$ and $\dot{e}(0) = u$,

$$e = \frac{u}{\omega}\sin\omega t.$$

Ask yourself Why is the angular frequency ω and not $\omega/\sqrt{2}$ or $\sqrt{2}\,\omega$?

Each particle feels the force $\frac{1}{2}m\omega^2 e$, but e changes because *both* particles move: their accelerations add when we subtract the equations. Equivalently, the reduced mass of two equal masses m is $\frac{1}{2}m$, and the string's stiffness is $\lambda/a = \frac{1}{2}m\omega^2$, so $\omega^2 = \frac{1}{2}m\omega^2 / \frac{1}{2}m$. The modulus was chosen to make this come out neatly.

The string is next of length a when $e = 0$ again, that is when $\sin\omega t = 0$ for the first time with $t > 0$:

$$t_1 = \frac{\pi}{\omega}.$$

Throughout $0 < t < \pi/\omega$ we have $e > 0$, so the string really is taut and the equations above hold.

Distance travelled

Writing each position as the centre of mass plus or minus half the separation,

$$x_P = \bar{x} + \tfrac{1}{2}(a + e), \qquad x_Q = \bar{x} - \tfrac{1}{2}(a + e).$$

At t_1, $e = 0$ again, so each particle has moved by exactly the same amount as G, namely

$$\tfrac{1}{2}u \cdot \frac{\pi}{\omega} = \frac{\pi u}{2\omega}.$$

To be sure this displacement is the distance travelled, we check that neither particle ever turns back. Differentiating,

$$\dot{x}_P = \tfrac{1}{2}u + \tfrac{1}{2}\dot{e} = \tfrac{1}{2}u(1 + \cos\omega t), \qquad \dot{x}_Q = \tfrac{1}{2}u - \tfrac{1}{2}\dot{e} = \tfrac{1}{2}u(1 - \cos\omega t),$$

and both are ≥ 0 for all t. So each particle moves only forwards, and each has travelled a distance $\dfrac{\pi u}{2\omega}$, as required.

The speeds

At $t_1 = \pi/\omega$, $\cos \omega t_1 = -1$, so

$$\dot{x}_P = 0, \qquad \dot{x}_Q = u.$$

So P is momentarily at rest and Q moves with speed u towards P. The particles have exactly swapped velocities, just as in a perfectly elastic collision of equal masses. That's no accident: momentum is conserved, and the elastic energy stored has been completely returned as kinetic energy.

The time to collision

After t_1, Q is moving towards P, so the separation is less than a. The string is slack and exerts no force, and neither particle feels any horizontal force. Hence P stays at rest and Q moves at constant speed u, starting a distance a from P. It reaches P after a further time a/u. The total time from the impulse to the collision is

$$\frac{\pi}{\omega} + \frac{a}{u}.$$

Checks

- Momentum: at t_1 the total momentum is $m \cdot 0 + mu = mu$, as at the start. Energy: kinetic energy $\frac{1}{2}mu^2$ at both moments, with the string at its natural length both times.
- Dimensions: u/ω is a length and a/u is a time, as they should be.
- We stepped the equations of motion numerically for $m = a = \omega = u = 1$. The string went slack at $t = 3.14159$ with each particle having moved $1.57080 = \pi/2$; the speeds were then 0 and 1, and the collision came at $t = 4.14159 = \pi + 1$.

Common slips

- Treating P as attached to a fixed point, which gives angular frequency $\omega/\sqrt{2}$ and the wrong time. Both ends of the string move.
- Continuing the simple harmonic motion after t_1. Once the string is shorter than a it is slack, so the motion is uniform.
- Forgetting the final stage altogether: the collision happens a further a/u after the string returns to its natural length.

Beyond STEP

Splitting a two-body problem into motion of the centre of mass and motion of the separation, governed by the reduced mass $\mu = m_1 m_2/(m_1 + m_2)$, is one of the most

useful tricks in first-year mechanics. It turns planetary orbits, vibrating diatomic molecules and colliding particles into one-body problems. Here the elastic string acts as a "soft" collision that lasts a time π/ω but has exactly the same outcome as an instantaneous elastic one. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the two-body reduction and the reduced mass carefully.

WORKED SOLUTION

STEP III 2011, Question 11

Mechanics • Rigid Body Rotation

STEP III 2011, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q11

The key idea

After the twist, every string still runs from its fixed ceiling point to the point of the rim that has moved round by θ. The horizontal part of each string is therefore a *chord* of the circle, so the geometry comes from looking down from above. By symmetry every string behaves the same way, so the vertical equilibrium fixes the tension, and the moments of the horizontal parts of the tensions about the axis give the couple. Once the disc is released, energy conservation does the rest.

Solution

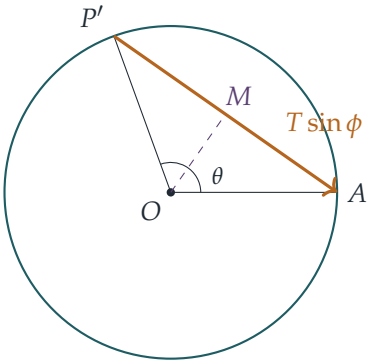
We model the disc as a rigid uniform lamina and the strings as light and inextensible. By symmetry, all the strings are taut with the same tension T and the same inclination ϕ, and the disc stays horizontal with its centre on the z-axis. When the disc is turned through θ, it rises to a depth h below the ceiling.

The angle of the string

The string through P is attached to the ceiling at $A = (a, 0, 0)$. After the rotation its lower end is at

$$P' = (a\cos\theta,\ a\sin\theta,\ -h).$$

Seen from above, A and P' both lie on the circle of radius a centred on the axis, with angle θ between them at the centre O. So the horizontal projection of the string is a chord of length $2a\sin\frac{1}{2}\theta$.



The view from above. M is the midpoint of the chord $P'A$, so OM bisects the angle θ, is perpendicular to the chord and has length $OM = a\cos\frac{1}{2}\theta$. The string itself has length b and makes angle ϕ with the vertical, so its horizontal projection has length $b\sin\phi$. Equating the two,

$$b\sin\phi = 2a\sin\tfrac{1}{2}\theta,$$

as required. Its vertical projection gives the new depth of the disc:

$$h = b\cos\phi = \sqrt{b^2 - 4a^2\sin^2\tfrac{1}{2}\theta}.$$

This is real because $b > 2a$. Since $0 < \theta < \pi$, we have $0 < \sin\frac{1}{2}\theta < 1$, so $0 < \phi < \frac{1}{2}\pi$.

The couple

Resolving vertically for the disc, the n equal tensions support the weight:

$$nT\cos\phi = mg.$$

The couple C acts in the plane of the disc, so it has no effect on this balance.

Each tension has a vertical component and a horizontal component $T\sin\phi$ along the chord from P' towards A. The vertical components are parallel to the axis, so they have no moment about it. The horizontal component acts along the chord, whose perpendicular distance from O is $OM = a\cos\frac{1}{2}\theta$. Every string's horizontal pull tends to turn the disc back towards $\theta = 0$, so all n moments have the same sense. Taking moments about the axis,

$$\begin{aligned} C &= n\,T\sin\phi\cdot a\cos\tfrac{1}{2}\theta = mg\tan\phi\; a\cos\tfrac{1}{2}\theta \\ &= mg\,a\cos\tfrac{1}{2}\theta\cdot\frac{2a\sin\frac{1}{2}\theta}{\sqrt{b^2-4a^2\sin^2\frac{1}{2}\theta}} = \frac{mga^2\sin\theta}{\sqrt{b^2-4a^2\sin^2\frac{1}{2}\theta}}, \end{aligned}$$

using $nT = mg/\cos\phi$, then $\tan\phi = b\sin\phi/(b\cos\phi)$, and finally $2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta = \sin\theta$. This is the required magnitude.

Release from rest

Once the couple is removed, the disc untwists and falls. The strings are inextensible and stay taut, so they do no work: each string's tension is perpendicular to the velocity of its attachment point relative to the fixed ceiling point. Gravity is the only force that does work, so mechanical energy is conserved.

By the geometry above, when the disc has turned through α from the vertical-string position, its depth is

$$z(\alpha) = \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\alpha} = \sqrt{b^2 - 2a^2(1 - \cos\alpha)}.$$

The disc's kinetic energy is $\frac{1}{2}m\dot{z}^2 + \frac{1}{2}I\omega^2$, with $I = \frac{1}{2}ma^2$ about the axis.

Ask yourself When the strings are vertical, does the centre of the disc still have a vertical velocity?

No. Differentiating,

$$\dot{z} = \frac{\mathrm{d}z}{\mathrm{d}\alpha}\,\dot{\alpha} = -\frac{a^2 \sin\alpha}{z}\,\dot{\alpha},$$

which is zero at $\alpha = 0$. The strings are vertical at that instant, and the disc is at its lowest point, so all the kinetic energy is rotational there.

The disc falls from depth $\sqrt{b^2 - 4a^2 \sin^2 \frac{1}{2}\theta}$ to depth b. Energy conservation gives

$$\tfrac{1}{2} \cdot \tfrac{1}{2}ma^2\,\omega^2 = mg\left(b - \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\theta}\right),$$

that is,

$$\frac{a^2\omega^2}{4g} = b - \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\theta},$$

as required.

Checks

- *Virtual work.* Turning the disc through a further $\delta\theta$ against the couple raises it by $-\delta z$, so $C = mg\,|\mathrm{d}z/\mathrm{d}\theta| = mg\,a^2 \sin\theta / z$. This is exactly the couple we found by taking moments.
- *Small twists.* For small θ, $C \approx mga^2\theta/b$, which is the familiar restoring couple of a bifilar (here n-filar) suspension.
- *Numerical.* We placed the n strings explicitly in 3D and summed the moments of the tensions for random a, b, θ, m, n. The result agreed with the formula. We also integrated the motion numerically after release with $a = 1$, $b = 3$, $\theta = 2$, $g = 9.81$. That gave $a^2\omega^2/(4g) = 0.51651$ when the strings became vertical, matching $3 - \sqrt{9 - 4\sin^2 1} = 0.51651$.

Common slips

- Taking the lever arm of the horizontal tension as a rather than $a\cos\frac{1}{2}\theta$. The force acts along the chord, not along a tangent.
- Forgetting that the disc *rises* when it is twisted. This change of height is exactly where the energy comes from in the last part.
- Including the translational kinetic energy $\frac{1}{2}m\dot{z}^2$ without noticing that it vanishes when the strings are vertical.

Beyond STEP

This is a system with one degree of freedom whose motion is constrained. The depth is a fixed function of the twist angle, so the energy depends only on α and $\dot{\alpha}$. Lagrangian mechanics turns that observation into equations of motion. It also explains why the couple is just $mg\,\mathrm{d}z/\mathrm{d}\alpha$, which is the principle of virtual work. Linearising about $\alpha = 0$ gives simple harmonic torsional oscillations, which is how a bifilar suspension is used to measure moments of inertia. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation and energy methods at this level.

WORKED SOLUTION

STEP III 2011, Question 12

Probability and Statistics • Compound Distributions via PGFs

STEP III 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q12

The key idea

Everything we need about a distribution's mean and variance sits in the first two derivatives of its pgf at $t = 1$. So we differentiate the composite $G(H(t))$ with the chain rule (twice), put $t = 1$, and use $H(1) = 1$. For the coin, N is geometric and each X_i is a single fair-coin indicator, so both pgfs are simple and $G(H(t))$ is a rational function we can expand as a geometric series.

Solution

The general results

For any random variable Z taking non-negative integer values with pgf $P(t) = \mathrm{E}(t^Z)$, we have $P(1) = 1$,

$$P'(1) = \mathrm{E}(Z), \qquad P''(1) = \mathrm{E}\big(Z(Z-1)\big),$$

and therefore

$$\mathrm{Var}(Z) = P''(1) + P'(1) - \big(P'(1)\big)^2.$$

Write $\mu = \mathrm{E}(X_i) = H'(1)$, $\sigma^2 = \mathrm{Var}(X_i)$, and $K(t) = G(H(t))$ for the pgf of Y.

By the chain rule, $K'(t) = G'(H(t))\,H'(t)$. Since $H(1) = 1$,

$$\mathrm{E}(Y) = K'(1) = G'(1)H'(1) = \mathrm{E}(N)\,\mathrm{E}(X_i),$$

which is the first result.

Differentiating again (product rule, then chain rule),

$$K''(t) = G''(H(t))\big(H'(t)\big)^2 + G'(H(t))\,H''(t),$$

so, again using $H(1) = 1$,

$$K''(1) = G''(1)\mu^2 + \mathrm{E}(N)\,H''(1).$$

From the variance formula above applied to N and to X_i,

$$G''(1) = \mathrm{Var}(N) - \mathrm{E}(N) + \big(\mathrm{E}(N)\big)^2, \qquad H''(1) = \sigma^2 - \mu + \mu^2.$$

Ask yourself Why not just compute $\mathrm{Var}(Y)$ from $K''(1)$ directly, without rewriting $G''(1)$ and $H''(1)$?

Because the answer we want is in terms of variances, not factorial moments; expressing $G''(1)$ and $H''(1)$ through $\mathrm{Var}(N)$ and σ^2 lets the unwanted terms cancel. Writing $m = \mathrm{E}(N)$,

$$\begin{aligned}\mathrm{Var}(Y) &= K''(1) + K'(1) - \big(K'(1)\big)^2 \\ &= \big(\mathrm{Var}(N) - m + m^2\big)\mu^2 + m\big(\sigma^2 - \mu + \mu^2\big) + m\mu - m^2\mu^2 \\ &= \mathrm{Var}(N)\mu^2 + m\sigma^2 + \big(-m\mu^2 + m\mu^2\big) + \big(m^2\mu^2 - m^2\mu^2\big) + \big(-m\mu + m\mu\big) \\ &= \mathrm{Var}(N)\big(\mathrm{E}(X_i)\big)^2 + \mathrm{E}(N)\,\mathrm{Var}(X_i),\end{aligned}$$

as required.

The coin

The pgf of N. The first head comes on toss n when the first $n-1$ tosses are tails and the nth a head, so $\mathrm{P}(N = n) = \left(\frac{1}{2}\right)^n$ for $n = 1, 2, 3, \ldots$ (a geometric distribution). Hence, for $|t| < 2$,

$$G(t) = \sum_{n=1}^{\infty} \left(\frac{t}{2}\right)^n = \frac{t/2}{1 - t/2} = \frac{t}{2-t}.$$

The pgf of each X_i. Let $X_i = 1$ if the ith of the further tosses is a head and $X_i = 0$ otherwise. Then $Y = X_1 + \cdots + X_N$, and $H(t) = \frac{1}{2} + \frac{1}{2}t = \frac{1}{2}(1+t)$. The further tosses are independent of each other and of the tosses that decided N, so the conditions of the general result hold.

The pgf of Y.

$$G(H(t)) = \frac{\frac{1}{2}(1+t)}{2 - \frac{1}{2}(1+t)} = \frac{1+t}{3-t}.$$

Mean and variance. For the geometric distribution with $p = \frac{1}{2}$, $\mathrm{E}(N) = 1/p = 2$ and $\mathrm{Var}(N) = (1-p)/p^2 = 2$ (or differentiate G: $G'(t) = 2/(2-t)^2$, $G''(t) = 4/(2-t)^3$, giving $G'(1) = 2$, $G''(1) = 4$ and $\mathrm{Var}(N) = 4 + 2 - 4 = 2$). Each X_i is Bernoulli$\left(\frac{1}{2}\right)$, so $\mathrm{E}(X_i) = \frac{1}{2}$ and $\mathrm{Var}(X_i) = \frac{1}{4}$. Therefore

$$\mathrm{E}(Y) = 2 \cdot \tfrac{1}{2} = 1, \qquad \mathrm{Var}(Y) = 2 \cdot \tfrac{1}{4} + 2 \cdot \tfrac{1}{4} = 1.$$

The distribution of Y. For $|t| < 3$ we expand

$$\frac{1+t}{3-t} = \frac{1+t}{3} \cdot \frac{1}{1 - t/3} = \frac{1+t}{3} \sum_{k=0}^{\infty} \frac{t^k}{3^k} = \sum_{k=0}^{\infty} \frac{t^k}{3^{k+1}} + \sum_{k=0}^{\infty} \frac{t^{k+1}}{3^{k+1}}.$$

The coefficient of t^0 comes only from the first sum. For $r \geq 1$ the coefficient of t^r is $\dfrac{1}{3^{r+1}} + \dfrac{1}{3^r} = \dfrac{4}{3^{r+1}}$. So

$$\mathrm{P}(Y = 0) = \frac{1}{3}, \qquad \mathrm{P}(Y = r) = \frac{4}{3^{r+1}} \quad (r = 1, 2, 3, \ldots).$$

Checks

- The probabilities sum to 1: $\frac{1}{3} + 4\sum_{r\geq 1} 3^{-(r+1)} = \frac{1}{3} + 4 \cdot \frac{1}{6} = 1$.
- $\mathrm{P}(Y = 0)$ directly: $\sum_{n\geq 1} \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = \sum_{n\geq 1} 4^{-n} = \frac{1}{3}$. Similarly $\mathrm{P}(Y = 1) = \sum_{n\geq 1} n\, 4^{-n} = \frac{4}{9}$, agreeing with $4/3^2$.
- Summing $\mathrm{P}(N = n)\binom{n}{r}2^{-n}$ exactly over $n \leq 200$ reproduced $\mathrm{P}(Y = r)$ for $r \leq 7$, and gave mean and variance both 1.000; a simulation of 200 000 runs gave 0.996 and 0.994.

Common slips

- Taking N to start at 0 (pgf $1/(2 - t)$): here N counts the toss that gives the head, so $N \geq 1$.
- Forgetting that $P''(1)$ is $\mathrm{E}(Z^2) - \mathrm{E}(Z)$, not $\mathrm{E}(Z^2)$, and so losing the $+P'(1)$ in the variance formula.
- Giving $4/3^{r+1}$ for all $r \geq 0$: the $r = 0$ term is different.

Beyond STEP

Y is a *random sum* (a compound distribution), and the variance formula is a special case of the law of total variance, $\mathrm{Var}(Y) = \mathrm{E}\big(\mathrm{Var}(Y \mid N)\big) + \mathrm{Var}\big(\mathrm{E}(Y \mid N)\big)$. Compositions of pgfs are also the engine of branching processes, where $G(G(\cdots G(t)))$ gives the size of the nth generation and the chance of extinction is a fixed point of G. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions, random sums and branching processes at exactly this level.

WORKED SOLUTION

STEP III 2011, Question 13

Probability and Statistics • Mode of Discrete Distributions

STEP III 2011, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q13

The key idea

To find the most likely value of a discrete random variable, we don't differentiate: we compare neighbouring probabilities. The ratio $\mathrm{P}(X = r+1)/\mathrm{P}(X = r)$ is a simple rational function of r, and asking when it exceeds 1 turns into a linear inequality in r. The probabilities rise while the ratio is above 1 and fall once it drops below, so the peak sits at the last r before the switch. The same method works with and without replacement; only the ratio changes.

Solution

Part (i)

With replacement, the draws are independent and each is black with probability $p = b/n$, so $X \sim \mathrm{B}(k, p)$:

$$\mathrm{P}(X = r) = \binom{k}{r} p^r (1-p)^{k-r}, \qquad r = 0, 1, \ldots, k.$$

Assume first that $0 < b < n$, so that every one of these probabilities is positive. For $0 \le r \le k-1$,

$$\frac{\mathrm{P}(X = r+1)}{\mathrm{P}(X = r)} = \frac{k-r}{r+1} \cdot \frac{p}{1-p} = \frac{(k-r)\,b}{(r+1)(n-b)}.$$

Since the denominator is positive, this ratio exceeds 1 exactly when

$$\begin{aligned} & (k-r)b > (r+1)(n-b) \\ \iff \quad & kb - rb > rn - rb + n - b \\ \iff \quad & (k+1)b > (r+1)n \\ \iff \quad & r + 1 < \frac{(k+1)b}{n}. \end{aligned}$$

In the same way the ratio equals 1 exactly when $r+1=m$ and is less than 1 exactly when $r+1>m$, where

$$m=\frac{(k+1)b}{n}.$$

So, as r increases, $\mathrm{P}(X=r)$ strictly increases at each step that ends at a value $r+1<m$, and strictly decreases at each step ending at $r+1>m$. Note that $0<m<k+1$ because $0<b<n$, so the switch happens inside the range $0,\ldots,k$.

Ask yourself Where exactly is the largest probability if m is not a whole number?

If m is not an integer, write $M=\lfloor m\rfloor$, so $M<m<M+1$ and $0\le M\le k$. The probabilities strictly increase up to $\mathrm{P}(X=M)$ (every step ends at a value $\le M<m$) and strictly decrease after it (every later step ends at a value $\ge M+1>m$). Hence the unique most probable number of black balls is

$$\left\lfloor\frac{(k+1)b}{n}\right\rfloor,$$

as required.

When it isn't unique. If m is an integer, the step from $r=m-1$ to $r=m$ has ratio exactly 1, so $\mathrm{P}(X=m-1)=\mathrm{P}(X=m)$, and these two equal values are larger than all the others (the probabilities strictly rise to $m-1$ and strictly fall after m). Since $1\le m\le k$, both values are possible. So the answer is not unique precisely when $(k+1)b/n$ is an integer (with $0<b<n$), and then $m-1$ and m are equally most probable. For example, $n=3$, $b=1$, $k=5$ gives $m=2$, and indeed $\mathrm{P}(X=1)=\mathrm{P}(X=2)=\frac{80}{243}$.

The trivial cases sit outside this: if $b=0$ then $X=0$ for certain, which agrees with the formula; if $b=n$ then $X=k$ for certain, and the formula (which would give $k+1$) doesn't apply, because the ratio above involves dividing by $n-b$.

Part (ii)

Without replacement, X has the hypergeometric distribution: choosing the sample as an unordered set of k balls, all $\binom{n}{k}$ sets equally likely,

$$\mathrm{P}(X=r)=\frac{\binom{b}{r}\binom{n-b}{k-r}}{\binom{n}{k}}.$$

Because $k\le b$, the largest possible value is $r=k$; the smallest is $r_0=\max(0,\,k-(n-b))$, since we can't draw more than $n-b$ non-black balls. For $r_0\le r\le k$ the probability is positive, and for $r_0\le r\le k-1$, using $\binom{a}{r+1}=\binom{a}{r}\frac{a-r}{r+1}$ and $\binom{a}{s-1}=\binom{a}{s}\frac{s}{a-s+1}$,

$$\frac{\mathrm{P}(X=r+1)}{\mathrm{P}(X=r)}=\frac{b-r}{r+1}\cdot\frac{k-r}{n-b-k+r+1}.$$

Both denominators are positive in this range, so the ratio exceeds 1 exactly when

$$(b-r)(k-r) > (r+1)(n-b-k+r+1).$$

Expanding, the left side is $bk - r(b+k) + r^2$ and the right side is $(r+1)(n-b-k+1) + r^2 + r$. The r^2 terms cancel, and the inequality becomes

$$\begin{aligned} & bk - r(b+k) > r(n-b-k+1) + (n-b-k+1) + r \\ \iff \quad & bk + b + k + 1 > r(n+2) + (n+2) \\ \iff \quad & r + 1 < \frac{(b+1)(k+1)}{n+2}. \end{aligned}$$

Ask yourself Does the switch point actually lie inside the range of possible values $r_0, \ldots, k$?

Let $m = \dfrac{(b+1)(k+1)}{n+2}$. Since $b+1 < n+2$ we have $m < k+1$. For the lower end, $m > 0$ trivially, and if $r_0 = k+b-n > 0$ we need $(b+1)(k+1) > (n+2)(k+b-n)$. Writing $x = n-b$ and $y = n-k$, the left side minus the right side is

$$(n+1-x)(n+1-y) - \left[(n+1)^2 - 1 - (n+2)(x+y)\right] = 1 + x + y + xy > 0.$$

So $r_0 < m < k+1$ in every case, and the argument of part (i) goes through word for word: the probabilities strictly rise while $r+1 < m$ and strictly fall once $r+1 > m$. Hence, when it is unique, the most probable number of black balls is

$$\left\lfloor \frac{(b+1)(k+1)}{n+2} \right\rfloor.$$

It is not unique exactly when $(b+1)(k+1)/(n+2)$ is an integer; then $m-1$ and m (both possible values, since $r_0 \le m-1$ and $m \le k$) are equally likely and more likely than anything else. For example, $n = 10$, $b = 7$, $k = 5$ gives $m = 4$, and $X = 3$ and $X = 4$ are tied, each with probability $\frac{105}{252} = \frac{5}{12}$.

Checks

- By computer we computed both distributions exactly with fractions for every $n \le 24$, $0 \le b \le n$ and $1 \le k \le 19$ (with $k \le b$ in part (ii)), found the modes by brute force and confirmed both formulas and both tie conditions in all cases (2565 cases for part (ii)).
- For $n = 10$, $b = 6$, $k = 4$: $m = \frac{35}{12}$, so the formula gives 2, and $X = 2$ is indeed the unique mode.
- If n is huge compared with k, removing a few balls barely changes the proportion of black balls, so part (ii) should look like part (i). It does: $(b+1)(k+1)/(n+2) \approx (k+1)b/n$.

Common slips

- Concluding that the mode is $\lfloor kb/n \rfloor$, the integer part of the mean. The mode is close to the mean but not always equal to its integer part.
- Forgetting that when the ratio is exactly 1 there are two modes, not one.
- In part (ii), slipping a sign when expanding $(r+1)(n-b-k+r+1)$; the cancelling r^2 terms are a useful check.

Beyond STEP

Comparing successive probabilities is the standard way to locate the mode of any discrete distribution, and the same calculation gives the Poisson mode $\lfloor \lambda \rfloor$ as a limiting case of part (i). The hypergeometric distribution is central to sampling theory and to Fisher's exact test in statistics. Distributions whose probabilities rise and then fall like this are called unimodal; for the binomial and hypergeometric this comes from a stronger property, log-concavity. For more, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

This book takes the most interesting STEP questions, from the specimen papers that preceded the first examination in 1987 to the papers of 2025, and uses each one as a doorway. Every question is solved carefully, with the questions a good student asks along the way, and then the book steps through the door: where the idea comes from, what it becomes at university, and what to read next.

It is written for talented aspiring mathematicians preparing for STEP who want more than practice papers: to start thinking like a mathematics undergraduate, and to find out what is waiting for them.

- 13 chapters across pure mathematics, mechanics, probability and statistics
- 136 past STEP questions, each with a guided solution
- The university mathematics behind every question
- Follow-up exercises with answers, and further reading

2011 Worked Solutions. The whole book, every chapter and all the worked solutions are free to read, download and share at **beyondstep.keithmcnulty.org**

Text dedicated to the public domain (CC0 1.0). STEP questions © Cambridge University Press & Assessment.

